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REVIEW PAPER

Casimir forces: from Au to time crystals

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Abstract

The Casimir effect, a fundamental manifestation of quantum vacuum fluctuations, has evolved from a curiosity between idealized metallic plates to a versatile probe of quantum and material properties in structured and time-dependent media. This review traces the progression of Casimir force research from its classical formulation in noble metals like gold (Au) to cutting-edge developments involving metamaterials and time crystals. Beyond surveying recent experimental and theoretical advances, we provide a unifying framework that bridges Casimir physics with essential solid-state principles—including band theory, dielectric response, and temporal periodicity—critical for understanding forces in real and engineered materials. Special attention is given to non-equilibrium and dynamically modulated systems, where time-periodic structures can profoundly alter vacuum-induced interactions. By synthesizing insights from quantum field theory and condensed matter physics, this review offers a comprehensive perspective on the rich interplay between quantum fluctuations, material response, and temporal symmetry breaking.

Keywords Casimir · Vacuum fluctuations · MEMS · Optical properties

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1 Introduction

The year 2025 has been designated as the International Year of Quantum Science and Technology, offering a timely occasion to reflect on the breadth and impact of quantum phenomena across modern physics. Among the most striking consequences of quantum field theory is the Casimir effect, a force arising from the quantum fluctuations of the vacuum electromagnetic field in the presence of material boundaries. First predicted in 1948 by H. B. G. Casimir [1] in the context of perfectly conducting plates, the Casimir force has evolved from a theoretical curiosity to an essential feature in nanotechnology, microelectromechanical systems (MEMS), and fundamental tests of quantum theory. Initially derived for two idealized perfectly conducting plates in vacuum, the Casimir force is now understood as a broader class of dispersion interactions governed by fluctuations in the quantum electromagnetic field, as formalized by the Lifshitz theory[2–4] for real materials at finite temperature.

A renewed series of modern Casimir experiments began in the late 1990 s with the measurement by Lamoreaux [5], who used a torsional pendulum with electromechanical feedback and achieved agreement at the 5% level with Lifshitz theory. Shortly afterwards, Mohideen and Roy [6] introduced AFM-based techniques that enabled submicron measurements with picoNewton sensitivity, marking the beginning of high-precision Casimir metrology. Subsequent studies, including those by Sushkov et al. [7], offered results that were interpreted in different ways by various authors and helped stimulate a broader examination of finite-temperature effects and material modeling. Previously, Decca et al. [8] established the first systematic high-accuracy constraints on theoretical descriptions of the Casimir force, further motivating refinements in the treatment of material response.

Experiments in the early 2000s included different material properties, geometry, and surface effects. Experiments by Chan et al. [9] integrated Casimir forces into MEMS, showing that quantum forces could actuate mechanical components at the nanoscale. Decca and collaborators made important contributions to precision measurements of the Casimir force using a micromechanical torsional oscillator. In pioneering experiments dating back to 2003, they achieved the first precise measurement of the Casimir force between dissimilar metals [8] (a copper-coated oscillator and a gold-coated sphere), with a noise floor as low as $\sim 6\,fN/\sqrt{Hz}$ over separations of 0.2-2 μ m; their results agreed with theoretical predictions including finite conductivity and surface roughness to better than 1% at the closest separations. Building on this, their measurements using two gold-coated parallel plates measured via the force gradient, and proximity-force approximation achieved an unprecedented experimental relative error of about 0.5% across a broad separation range, and enabled exclusion—at the 99% confidence level—of the Drude model in favor of the plasma or impedance approaches in Lifshitz theory. These results also imposed significantly tighter constraints on hypothetical Yukawa-type deviations from Newtonian gravity in the nanometer range [10–13].

In 2002, Bressi et al. [14] performed a parallel plate Casimir measurement using a MEMS-based setup. Although challenging due to alignment and stability issues, this work confirmed Casimir's original geometry and emphasized the importance of precise micro fabrication. An interesting breakthrough came in 2009, when it was

demonstrated that Casimir–Lifshitz repulsion in a liquid medium is possible[15, 16]. By placing a gold sphere and a silica plate in bromobenzene, they reversed the sign of the force, showing that it can be repulsive. This opened possibilities for Casimir engineering, where interactions can be tuned or canceled altogether [17–20].

Beyond noble metals, contemporary research has extended Casimir studies into low-dimensional and topologically nontrivial systems. Graphene, with its gapless Dirac spectrum and tunable conductivity, exhibits a Casimir interaction that can be modulated via doping and gating [21–29]. Topological insulators [30, 31] and Weyl semi-metals [32, 33] introduce additional complexity through axion electrodynamics and anomalous Hall responses, leading to quantized or anisotropic Casimir interactions under certain conditions [18, 34]. Recent theoretical work has predicted repulsive Casimir forces between topological insulator slabs separated by a vacuum gap when time-reversal symmetry is broken at the surfaces [35].

Metamaterials and photonic crystals further enrich the phenomenology by enabling bandgap engineering of vacuum modes [36, 37]. Studies have shown that hyperbolic metamaterials [38] can dramatically alter the spectral density of vacuum fluctuations, leading to non-monotonic and even sign-reversed Casimir forces [39].

Emerging interest in non-equilibrium and time-dependent Casimir effects has introduced an entirely new paradigm. Systems with time-modulated dielectric properties can produce dynamical Casimir effects (DCE), where photon pairs are generated from vacuum due to parametric modulation [40]. Time crystals—phases of matter characterized by spontaneously broken discrete time-translation symmetry [41]—have been proposed as platforms to study temporally modulated Casimir forces. Recent theoretical models suggest that periodic temporal ordering in such systems could lead to Floquet-modified vacuum spectra and time-dependent Casimir interactions [39]. These developments open intriguing possibilities for tuning vacuum forces in both amplitude and phase, and for harnessing time-periodic materials in active Casimir-based devices.

This review surveys the landscape of Casimir force research in modern materials science, beginning with classical metal systems and extending to the frontier of non-equilibrium quantum materials. We emphasize the interplay between material properties and vacuum fluctuations, with a focus on experimental realizations, theoretical advancements, and emerging applications.

While several excellent reviews [42–44] and books [45] on the Casimir effect are available, this article aims to highlight the role of the optical response of materials, approached from the perspective of solid-state physics and contribute to the ongoing discussion and applications of Casimir physics. Inevitably, as with any review, it is not possible to provide an exhaustive account of all the literature on the subject.

2 Lifshitz theory

The original derivation of the Casimir force [1] assumed two infinite metallic parallel plates separated a distance d, with the following assumptions: a) the separation was larger than the skin depth δ, so the plates can be treated as perfect reflectors and b) there is a cutoff frequency ω_c, limiting the contribution of high-frequency modes.

Casimir calculated the change in the zero-point energy of the quantized electromagnetic field due to the presence of the cavity, by summing over allowed modes between the plates and subtracting the infinite vacuum energy outside the plates. The resulting net attractive force per unit area between the plates is

$$F_0 = -\frac{\hbar c \pi^2}{240 d^4}. \tag{1}$$

For realistic metallic plates, such as gold or aluminum, the dielectric response must be modeled via frequency-dependent dielectric functions $\varepsilon(\omega)$, incorporating dissipation and a finite plasma frequency. For metals modeled by this free-electron plasma, at frequencies above a characteristic cutoff ω_c, (determined by the metal's plasma response) related to the plasma frequency, the reflectivity drops and the material becomes increasingly transparent to electromagnetic waves. The Casimir force for such materials is no longer given by Eq. (1), but instead must be calculated using other methods like the Lifshitz theory, which generalizes Casimir's result to arbitrary, frequency-dependent dielectric functions. In this framework, the force depends on both the dielectric properties of the materials and the temperature, and can be expressed in terms of the reflection coefficients of the electromagnetic modes between the plates.

To briefly recall the essentials of Lifshitz theory, we consider two infinite, parallel plates separated by a distance d, characterized by dielectric functions $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$. The medium between the plates is described by a dielectric function $\varepsilon_3(\omega)$. Throughout this discussion, we primarily consider vacuum between the plates, i.e., $\varepsilon_3(\omega) = 1$, and assume nonmagnetic materials ($\mu = 1$) without loss of generality. Based on Rytov's theory of thermally fluctuating electromagnetic fields [46, 47], Lifshitz derived an expression for the Casimir force between two planar surfaces at finite temperature [4], commonly referred to as the Lifshitz formula. Later Dzyaloshinskii, Lifshitz, and Pitaevskii [3] made another derivation of the same formula. A pedagogical summary of these developments is presented in classical textbooks such as Ref. [2]. The expression for the Lifshitz formula is given by:

$$F(T,d) = \frac{\hbar}{2\pi^2} \int_0^\infty d\omega \coth\left(\frac{\hbar\omega}{2k_B T}\right) \mathrm{Re}\left[\int_0^\infty dQ\ Q\ k_0\ g(Q,\omega,d)\right], \tag{2}$$

where k_B is the Boltzmann constant and the wave vector in the gap has a component parallel to the slabs Q, contained in the xy-plane and a normal component or the z-component $k_0 = \sqrt{\omega^2/c^2 - Q^2}$. The function $g(Q,\omega,d)$ is expressed as

$$g(Q,\omega,d) = \sum_{\nu=s,p} \frac{r_1^\nu r_2^\nu e^{2ik_0 d}}{1 - r_1^\nu r_2^\nu e^{2ik_0 d}}. \tag{3}$$

Here, $r_{1,2}^\nu$ are the Fresnel reflection coefficients from the inner surfaces of the plates (index 1 or 2) for the polarizations $\nu = s$ (transverse electric (TE)) and $\nu = p$ (transverse magnetic (TM)). The frequency-dependent factor $\coth(\hbar\omega/2k_B T)$ originates from the fluctuation-dissipation theorem.

For two half-spaces, the reflection coefficients are expressed in terms of the z-component of the wave vectors in the gap k_0 and in the ith body k_i as

$$r_i^s = \frac{k_0 - k_i}{k_0 + k_i}, \quad r_i^p = \frac{\varepsilon_i k_0 - k_i}{\varepsilon_i k_0 + k_i}, \tag{4}$$

where

$$k_i = \sqrt{\varepsilon_i(\omega)\frac{\omega^2}{c^2} - Q^2}. \tag{5}$$

If instead of having two half-spaces, we had finite-width plates, layered media such as photonic crystals, or any other system, the Fresnel coefficients, in Eq.(4), have to be replaced by the appropriate reflection coefficients [48–50].

From the definition of k_0, for $Q < \omega/c$, we see that we have propagating waves, and the factor $g(Q, \omega)$ is oscillatory. If $Q > \omega/c$, we have evanescent waves and the exponential factors in $g(Q, \omega)$ decay exponentially. So, we see that the Casimir force has contributions from both propagating and evanescent waves [51–53].

2.1 Force at imaginary frequencies

Working with the Lifshitz formula in real frequency representation ω is not very convenient because the integrand in Eq. (2) is a fast oscillating function due to the factor $e^{ik_0 d}$, and care should be taken with fast oscillating integrals [54]. However, the integrand over frequency in Eq. (2) is analytic, and a contour rotation to the imaginary-frequency axis can be made, that is, $\omega \to i\zeta$. Using Cauchy's theorem to integrate over frequency, we obtain

$$F(T, d) = \frac{k_B T}{\pi} {\sum_{n=0}^{\infty}}' \int_0^{\infty} dQ\, Q\, |k_0|\, g\,(Q, i\zeta_n, d)\,. \tag{6}$$

The "prime" at the sum means that the $n = 0$ term must be taken with the weight 1/2, and $\zeta_n = 2\pi n \frac{k_B T}{\hbar}$ ($n \in \mathbb{N}$) are the Matsubara frequencies that result from the poles of $\coth(\hbar\omega/2k_B T)$. In the rotated-frequency space, $|k_0| = \sqrt{\zeta_n^2/c^2 + Q^2}$ and the function $g\,(Q, i\zeta_n, d)$ is no longer oscillating. Also, $k_i = \sqrt{\varepsilon_i(i\zeta)\zeta_n^2/c^2 + Q^2}$.

In the real frequency domain, the Kramers–Kronig relations have to be satisfied by the dielectric function, which, after rotation to the imaginary-frequency axis, is given by

$$\varepsilon(i\zeta) = 1 + \frac{2}{\pi}\int_0^{\infty} d\omega \frac{\omega \varepsilon''(\omega)}{\omega^2 + \zeta^2}, \tag{7}$$

where $\varepsilon''(\omega)$ is the imaginary part of the dielectric function. In this space, $\varepsilon(i\zeta)$ is real, positive, and monotonously decreasing with increasing ζ. The relation (7) shows that the Casimir–Lifshitz force is completely determined by $\varepsilon''(\omega)$.

A further simplification of Eq.(6) can be achieved by introducing the variable p with $p \in [1, \infty)$ defined by $Q^2 = (\zeta_n/c)^2(p^2 - 1)$. In this notation, the wave vectors take the form $k_i = (\zeta_n/c)\sqrt{\varepsilon_i(\zeta_n) - 1 + p^2} = (\zeta_n/c)s_i$ and $k_0 = p$, leading to the Fresnel coefficients in Eq. (38) as

$$r_i^s = \frac{p - s_i}{p + s_i}, \quad r_i^p = \frac{\varepsilon_i p - s_i}{\varepsilon_i p + s_i}, \tag{8}$$

and the Casimir pressure can be written as

$$F(T, d) = -\frac{k_B T}{\pi c^3} {\sum_{n=0}^{\infty}}' \zeta_n^3 \int_1^{\infty} p^2 g(p, i\zeta_n, d) dp, \tag{9}$$

with

$$g(p, i\zeta_n, d) = \sum_{\nu=s,p} \frac{r_1^\nu r_2^\nu e^{-2d(\zeta_n/c)p}}{1 - r_1^\nu r_2^\nu e^{-2d(\zeta_n/c)p}}. \tag{10}$$

From the term in the exponent, we can define a characteristic frequency of the system, which is $\zeta_c = c/2d$. Since $\zeta_n = \zeta_1 n$, the factor $g(p, i\zeta_n)$ takes the form

$$g(p, n) = \sum_{\nu=s,p} \frac{r_1^\nu r_2^\nu e^{-n(\zeta_1/\zeta_c)p}}{1 - r_1^\nu r_2^\nu e^{-n(\zeta_1/\zeta_c)p}}, \tag{11}$$

and Eq. (9) is

$$F(T, d) = -\frac{k_B T}{8\pi d^3} \left(\frac{\zeta_1}{\zeta_c}\right)^3 {\sum_{n=0}^{\infty}}' n^3 \int_1^{\infty} p^2 g(p, n) dp. \tag{12}$$

In the limit of $T \to 0$, and assuming perfect conductors, the previous equation reduces to the known expression for the Casimir force between two ideal metallic plates F_0.

Another relevant limiting case is that of small d. In this case, the force between the semispaces goes as d^{-3} or, more exactly,

$$F(d) = -\frac{H}{6\pi d^3}, \tag{13}$$

where H denotes the Hamaker constant [55], whose value depends upon the materials of the two interacting bodies and the medium in between. This limiting behavior is observed for d below a few nanometers and is also known as the non-retarded regime, because the propagation of the electromagnetic waves can be treated as instantaneous. See in Fig. 1 the results for the correction factor $\eta(d)$ defined above as Eq. (16). The force in Eq. (13) is also referred to as the van der Waals force between planes, because it was first obtained from the pairwise summation of van der Waals potentials between the atoms or molecules filling each semispace.

The previous discussion assumed an isotropic dielectric function for the plates. However, it can be generalized to anisotropic plates where the dielectric function is given by a tensor ε_{ij}. In this case, the anisotropy has to be taken into consideration as was done in Refs. [36, 56]. In the anisotropic case, the reflection coefficient must be changed by matrices

$$\mathbf{R}_i = \begin{pmatrix} r_i^s & 0 \\ 0 & r_i^p \end{pmatrix} \to \begin{pmatrix} r_i^{ss} & r_i^{sp} \\ r_i^{ps} & r_i^{pp} \end{pmatrix}. \tag{14}$$

Here $r_i^{\nu\mu}$ is the ratio of a reflected field with μ-polarization divided by an incoming field with ν-polarization for body i. In the Lifshitz formula, the product $r_1^{\nu} r_2^{\nu}$ must be changed by the matrix product $\mathbf{R}_1 \cdot \mathbf{R}_2$, and instead of the sum over polarizations, one has to take the matrix trace. The Lifshitz formula at $T = 0$ then takes the form

$$F(d) = \frac{\hbar}{2\pi^2} \mathrm{Tr} \int_0^\infty d\zeta \int_0^\infty dQ\, Q k_0 \frac{\mathbf{R}_1 \cdot \mathbf{R}_2 e^{-2k_0 d}}{1 - \mathbf{R}_1 \cdot \mathbf{R}_2 e^{-2k_0 d}}, \tag{15}$$

where $k_0 = \sqrt{\zeta^2/c^2 + Q^2}$. Detailed expressions for the ratios $r_i^{\nu\mu}$ and other information specific for anisotropy case can be found in Ref. [36].

The Casimir force between semispaces with real materials can be calculated by methods other than the Lifshitz theory. An import alternative method was proposed by van Kampen et al. [57], which simplifies quite considerably the task of obtaining Eq. (12) and similar equations valid for other planar or non-planar configurations, like the above result for anisotropic media. In this method, the eigenvalues of the electromagnetic field confined by the boundaries are found applying the boundary conditions at the surfaces, and then summed up. Appropriate regularization and renormalization procedures are then used for obtaining a finite result [45]. The method proposed in Ref. [57] was used, for instance, to generalize Eq. (12) to the case in which the semispaces are replaced by planar multilayers. This result was achieved in Refs. [58, 59]. The expressions for planar multilayers have allowed the investigation of the Casimir force in many relevant physical systems such as finite thickness slabs and surfaces covered with 2D materials.

Because the Casimir force calculated via the Lifshitz theory results in a force that is always smaller than that between two perfect conductors, it is customary in the Casimir literature to report the correction factor defined as the ratio

$$\eta(d, T) = \frac{F(T, d)}{F_0}. \tag{16}$$

This factor contains the information regarding the correction to the Casimir force due to the finite conductivity of the materials involved, and the effect of the temperature. Examples of η for gallium arsenide (GaAs) and amorphous silicon dioxide (a-SiO_2) are presented in Fig. 1. This correction factor can also include other effects like the contribution of the surface roughness discussed in Sect. 3.3 and the finite thickness effect [59].

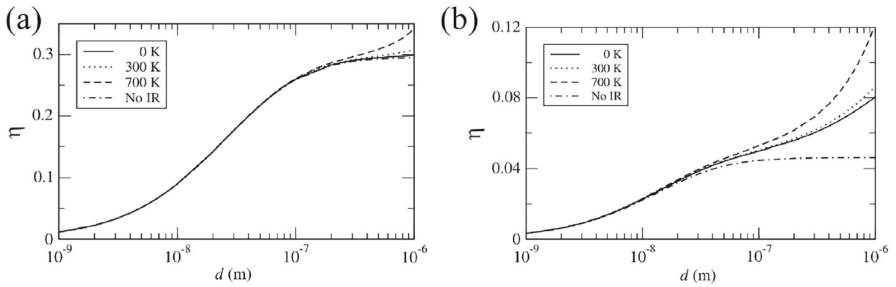


Fig. 1 The correction factor, Eq. (16), obtained with experimental optical data for (**a**) crystalline GaAs and (**b**) amorphous SiO_2. Results for different temperatures and with inclusion of the contributions from the IR. Also shown in the result at 0 K ignoring the contributions from the IR. Reproduced from Ref. [60]

3 Beyond the planar geometry

3.1 Approximate methods

The original Casimir force formulation as well as the Lifshitz theory are formulated for parallel plates. However, most experiments measure the force between a plate and a large sphere, using the proximity-force approximation (PFA) [6] to model the force predicted theoretically. In this particular case, this approximation is valid, provided the radius of the sphere is much larger than the smallest separation between the sphere and the plane [61]. The PFA can be used to calculate the Casimir force whenever the interacting surfaces are sufficiently smooth [45] and the bodies are in close proximity. The general expression for the force between two bodies is obtained assuming that the surfaces in close proximity can be considered as piecewise plane. It is then given by

$$F(d) = \int_S F(d')dA\,, \tag{17}$$

where $F(d')$ denotes the pressure between two infinitesimal planes separated by the distance d' given by Eq. (9), and the integral runs over the area of the two bodies. A similar expression can be defined to calculate the interaction energy.

The PFA has been used, for instance, to calculate the force between micro and nanodevices like torsional actuators, based on the tilting of large plates, or suspended beam resonators, which are curved by the actuating forces (Casimir, electrostatic, optical, etc) [62–64]. When the required conditions for application of the PFA are not satisfied, or when we have objects of arbitrary shape, other methods for calculating the Casimir force are needed.

Historically, a method first applied in the context of the limiting case of the van der Waals force, and later extended to the Casimir force, is the pairwise summation (PWS) [65]. It was applied initially to estimate the interaction between atoms within molecules. In its modern formulation, it generally assumes that small pieces of the bodies interact via a potential energy that depends upon the polarizability of the bulk material [66, 67]. The total energy is calculated integrating the interaction energy

over the volume of the bodies, and is a function of a separation d (closest approach, between centers, etc), or some other parameter defining the changes in the separation between these bodies, like a tilting angle. The potential energy between two small portions of each body is expected to be given approximately by the Casimir–Polder potential, $U = C_{12}|\mathbf{r}_2 - \mathbf{r}_1|^{-7}$, where the constant C_{12} is going to depend upon the materials that body 1 and 2 are made from. The total energy of interaction is obtained by integrating over the volumes V_1 and V_2 of the bodies

$$E(d) = C_{12} \int_{V_1} d\mathbf{r}_1 \int_{V_2} d\mathbf{r}_2 |\mathbf{r}_2 - \mathbf{r}_1|^{-7} . \tag{18}$$

The force can be calculated from the energy as

$$F(d) = -\frac{\partial E(d)}{\partial d} . \tag{19}$$

Alternatively, one can calculate the total force by integrating the force between the infinitesimal volumes in each body.

The PWS has been used to calculate the force for a great variety of complex geometries like in the interaction between spheres, plates, and cylinders or combinations of such objects [65, 68]. While comparison with results obtained using more physically consistent methods have shown that the PWS delivers good results when used for materials like insulators [69, 70], the method can fail significantly even for planar geometries for materials like semiconductors and metals [69, 71].

Recent experiments that measured the Casimir force between bodies of complex geometries have probed the limits of applicability of both PFA and PWS. They have measured the force between a metallic sphere and periodic rectangular gratings [72, 73], and between interpenetrating gratings of different geometries [74, 75]. Significant deviations from the forces predicted by the PWS or PFA have been observed. However, the measurements agreed well with the forces predicted using more general theoretical methods of calculation, like the ones described next.

3.2 Exact methods

As an example of the different approaches taken to calculate the Casimir force for more general geometries, we first consider the work of Bimonte and Emig [76, 77]. They calculated the Casimir force using scattering operators localized at the surface of the interacting bodies. The Casimir interaction can be expanded in the number of surface scatterings. A fundamental idea behind this approach is the proof by Balian and Duplantier of the convergence of a multiple scattering expansion to calculate the Casimir force between perfect conductors [78, 79]. The Casimir force between two objects of arbitrary shape is (See Fig. 2)

$$F = k_B T \sum_{n=0}^{\infty} Tr[(\mathbb{I} - \mathbb{K})^{-1} \nabla_r \mathbb{K}], \tag{20}$$

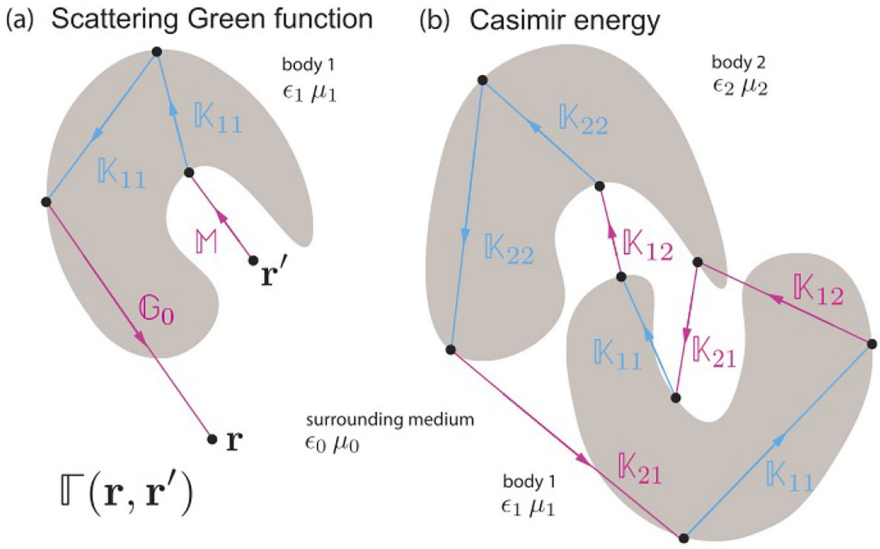


Fig. 2 Illustration of the scattering method: (**a**) scattering from a single body, (**b**) scattering operators between the bodies (red lines) and within each body (blue lines). Reprinted with permission from G. Bimonte et al. Physical Review A, **108**, 052807 (2023), Copyright 2023 the American Physical Society [76]

where $\mathbb{K}$ describes a free propagator. Multiple scattering is taken into account by expanding $(\mathbb{I} - \mathbb{K})^{-1}$. Figure. 2 shows the process of scattering for a single body and for two bodies. The components $\mathbb{K}_{ij}$ are the propagators between body i and j.

The surface scattering expansion closes the gap between exactly solvable planar geometries and realistic curved/structured magneto-dielectric surfaces, enabling systematic evaluation of Casimir and Casimir–Polder interactions that include both electric and magnetic material responses, beyond the limits of the proximity-force approximation. Other formulations of the scattering theory also exist in which the Casimir force is viewed as having its origin in the difference of the radiation pressure of vacuum fields between the two mirrors and in the outer free field vacuum. This vacuum radiation pressure can be written as an integral over all modes, each mode being associated with reflection amplitudes on the two mirrors [80].

Methods like the scattering theory can provide us with analytical expressions for the Casimir force in certain limiting cases. In more general configurations, however, the calculation of the Casimir force demands significant computational effort. Other methods have been developed, which are not intended for an analytical assessment of the force, but, instead, to allow its numerical calculation for arbitrary geometries and optical properties of the interacting systems. Two such numerical methods are based on Boundary Element Methods (BEM) and Finite-Difference Time Domain (FDTD). Both are general-purpose electromagnetic solvers adapted to compute fluctuation-induced forces, but they differ in formulation and computational strategy, as we now describe.

Table 1 Comparison of BEM and FDTD methods for Casimir force calculations

Feature	BEM	FDTD
Discretization	2D meshes	3D volumetric grid
Material	Piecewise-homogeneous	Arbitrary materials
Geometry	Smooth objects	Arbitrary
Efficiency	Fewer degrees of freedom	Memory/time demanding
Approach	Scattering/T-matrix	Stress tensor
Software	SCUFF- EM	MEEP

BEM reduces Maxwell's equations to integral equations on the surfaces of the objects [81]. It solves for induced surface currents or charges using the vacuum Green's function and computes the Casimir energy via scattering or stress-tensor formulations. Because only surfaces are discretized, the dimensionality is reduced (2D meshes instead of 3D volumes), making BEM efficient for piecewise-homogeneous objects [82–84]. BEM is implemented in open-source packages such as SCUFF- EM [85] (http://github.com/homerreid/scuff-EM).

In comparison, the FDTD method offers an alternative, highly flexible route to the computation of Casimir forces. Unlike BEM, FDTD discretizes Maxwell's equations on a volumetric grid in both space and time [86]. This allows the direct simulation of field fluctuations in arbitrarily complex geometries, including inhomogeneous, anisotropic, and dispersive media. Casimir forces can be expressed in terms of field correlation functions, which are in turn obtained from the fluctuation–dissipation theorem. In the time-domain formulation introduced by Rodriguez, Johnson, and co-workers [87, 88], the electromagnetic Green's functions are computed by evolving Maxwell's equations with impulsive sources and recording the system's temporal response. After a Wick rotation to imaginary frequencies, the Casimir energy is recovered from stress-tensor integrals or equivalently from spectral densities of the field fluctuations. This approach exploits the fact that time-domain solvers already accommodate dispersive materials and complex geometries without modification. The FDTD procedure can be implemented using the open source code MEEP (https://meep.readthedocs.io/en/latest/).

Table 1 summarizes the main features of both BEM and FDTD.

3.3 Surface roughness effect

So far, we have seen that the interaction between two bodies is determined by their optical properties and geometry. In principle, the dependence of the Casimir force on these characteristics of the bodies cannot be separated. However, under certain conditions, this separation is possible as an approximation, as it is more clearly done in the case of PFA and PWS. Another physical property that influences the Casimir force is the surface roughness. While it can be considered as part of the geometry of the interacting bodies, in practice, it has been convenient to treat the effect of surface roughness separately.

Here, we refer to the surface roughness as the formation of random patterns in a smooth background surface, like a plane or a sphere. Roughness is present, for instance, in surfaces of micro- and nanodevices, probe tips used in atomic force microscopy, and in colloidal particles. Rough surfaces can assume a great variety of forms, and many different parameters are required to characterize their properties like the dispersion, skewness, kurtosis, correlation length, the Hurst exponent and fractal dimension, among others [89].

In spite of fact that theoretical and experimental evidences suggest that the contribution of the surface roughness to the actual Casimir force can be quite relevant, there is no general method to calculate its effect for arbitrary surface roughness profiles and surface separation. The first attempts to model the effect of surface roughness on the van der Waals force were made by van Bree et al. [90] and relied on the use of PWS. This work was extended to the Casimir force by Bordag et al. [91]. Both works approached the effect of the surface roughness perturbatively. The force between two plane semispaces covered with roughness with dispersion σ_1 and σ_2 up to second order in the small parameters σ/d can be obtained evaluating Eq. (18) for the energy and then substituting in Eq. (19) [91] giving

$$F(d) = F_0(d)\left\{1 + 10\left[\left(\frac{\sigma_1}{d}\right)^2 + \left(\frac{\sigma_2}{d}\right)^2\right] + 105\left[\left(\frac{\sigma_1}{d}\right)^2 + \left(\frac{\sigma_2}{d}\right)^2\right]^2\right\}, \quad (21)$$

where $F_0(d)$ denotes the force between the two planes separated by the mean surface distance d including the finite conductivity correction. The term between braces can be seen as the surface roughness correction to the force between two semispaces with plane boundaries. Due to the simplicity of the method, it is possible to derive exact, nonperturbative, expressions for the roughness effect using the PWS [69, 92]. The PFA can also be used to obtain both exact and perturbative expressions for the roughness correction [45]. However, the results obtained with PWS and PFA have strict ranges of applicability, for either the exact or the perturbative results. PFA provides reliable results only for surfaces with very smooth deformations [93]. This can be translated on conditions over the correlation length λ, the height dispersion σ, and the mean surface separation d. The results obtained from the PFA are reliable when $\sigma \ll d \ll \lambda$. The limit of application of PWS is not so clear. It is expected to be valid under conditions similar to those for the PFA [45]. However, as argued by Gusso and Burnham in Ref. [69], it could also be valid for very rough surfaces under certain conditions. Their results for the interaction between planar rough surfaces and those of Ref. [70] for the interaction between small particles with rough geometry indicate the validity of PWS for materials with the optical properties of insulators, like silicon dioxide.

The roughness effect was also investigated using the scattering method by Maia-Neto et al. [93]. The second-order perturbative calculations have a range of validity that extends well beyond those of the PFA. However, the results in Ref. [93] are still limited to satisfy the condition $\sigma \ll \lambda$. The main improvement is that the predictions are valid for any separation d. Also, rougher surfaces can be assessed because λ now can be smaller than d. More recently, Wu and Schaden [94] calculated the roughness effect resorting to an effective low-energy field theory.

It is interesting to note that while the results obtained using the scattering theory predict an increase in the force with the decrease of λ, the predictions of Wu and Schaden [94] go in the opposite direction. These contradictory results have established a controversy that deserves further theoretical and experimental investigation.

Another method based on the Lifshitz theory has been developed that can be applied to surfaces with short scale roughness, that means, surfaces with λ and σ of the same order. The method proposed in Refs. [95, 96] models the rough surface as a layered effective optical medium whose effective dielectric functions are calculated using different mixing rules like Clausius–Mossoti and Bruggeman mixing rules. The rough region is subdivided as a multilayer and the Casimir force calculated using known generalizations of the Lifshitz formula for planar multilayers. This method predicts a surface roughness effect that is larger than that obtained using the scattering method.

Finally, resorting to the numerical method proposed by Reid et al., based on the computation of surface currents, Liu et al. [70] have investigated the interaction between small particles with rough surfaces.

While the roughness effect has been extensively investigated both theoretically and experimentally [95, 97], there are still many open questions regarding, particularly, the case for which σ, λ, and d are of the same magnitude.

4 Optical properties of materials

4.1 General description

The Casimir–Lifshitz force depends critically on the electromagnetic response of the interacting media. Within the Lifshitz formalism, this response enters through the complex permittivity $\varepsilon(\omega)$ and permeability $\mu(\omega)$ of the materials and the intervening medium. For most materials relevant to Casimir experiments, $\mu(\omega) \approx 1$ over the contributing spectral range, rendering the magnetic contribution negligible. Consequently, the dielectric response dominates the magnitude and sign of the force.

The dielectric tensor of a general anisotropic material may be written as.[1]

$$\boldsymbol{\varepsilon}(\mathbf{k}, \omega) = \varepsilon'_{ij}(\mathbf{k}, \omega) + i\, \varepsilon''_{ij}(\mathbf{k}, \omega), \tag{22}$$

where nonlocality (dependence on $\mathbf{k}$) accounts for spatial dispersion. In most situations, this effect is small, and one may employ the local approximation $\boldsymbol{\varepsilon}(\omega)$, as we discuss later.

The concept of dielectric function comes from the linear response theory of electrodynamics in continuum media [2], but the origin of the macroscopic optical response comes from the quantum nature of materials [98]. This means that different spectral regimes can explain the optical response. For crystalline solids, distinct physical processes dominate different regions of the spectrum:

- Infrared (IR): absorption due to optical phonons and lattice vibrations

[1] In the rest of this review, we are going to use the equivalent notations to refer to the components of the complex dielectric function: $\varepsilon' = \varepsilon_1$ and $\varepsilon'' = \varepsilon_2$.

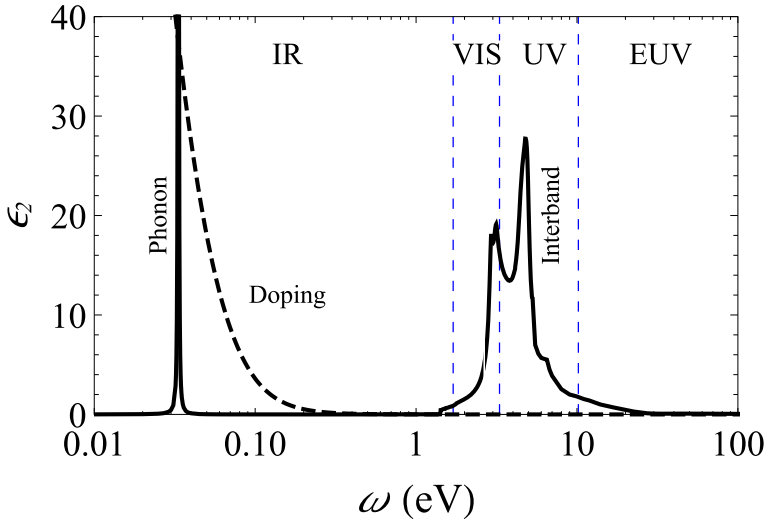


Fig. 3 Imaginary part of the dielectric function for crystalline GaAs. The contributions of the optical phonons, interband transitions, and doping (Drude model with $\omega_p = 0.3$ eV and $\gamma = 0.05$ eV) are indicated. The different spectral regions: infrared (IR), visible (VIS), ultraviolet (UV), and extreme ultraviolet (EUV) are separated by vertical lines

- Visible–ultraviolet (VIS/UV): interband electronic transitions
- Extreme UV (EUV): onset of transparency where $\varepsilon''(\omega) \to 0$

Figure 3 illustrates these features for GaAs, highlighting the phonon and interband contributions.

While the examples above refer to crystalline semiconductors and insulators, the same general principles extend to molecular materials, polymers, and liquids. In these systems, characteristic resonances arise from molecular vibrations (IR) and electronic or vibronic transitions (VIS/UV). The optical response of the intervening medium—such as water or bromobenzene—can even invert the sign of the Casimir–Lifshitz force when the dielectric functions of the bounding media satisfy suitable inequalities.

In the following, we summarize the principal models used to describe the dielectric response of conductors and semiconductors across the relevant frequency ranges.

4.2 Plasma, Drude, and Drude–Lorentz models

The optical response of metals and doped semiconductors is largely governed by the dynamics of free carriers undergoing intraband transitions. In non-conductive systems such as insulators or wide-bandgap semiconductors, interband transitions dominate. The simplest and most widely used description of the free-electron contribution is provided by the classical Drude model.

The Drude model assumes a free-electron gas with damping γ, the dielectric function is

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega}, \tag{23}$$

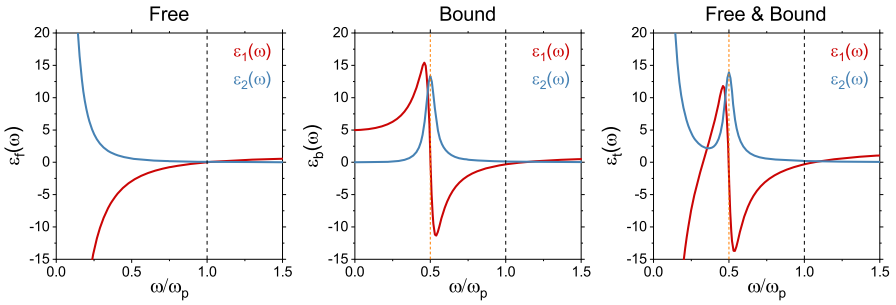


Fig. 4 Real ($\varepsilon_1(\omega)$) and imaginary ($\varepsilon_2(\omega)$) parts of $\varepsilon(\omega)$ for free (left panel), bound (middle panel), and the total (right panel) contributions as functions of ω/ω_p. Here, we set the values $\omega_0 = 0.5$ (dashed orange line), $\omega_p = 1.0$ (dashed black line)

where $\omega_p = \sqrt{ne^2/(\varepsilon_0 m)}$ is the plasma frequency determined by charge carrier density n and effective mass m. This form successfully describes alkali metals but deviates for noble metals (Au, Ag, Cu), where d-band electrons contribute to interband absorption. The Drude model can also describe the response of the electrons or holes added by the doping of semiconductors. The damping parameter may depend on both temperature and, in some cases, frequency, $\gamma = \gamma(\omega, T)$. Such dependencies arise in frequency ranges where electron–electron and electron–phonon scattering processes play a significant role[99–101]. The extent to which γ varies with frequency or temperature is strongly material dependent. For example, thin films of TiN exhibit pronounced temperature-induced changes in the imaginary part of the dielectric function [102].

4.2.1 Drude–Lorentz model

To incorporate the contribution of bound charges (phonons) and interband resonances, the dielectric function is generalized as

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega} + \sum_j \frac{S_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega}, \tag{24}$$

where S_j, ω_j, and γ_j are the oscillator strength, resonance frequency, and damping of the j-th Lorentzian term, respectively. The Drude–Lorentz form accurately reproduces measured dielectric spectra of metals and semiconductors across a broad frequency range. For a material described by a single Lorentz term, we show in Fig. 4 the Drude contribution (free or intraband), the Lorentz contribution (bound or interband), and the sum of both.

Applying Eq. (7), the Drude–Lorentz dielectric function at imaginary frequencies takes the form

$$\varepsilon(i\zeta) = 1 + \frac{\omega_p^2}{\zeta^2 + \gamma\zeta} + \sum_j \frac{S_j}{\omega_j^2 + \zeta^2 + \gamma_j\zeta}. \tag{25}$$

In the rotated-frequency space, the dielectric function is real and the free, bound and the total dielectric function, that are now real and uniform, as shown in Fig. 5.

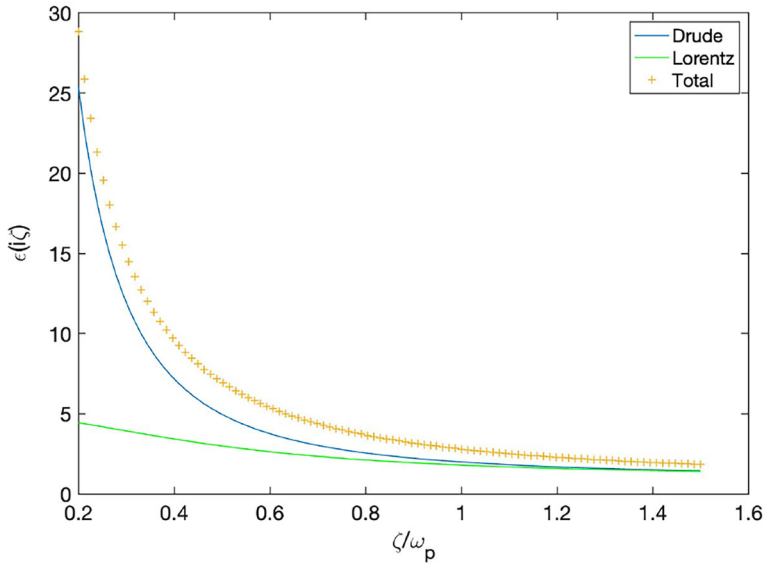


Fig. 5 Same as Fig. 4 but evaluated in the rotated-frequency space

4.2.2 Plasma model

Setting $\gamma = 0$ in the Drude model leads to the plasma model $\varepsilon(\omega) = 1 - \omega_p^2/\omega^2$, which is often employed for analytical estimates at low frequencies. However, this idealized form shows an infinite DC conductivity that does not possess a finite static limit, and therefore is not physical in many applications. For example, in plasmonics, the extinction factor of metal nanoparticles does not match experimental results unless a finite damping is introduced [103]. However, in the case of the Casimir force, describing the free electrons by a plasma model at low frequencies provides a better match to the experimental data and is consistent with the Nernst's heat theorem [104, 105], as we will show in Sect. (5.3). The plasma model presents a challenge when rotating to the imaginary-frequency space due to the second-order pole at $\omega = 0$.

Klimchitskaya, Mohideen, and Mostepanenko [106] showed that when the second-order pole at $\omega = 0$ is treated correctly, the dielectric function determined by the bound charges extrapolated to low frequencies by the plasma model obeys generalized Kramers–Kronig relations and remain fully consistent with causality. The rotation to the imaginary-frequency space through the generalized relation:

$$\varepsilon(i\zeta) = 1 + \frac{2}{\pi}\int_0^\infty \frac{\omega\,\varepsilon''(\omega)}{\omega^2 + \zeta^2}\,d\omega + \frac{\omega_p^2}{\zeta^2}. \tag{26}$$

4.2.3 Physical regimes

The Drude model accurately describes metals in the microwave and terahertz ranges, where reflectivity is nearly unity and absorption negligible. Deviations appear in the infrared due to electron–phonon coupling, and at visible and ultraviolet frequencies, the interband contributions become significant. In typical metals, ω_p lies between 4 eV and 15 eV, while in heavily doped semiconductors, it may range from tens of meV to close to an electronvolt. The precise values of ω_p and γ strongly influence Casimir-force predictions, and these precise measuremements became an important issue when the "Drude versus plasma" discussion was ongoing. This will be discussed later in Sect. 5.3.

Dielectrics also present strong optical response in the VIS and UV spectral region as illustrated in Fig. 3. This response is due to the interband transitions of electrons [107]. The optical response at energies close to the interband transitions can also receive contributions from excitons. The optical activity starts at frequencies corresponding to the gap energy. However, it is usually weak close to this threshold and becomes significant only at higher energies, for which the density of states in the bands is larger. For instance, as illustrated in Fig. 3, the peak absorption (peak on ε_2) for GaAs occurs between approximately 3 eV and 6 eV, while its gap energy is 1.4 eV. The highest energies where the peak of interband transitions can be found are around 10 eV, and occurs for large band gap dielectrics (insulators) like silicon carbide (SiC), silicon dioxide (SiO2), and diamond. For energies below that of the lowest energy optical phonon and above the interband transition region, in the EUV, the dielectrics become transparent, meaning that ε_2 vanishes.

Metals or semi-metals also present the contribution of the interband transitions in the VIS and UV seen in pristine semiconductors and insulators. In addition, they always have the contribution of the free-electron plasma, that can also be very well approximated by the Drude model. In the case of semi-metals, ω_p is below that of typical metals, ranging from a few hundredths of an electronvolt up to a one electronvolt [108]. It is important to note that there is a significant variation in the data on ω_p and γ for the different metals and semi-metals. The particular values used are going to impact the calculation of the Casimir force, as discussed in more detail for gold in Sect. 5.2.

It is worth to note that in the microwave and terahertz region of frequencies, the Drude model works well in the description of metals because they behave as nearly perfect conductors with a high reflectivity and negligible absorption. As we move into the IR region, the Drude model still holds, though some deviations are present due to electron–phonon interactions. Due to the large plasma frequencies in metals, in the VIS and UV, the interband and plasma contributions to ε_2 can be both significant. In Fig. 6, experimental data for gold are presented that illustrates the superposition of both contributions.

For both doped semiconductors, metals and semi-metals, it has been proven convenient in the calculations of the Casimir force to represent the experimental data for the peaks of the interband transitions by an analytical expression like Eq. (24) rather than an interpolation function of the data points.

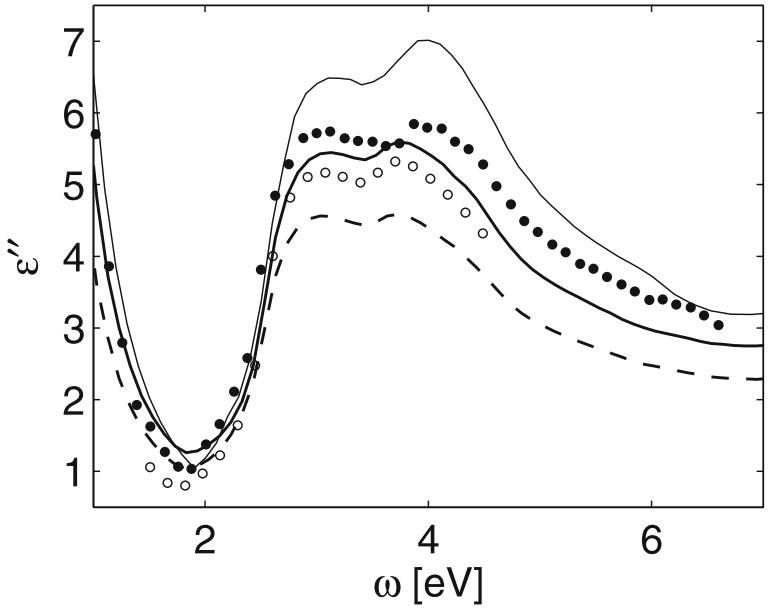


Fig. 6 Imaginary part of the dielectric function reported by different authors. The thick solid lines are from Au films [109], the thin solid line is the data from Theyé [110], and the dashed line is the measurements from [111]. Finally, the open circles are from Wang et al. [112] and the closed circles are from Palik [108]. Reprinted with permission from V. B. Svetovoy et al. Physical Review B **77**, 035439 (2008). Copyright 2008 the American Physical Society [109]

The materials mentioned above can also, in general, be found as microcrystalline or nanocrystalline solids or even with an amorphous structure. As the solid losses its spatial symmetry, its optical response changes. One effect in going from the crystalline to the amorphous material is the loss of the optical phonons. However, IR activity can still exist due to molecular vibrations. That is the case, for instance, for silicon dioxide (SiO_2). While it has a single active optical phonon in the crystalline phase, in amorphous SiO_2, three IR absorption peaks are observed [113] which are due to Si-O-Si stretching and bending modes. In its turn, sharp interband transitions are smeared out, resulting in a smoother and broader peak in ε_2 [60, 114].

It is important to note that interband transitions in the visible and ultraviolet primarily affect the optical response for propagating modes. These modes penetrate tens to hundreds of nanometers into the material, and therefore sample the full high-frequency dielectric function. In contrast, the evanescent waves are confined to nanometric distances from the interface and occur at infrared frequencies set by thermal fluctuations. As a result, the VIS/UV interband contributions play a negligible role in the evanescent channel relevant to Casimir forces. In contrast, the intraband (Drude) response remains fully relevant for evanescent waves if one does not take into account possible dependence of the dielectric function on the wave vector. The Drude term dominates the dielectric function in the infrared and sub-THz range, i.e., precisely in the frequency window where thermally excited evanescent modes and surface plasmon polaritons. Recall that surface plasmons in metals are only excited for p-polarized

waves [115]. Interestingly, in a recent paper, it was shown a discrepancy in the Drude model when measuring the reflection of s-polarized evanescent waves [116], rising questions on how appropriate it is to use the Drude model extrapolation in Casimir force calculations.

4.3 Molecular materials

The Casimir force has been calculated or measured between several other materials of interest besides dielectrics and metals, particularly at short separations, where it turns into the van der Waals force [55, 68, 113]. These include organic materials, like cellular tissues, or polymers. Simple organic materials and polymers can be described as part of the broad class of molecular materials [107]. In these materials, the role of the atoms as the basic constituents of the material is substituted by that of the molecules. The small interaction between neighboring molecules implies that the optical properties of the bulk are close to those of the free molecules. These materials present three relevant channels of interaction with the electromagnetic radiation: through molecular vibrations, electronic transitions, and vibronic transitions in which both electronic and vibrational states are excited simultaneously [107]. A relevant difference in the electronic transitions compared with solid crystals is that, while in the last materials, they occur between the valence and conduction bands, in molecular materials, they take place between the highest occupied and the lowest unoccupied molecular orbitals. The molecular vibrations are typically observed in the IR, like for the phonons in solid crystals, while the electronic and vibronic interactions are observed in the VIS and UV.

The particular optical response of molecular materials can vary significantly depending upon specific properties of such materials [107]. However, typical dielectric functions are presented in Fig. 7, obtained experimentally for the solid phase of the polymer BBL [117]. In this figure, it is possible to see the typical contributions of the ionic vibrations in the IR and that of electronic transitions in the VIS and UV, common to most polymers used as dielectric materials [118]. The response in the IR is quite similar to that found for semiconductors and insulators with multiple active optical phonons. In its turn, the electronic transitions between different molecular levels result in a series of well-defined peaks in ε_2. Most generally, the peaks in the VIS and UV range could also involve vibronic transitions [107].

4.4 Liquids

In biological systems or in chemistry, it is quite common the presence of liquids between interacting structures like biological membranes or large particles in solutions [55]. Here we focus not in the liquid phase of metals like mercury, whose dielectric function is similar to that of a solid metal, but in liquids that have been relevant in experiments of the Casimir force or can be present in microdevices and nanodevices and biological systems, like water and bromobenzene.

Because water is the most important chemical solvent a-nd is present, for instance, in micro and nanofluidic devices and in organic systems on which the van der Waals

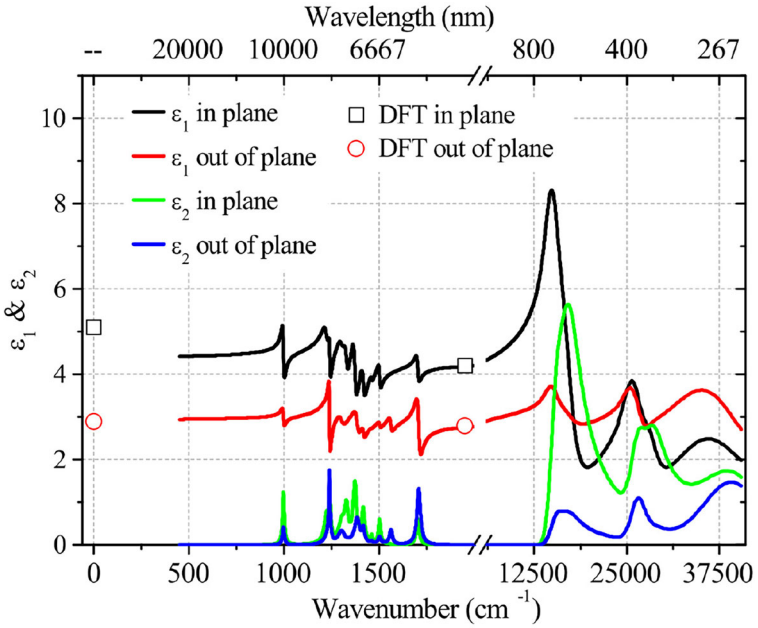


Fig. 7 The lines correspond to the experimental values of the real (ε_1) and imaginary (ε_2) part of the dielectric function of the BBL polymer. The peaks in ε_2 in the IR (wavelength $\sim 10\mu$m) are due to molecular vibrations, while those at the VIS and UV (wavelength $\lesssim 1\mu$m) are due to electronic transitions. Reproduced from Ref. [117]

force plays a significant role [55], we focus primarily on its properties. The optical properties of water have been extensively studied. A recent work is that of Fiedler et al. [119] which collected and used in its analysis the most recent optical data to model the dielectric function of water from microwave up to EUV. The components of ε obtained in Ref. [119] are shown in Fig. 8. Water presents a broad peak in ε_2 in the UV due to electronic transitions and two peaks in the near-IR associated to its three vibrational modes (two vibrational modes have very close frequencies and cannot be distinguished). However, its most striking feature is the strong nonresonant response in a broad range of frequencies, from microwave up to the IR. This response at low frequencies is a characteristic of polar liquids, of which water is an example. This response is considered as a relaxation process that involves movement of the dipoles of polar molecules. In nonpolar liquids, a similar response at low frequencies is absent. Similar peaks can also be observed in polymers, for which relaxation can also play a significant role in the definition of their dielectric constants [118, 120]. It is this process of dipole relaxation that results in the large static dielectric constant of water of about $\varepsilon(0) = 80$.

While, in its bulk form, water already presents a complex and interesting dielectric function, it has been discovered more recently that under certain conditions relevant, for instance, for microsystems and nanosystems, its dielectric properties can change in unexpected ways. It has been observed that confined thin water layers may present an

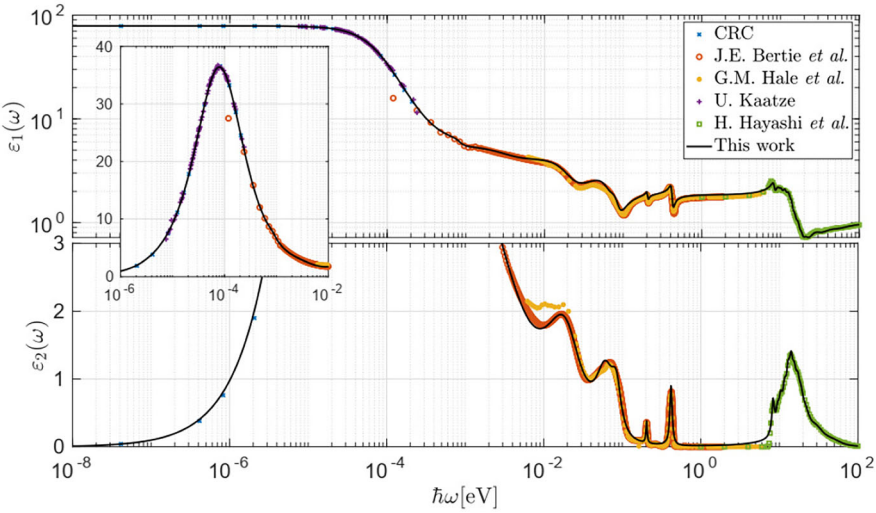


Fig. 8 The colored points correspond to the experimental values of the real (ε_1) and imaginary (ε_2) part of the dielectric function of water. The continuous line is a fit to the data proposed in Ref. [119]. Reproduced from Ref. [119]

effective dielectric constant that is as small as $\varepsilon(0) = 2$. While still a matter of debate [121, 122], it is usually considered that this decrease in $\varepsilon(0)$ is due to the freezing of the dipoles close to the solid–water interface. This effect of the confinement can have quite significant impact on the Casimir force when, for instance, water is an intervening medium [123].

We are later going to see that the presence of an intervening medium like water can significantly alter the Casimir force. Some nonpolar liquids, like bromobenzene, have been used to turn the usually attractive Casimir force into a repulsive force between two dissimilar materials [15]. In the case of nonpolar liquids, the dielectric function is similar to that of molecular materials without the nonresonant response of the relaxation in the microwave and far-IR frequencies [68].

4.5 Low-dimensional materials

So far, we have characterized the dielectric function of bulk materials. This leads to an accurate description of the optical properties of microsystems and nanosystems whose smaller dimensions are larger than a few nanometers. However, as the dimensions decrease to the atomic scale, this description may not be accurate any longer. In the emerging field of nanotechnology, we are faced with low-dimensional systems like quantum dots, nanowires, nanostrips or atomically thin films and membranes. The first effect of the small dimensions is the quantum confinement effect which can be explored in several applications [107]. However, more significant and fundamental changes of the optical properties generally occur when the systems achieve atomic dimensions. These changes in the optical properties should not be confused with the

effect of the effective dimension of the systems, that can be considered as an influence of the geometry on the Casimir force. The effects of the geometry can take place for systems sufficiently large that their optical properties can be well described by those of the bulk. As two examples, we can mention a large suspended membrane with a thickness of a few nanometers or a long circular nanorod with a radius of about a few nanometers. For electromagnetic waves with wavelengths much larger than a few nanometers, these systems behave optically as a 2D and an 1D system, respectively.

However, when any dimension of an approximately 0D, 1D, or 2D system becomes sufficiently small, a description using the optical properties of the bulk can no longer be accurate. Let us consider, for instance, the changes in passing from bulk layered materials (van der Waals materials) to a monolayer. The monolayers are examples of 2D materials, as their thickness is of the order of a few angströns, while the dimensions along the crystalline plane can be macroscopic. Ma et al. [124] have calculated the dielectric functions for materials like $MoSe_2$, $PtSe_2$, TiS_2 and graphite/graphene, for both the bulk and a monolayer, for investigations of near-field radiative heat transfer. Figure. 9 presents their results, which are limited to include the interband transitions and the contribution of the free-electron plasma, but not that of the optical phonons. We can see that for some materials, like $MoSe_2$ and TiS_2, there are significant changes in the dielectric function. However, their, respectively, semiconducting or metallic character, is not changed. In the case of $PtSe_2$, an even more significant change takes place, as the bulk material is metallic and the monolayer is a semiconductor. In the case of graphite, it goes from a semi-metal with a parabolic band for the bulk to a Dirac material, with zero gap and linear bands at the K point, for the graphene monolayer. While, in Ref. [124], only the frequency dependence of ε was considered, it is generally the case that the in-plane dielectric function depends significantly on the wavevector in 2D electronic systems [125]. However, within DFT, the calculations are usually performed in the long wavelength limit in which the wavevector dependence can be ignored. This approximation can limit the accuracy of the calculations of the Casimir force involving 2D systems. Due to its particular relevance, approximate analytical results for the optical response (conductivity or polarizability) of graphene including the dependence on the wavevector have been derived in the literature [126] and used in the calculation of the Casimir force [127, 128].

To conclude on the optical properties of 2D materials, we note that, in the relevant case of systems that comprised only a few atomic layers, their optical properties are going to vary with the thickness. In any case, they are going to present the general features of optical response observed for bulk materials: interband transitions, the free electron or hole plasma, and the optical phonons [129].

The dielectric response of materials governs the strength and sign of the Casimir interaction. To highlight the contrasting behaviors across the different classes of materials reviewed in this section, Table 2 summarizes the main characteristics of the complex dielectric function $\varepsilon(\omega)$ for metals, semi-metals, semiconductors, and two-dimensional (2D) materials. While metals exhibit the strongest broadband reflectivity and thus the largest Casimir forces, semiconductors and semi-metals offer tunability through carrier concentration, and 2D materials introduce an additional level of control via surface conductivity and chemical potential engineering.

Table 2 Comparative dielectric function characteristics by material class

Material Class	Typical ω_p (eV)	Main contributors to $\varepsilon(\omega)$	Spectral features (peaks in ε_2)
Metals	4–15 (Al ~15, Au ~9)	Drude free carriers + interband (Lorentz)	Broad VIS–UV absorption; $\varepsilon_1 < 0$ at low ω
Semimetals	0.1–1	Weak Drude + interband	IR–VIS overlap; smaller $\lvert\varepsilon_1\rvert$ than metals
Semiconductors	Undoped: none; doped: 0.01-−0.5	Interband (VIS–UV) + IR phonons; doping adds Drude term	Onset near E_g; IR phonon resonances (10–100 meV)
2D materials	(Effective via $\sigma(\omega)$) few meV–1	Intra/interband in surface conductivity (Kubo/Dirac)	Features set by chemical potential μ and bandgap; strong anisotropy/nonlocality

VIS visible, *IR* infrared, *UV* ultraviolet

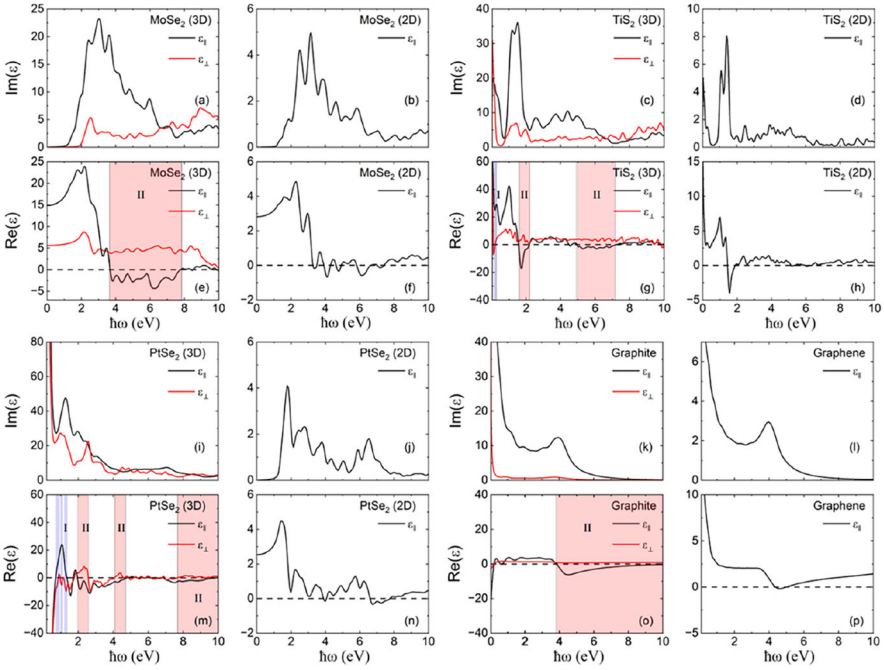


Fig. 9 Real and imaginary components of the dielectric function of $MoSe_2$, $PtSe_2$, TiS_2 and graphite/graphene for the bulk (3D) and monolayers (2D). Reproduced from Ref. [124]

4.6 Nonlocal response

In the quest of better understanding all the factors that come into play in the Casimir force, the role of nonlocality or spatial dispersion in the dielectric function has also been considered in the literature. The Lifshitz theory [3, 4] evaluates the Casimir interaction between bodies using local dielectric response functions $\varepsilon(\omega)$, assuming that the material's response at a given point depends only on the local electric field. However, this approximation becomes inadequate when the electromagnetic wavelength becomes comparable to microscopic length scales such as the electron mean free path, the Fermi wavelength, or interatomic distances.

4.6.1 Nonlocal dielectric function

Nonlocal electrodynamics must be employed to account for spatial nonlocality, wherein the dielectric function becomes spatially dispersive [130–132]: $\varepsilon(\mathbf{r}, \mathbf{r}'; \omega)$, meaning that the response to an electric field at the point $\mathbf{r}$ is also dependent upon the field at the point $\mathbf{r}'$. In the reciprocal space, the dielectric function is $\varepsilon(\mathbf{k}, \omega)$, resulting in the constitutive relation $\mathbf{D}(\mathbf{k}, \omega) = \varepsilon(\mathbf{k}, \omega)\mathbf{E}(\mathbf{k}, \omega)$. Nonlocal media accepts not only transverse fields, but also longitudinal fields [133]. To see this, the dielectric tensor can be written in terms of a transverse part $\varepsilon_T(\mathbf{k}, \omega)$ and a longitudinal part

$\varepsilon_L(\mathbf{k},\omega)$ as

$$\varepsilon_{ij}(\mathbf{k},\omega)=\varepsilon_T(\mathbf{k},\omega)(\delta_{ij}-\frac{k_ik_j}{k^2})+\varepsilon_L(\mathbf{k},\omega)\frac{k_ik_j}{k^2}, \tag{27}$$

where $k_{i,j}$ are the different components of the wave vector. For transverse waves, $k\cdot\mathbf{E}=0$ and Gauss law are satisfied if $\varepsilon_L(\mathbf{k},\omega)=0$, that implies the existence of longitudinal waves. Thus, with nonlocality, the $k-$dependence allows longitudinal waves with dispersion.

An example of a nonlocal approach is the Bohm–Gross model, also known as the hydrodynamic model, in which the plasma dispersion is given by the expression [134]

$$\omega^2=\omega_p^2+\beta^2k^2, \tag{28}$$

where β is a constant related to the Fermi velocity v_F. The hydrodynamic model [135] is an extension of the Drude model and is a semiclassical, continuum description of conduction electrons that introduces nonlocality through a pressure-like term proportional to $\beta^2\nabla(\nabla\cdot\mathbf{P})$, where $\mathbf{P}$ is the polarization vector, that satisfies the equation,

$$\frac{\partial^2\mathbf{P}}{\partial t^2}+\gamma\frac{\partial\mathbf{P}}{\partial t}-\beta^2\nabla(\nabla\cdot\mathbf{P})+\omega_p^2\mathbf{P}=\varepsilon_0\omega_p^2\mathbf{E}. \tag{29}$$

If the hydrodynamic term (the third term on the lhs) is zero, we recover the usual Drude model [133], and the only component of the dielectric function is the transverse part $\varepsilon_T(\omega)$. Otherwise, the longitudinal dielectric function $\varepsilon_L(\omega,\mathbf{k})$ is also present and is given by

$$\varepsilon_L(\mathbf{k},\omega)=1-\frac{\omega_p^2}{\omega^2+i\gamma\omega-\beta^2k^2}. \tag{30}$$

At this point, it is important to notice that nonlocal response is not exclusive to metals. Metals show it because of their free-electron plasma, but dielectrics and semiconductors exhibit nonlocality through excitons, phonons, and polarization waves, especially near resonances [131]. It is generally weaker and narrower in frequency range compared to metals, but essential in nanophotonics and Casimir physics [136]. For example, Eq. (30) is valid in dielectrics in the presence of excitons [137].

More elaborate methods beyond the hydrodynamic model have been developed such as the Linhard model [138], that uses a quantum-mechanical approach and the random phase approximation (RPA). In this approach, the response of the electron gas to the self-consistent oscillating electric field is identified with the response of a gas of independent fermions to an external perturbing field. The Lindhard longitudinal dielectric function

$$\varepsilon_L(\mathbf{k},\omega)=1+\frac{3\omega_p^2}{k^2v_F^2}f_1(u,w). \tag{31}$$

where $f_1(u,w)$ is a function that depends on the dimensionless variables $u=k/2k_F$, and $w=\omega/kv_F$ being k_F the Fermi wave vector [138, 139].

4.6.2 Reflectivity from nonlocal media

Having the extra longitudinal field poses a problem when a boundary is present. In the local theory of electrodynamics, the fields at an interface are determined by Maxwell's boundary conditions. These four conditions suffice because the electromagnetic fields in each medium satisfy a second-order wave equation with only two independent solutions (forward and backward transverse waves). However, in nonlocal media, due to the the presence of additional longitudinal modes, the number of propagating or evanescent solutions in each medium exceeds the number of Maxwell boundary conditions, the system of equations at an interface is underdetermined. This led to the problem of additional boundary conditions (ABC) [136, 140, 141] for a unique and physically consistent solution. Two commonly used ABC are the Pekar condition [136] $\mathbf{P} = 0$, and the Kliewer and Fuchs [142, 143] condition where the current density normal to the surface is set to zero, this is $\mathbf{J} \cdot \hat{n} = 0$. This ABC is used together with the usual Maxwell boundary conditions (continuity of the tangential E and H) and is implemented using a mirror-image technique to relate the longitudinal (nonlocal) modes inside the metal to the fields outside. As an example, let us consider a p-polarized wave incident from vacuum on a nonlocal half-space with tranverse and longitudinal dielectric function ε_L and ε_T. The components of the wave vector perpendicular to the surface are k_0 in vacuum and k_t in the nonlocal medium, while k_l represents the longitudinal component of the wave vector inside the nonlocal medium. In addition, Q will be the wave vector component parallel to the surface. The reflectivity in this case is given by [132]

$$r_p = \frac{\varepsilon_T k_0 - k_t + Q^2(\varepsilon_L - 1)/k_l}{\varepsilon_L k_0 + k_t - Q^2(\varepsilon_T - 1)/k_l}. \tag{32}$$

For comparison, recall that the local Fresnel coefficient is

$$r_p = \frac{\varepsilon_T k_0 - k_t}{\varepsilon_T k_0 + k_t}. \tag{33}$$

Another approach to calculate the reflectivity from a metal surface taking into account nonlocality is based on the jellium model. This model replaces the lattice by a positive background and adds an electron density n, and assumes that the surface is not a sharp interface, but a region where the electron density oscillates (Friedel oscillations) and decays, generating a surface dipole and finite work function. These ground-state properties are calculated using a density functional theory in the local density approximation [139, 144, 145].

Since the electron density does not end at the surface and spills-out, as shown in Fig. 10, where the electron density $n(z)$ is plotted as a function of the distance from the surface. The surface is located at $z = 0$. The different curves correspond to different values of the Wigner–Seitz radius r^s. The spatial extent of the spill-out increases with r_s, reflecting weaker binding in low-density metals. $r^s = 2$ is typical of alkali and noble metals and $r^s = 5$ for good Drude metals like K and Na.

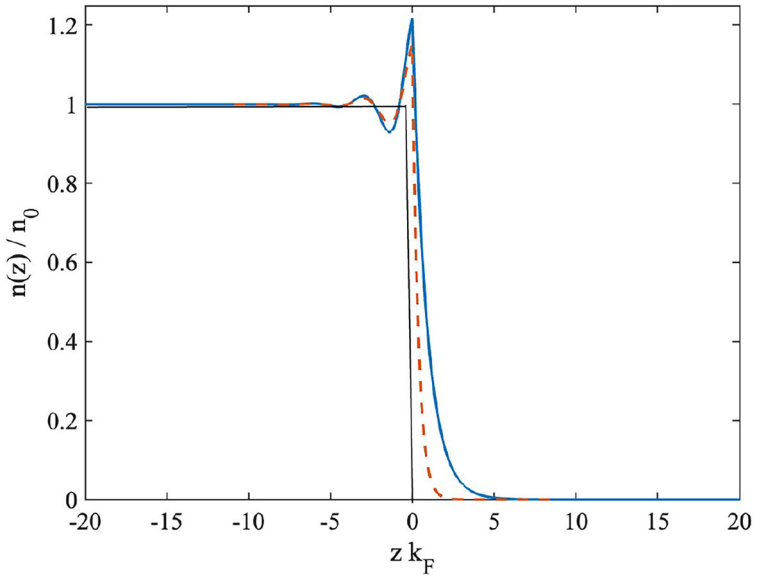


Fig. 10 Electron density in the jellium model. The surface is at $zk_F = 0$. The black curve assumes that the surface is at $z = 0$, the blue line for $r^s = 2$ and the dotted-red line $r^s = 5$. The separation is in units of the Fermi wave vector given by $(2.25\pi)^{1/3}1/(r^s a_0)$, where a_B is the Bohr radius [144]. The positive values of zk_F are indicative of the electron spillage outside the surface

The reflectivity for p-polarized light in the jellium model is given by [146]

$$r_p = r_p^0 \left[1 + \frac{2ik_0\varepsilon_T}{1 + \varepsilon_T k_0^2/Q^2} d_\perp \right], \tag{34}$$

where r_p^0 is the local Fresnel coefficient and $d_\perp$ is the average thickness of the spill-out of the electron density given in terms of the distribution of charge $\delta\rho(z)$ as

$$d_\perp = \frac{\int z\delta\rho(z)dz}{\int \delta\rho(z)dz}. \tag{35}$$

From this equation, we see that $d_\perp$ is the centroid of the charge distribution.

The influence that nonlocality has on the Casimir force is shown in Fig. 11 for the hydrodynamic model (left panel) and for the Jellium model (right panel). In this figure, we defined $\delta F/F = (|F_l| - |F_{nl}|)/|F_l|$, where the subindex refer to local and nonlocal forces. We see that the absolute value of the difference is larger for the jellium model. However, the nonlocal effect becomes relevant only at small separations and the differences do not exceed a few percent.

Calculated reflectivities depend on the choice of ABC [147, 148]. Ruppin and Englman [149] showed that the differences in the reflectivity for different ABC disappear when the surface has a constant potential. This shows that the calculation

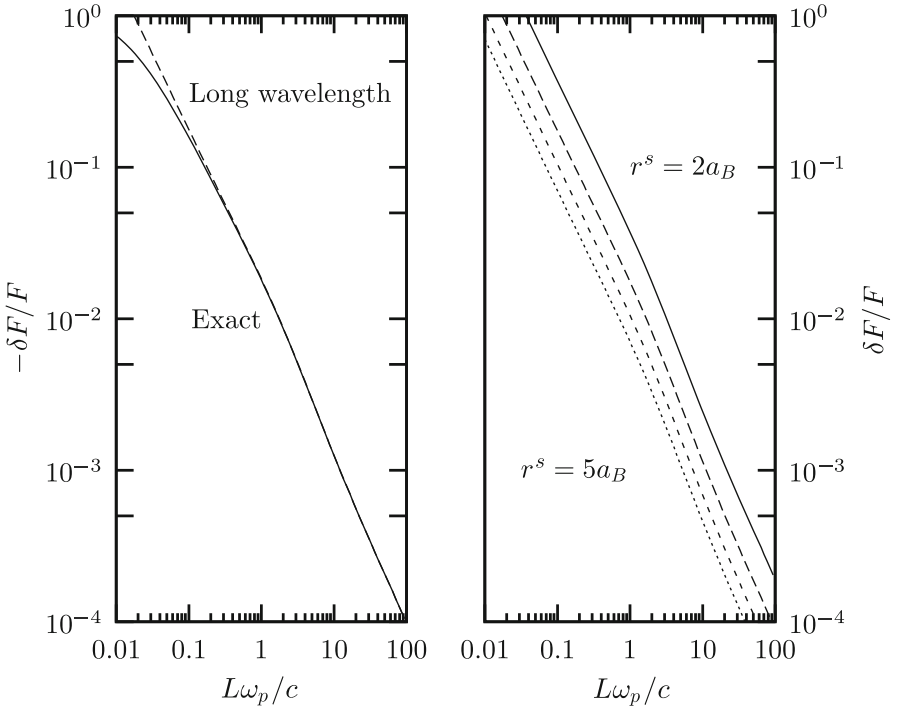


Fig. 11 Comparison between local and nonlocal calculations of the Casimir force using the hydrodynamic model for gold (left) and the jellium model (right) for different free-electron densities. In this figure, L is the separation between the plates. Reproduced from Ref. [139]

of the reflection from a nonlocal surface depends on the model and the underlying assumptions.

The importance of nonlocal response in the Casimir force has been explored for other systems. In Ref. [150], magnetic materials such as Ni are considered and the difference between the local and nonlocal response is no larger than 1%. In a microscopic theory proposed by Despoja and coauthors [151], it is argued that since the Casimir force comes from integrating over all frequencies and wave vectors, subtle effects as nonlocality should not be relevant.

The effect of anomalous skin effect [152] and film thickness [153] corrections also yield small changes in the force at large separations. From a thermodynamic point of view, nonlocality was included in an attempt to solve the Drude vs plasma controversy by calculating the entropy and see if the Nerst heat theorem was satisfied. However, the results were inconclusive [154, 155].

5 Casimir force for different materials

The Casimir force can vary significantly with the different materials that can make the interacting bodies and the intervening medium for a given geometric setup, because of

their particular dielectric functions. In this section, we review how the Casimir force is affected by the different contributions to the optical response of materials seen in the previous section. We focus on the planar geometry as it is the most studied case, and can serve as a basis to understand the role of the optical properties to other geometries. We present first a general discussion on what to expect for the Casimir force for various materials, to later review its behavior for materials relevant for modern experiments and applications, like gold and graphene.

5.1 General results for insulators, semiconductors, and conductors

An important feature of the force given by Eq. 12, for the planar geometry, is that the terms that contribute more significantly to the sum over the Matsubara frequencies are those whose frequencies correspond to wavelengths of the electromagnetic modes with values around that of the surface separation d [68]. As a consequence, at the separations for which the Casimir force is considered relevant, usually below a few hundreds of nanometers, the most important contributions to the force involve the modes with frequencies in the VIS and UV range. Therefore, for all undoped dielectrics (semiconductors, insulators, polymers, etc), the force is determined mainly by the electronic transitions, which occur in this frequency range. In its turn, the contribution from the optical phonons in the IR tends to be smaller. Their effect can usually be ignored in calculations for separations below a few tens of nanometers [60]. For larger separations, however, the contribution can be more significant. These facts are illustrated in Fig. 1 where the finite conductivity correction factor η is plotted for gallium arsenide (GaAS) and amorphous SiO_2. For GaAs, the contribution of the optical phonon in the IR is negligible at any separation. In the case of amorphous SiO_2, the contribution of the response in the IR becomes relevant for d about 100 nm and above. Gusso and Delben [60] have found that for silicon carbide and silicon nitride, the contribution of the phonons in the IR is similar to that found for SiO_2. These results indicate that the contribution from the phonons becomes more relevant for large gap materials with weak interband transitions. Anyhow, it is only for materials with very large infrared activity, like barium titanite, that the IR contribution could surpass that of the interband transitions [156] over a wide range of separations, being relevant even for d as small as a few nanometers [113].

For doped semiconductors with moderate doping levels ($n \sim 10^{16}-10^{18}\,\mathrm{cm}^{-3}$), the plasma frequency is in the far- to mid-IR range (from a few meV up to tens of meV) for semiconductors like silicon or GaAs. For such low plasma frequencies, the contribution of the free-electron (or hole) plasma to the Casimir force tends to be negligible for sufficiently small separations. As it was found in Ref. [62] in the theoretical analysis of torsional MEMS actuators, even for heavily doped silicon resulting in a $\omega_p = 0.4$ eV, similar to that of semi-metals, the corrections due to doping are relevant only for separations above approximately 100 nm. This conclusion is in line with the theoretical results obtained in the analysis of the experimental measurement of the Casimir force between a gold-coated sphere and a heavily doped silicon plate [157]. While the effect of heavy doping was small, this experiment was sufficiently sensitive to detect its effects on the force. In general, it can be concluded that for semiconductors with

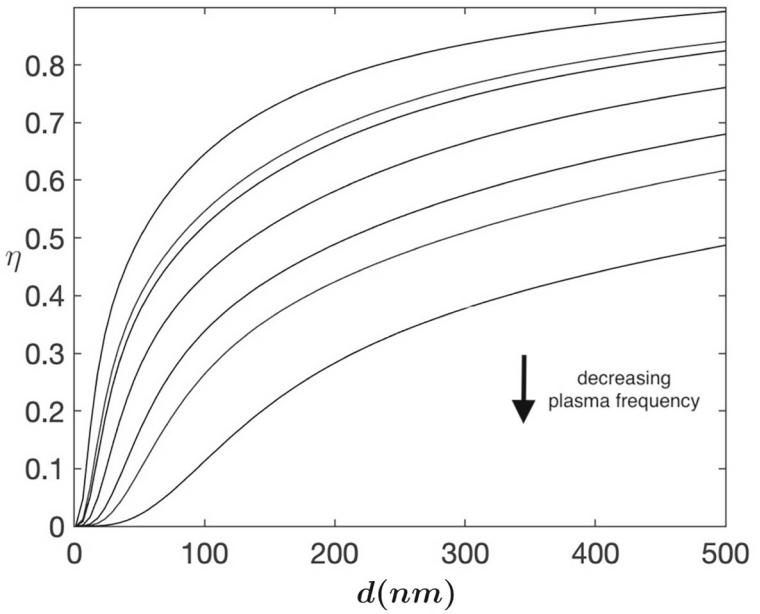


Fig. 12 Reduction factor as a function of separation at $T = 300$K for different metals. From top to bottom, the plasma frequency decreases. The metals are from top to bottom: Al, Ag, Au, Na, K, VSi_2, and H_fSi_2

moderate to low doping levels, the contribution from the Drude plasma has a negligible contribution to the Casimir force for d up to a few hundreds of nanometers. For heavily doped semiconductors, the effects are small for d below approximately 100 nm, but may be significant at larger separations.

In the case of metals, they present large plasma frequencies. As a consequence, both the interband and intraband (free-electron plasma) transitions contribute significantly to the Casimir force. For good conductors like Au and Al at separations above a few tens of nanometers, the contribution of the intraband transitions to the Casimir forces tends to prevail over the contribution from the interband. As the separation decreases, the contribution from the interband becomes more relevant. For a material like Au, the interband and intraband transitions contribute almost equally at a separation of 1 nm.

It is for metals that the largest Casimir forces are expected and have been measured, a consequence the high reflectivity of the free-electron plasma and the broadband nature of the Casimir force. This is evidenced in Fig. 12 where η is presented for good conductors and reflectors, like aluminum, silver, and gold. The factor η is much larger for these metals than for the semiconductor and the insulator in Fig. 1. Due to their practical, theoretical, and experimental relevance, we present a more detailed review of the Casimir force for metals.

5.2 Gold as a standard material

Gold has become the standard metal for Casimir force experiments. Its dielectric function, especially in the optical range, has been extensively studied and reported by many researchers over the years. The problem when measuring the Casimir force and calculating the theoretical values, is that the dielectric function of Au varies depending on several factors.

These variations are due to differences in measurement techniques, sample purity, surface morphology, film thickness, and temperature. Some of the most well-known and commonly cited data sets include those of Johnson and Christy [111] that report measurements on bulk crystalline gold. The Handbook of Optical properties of Solids by Palik [108] presents data in a wide range of energies from 0.125 eV to 9900 eV and is a compilation of data from several sources and samples. Other highly cited works on the dielectric function of Au are those of Rakic [158] and Olmon [159].

The data reported in the different references present some discrepancies. In the case of Palik's Handbook [108], the impressive range of measurements has several problems despite its common use. Depending on the frequency range, different experimental techniques were needed to measure the optical properties. These include spectroscopic ellipsometry [160], energy electron loss spectroscopy (EELS) [161], Fourier transformed infrared spectroscopy (FTIR) [162, 163]. On top of that, the data reported by Palik come from different samples and measurements. That means, there is no standard sample that has been used as a reference for the dielectric function of Au. To calculate the Casimir force, extrapolation to low frequencies ($\omega \to 0$) has to be done. At low energies, a Drude model adjustment yields the typically used values for the plasma frequencies and damping [164] of $\omega_p = 9$ eV and $\gamma = 0.035$ eV.

In the context of the Casimir force measurements, Svetovoy et al. [109] performed an analysis of the optical properties on different Au samples, measuring the dielectric properties using elipsometry. The reported plasma frequency range was $\omega_p \in [6.8, 8.4]$ eV.

In Fig. 6, the value of the imaginary part of the dielectric function is shown for different experimental data sets. The difference in $\varepsilon_2(\omega)$ implies that the use of bulk Au data may not be adequate when a thin film is placed on a substrate. As mentioned in the work by Svetovoy et al. [109], these variations also have an impact on the calculation of the force.

Additional factors can contribute to changes in the values of the dielectric function of Au, such as annealing and quenching [165]. Thermal processes have a significant influence on the structural and electronic properties of gold surfaces, which in turn affect their dielectric function. During annealing, the Au surface is heated to high temperatures and held there to allow atomic rearrangement, promoting grain growth, surface smoothing, and the reduction of defects. Quenching, the rapid cooling that follows, can lock in these structural changes or, depending on the rate, introduce new strain and defects. These thermal treatments alter the electron scattering rates and the density of states at the surface, thereby modifying the complex dielectric function of Au [166]. Typically, annealed surfaces exhibit reduced electron scattering and improved crystallinity, resulting in a decrease in the imaginary part of the dielectric function,

which corresponds to optical losses, particularly in the visible and near-infrared ranges. In contrast, quenched surfaces may show an increased density of disorder and defects, potentially enhancing optical absorption and increasing damping.

The consequences of these variations on the calculation of the Casimir force are significant: depending on the specific preparation and condition of the samples, measurable deviations can arise between experiments. As demonstrated by Svetovoy et al. [109], achieving an experimental precision better than about 10 % in the force determination requires in situ characterization of the optical response. We emphasize that the tabulated data of Palik—and particularly their low-frequency extrapolations—cannot be assumed to accurately represent the dielectric properties of the samples used in a given experiment.

5.3 Plasma or Drude

For more than 2 decades, the extension of the dielectric response of metals to low frequencies has been the subject of sustained discussion in the Casimir community. From the standpoint of solid-state physics, the Drude model is the natural low-frequency extrapolation because it incorporates dissipation and correctly reproduces the dc conductivity of metals. Nevertheless, several precision measurements have been interpreted as being more consistent with the dissipationless plasma model [43, 167]. The difference between these approaches originates from the treatment of the zero-frequency Matsubara term, and in particular the contribution (or lack thereof) of the TE (s-polarized) mode [168–170].

To discuss the issue, we consider the Au data reported by Palik and the extrapolation to zero frequency. In the rotated-frequency space, the dielectric function including interband transition is classically described by the Drude (or Plasma) contribution plus six Lorentz oscillators [45] as:

$$\varepsilon(i\zeta) = 1 + \frac{\omega_p^2}{\zeta^2 + \gamma\zeta} + \sum_{j=1}^{6} \frac{g_j}{\omega_j^2 + \zeta^2 + \gamma_j\zeta}. \tag{36}$$

If $\gamma = 0$, we obtain the plasma model. In Fig. 13, this dielectric function is plotted for three cases: the Drude model (without the Lorentz contribution), Drude+Lorentz, and plasma+Lorentz. As seen from this figure, at low frequencies, there is a significant difference between the Drude and Plasma extrapolations.

To calculate the force in the rotated-frequency space, the problem arises for the zero-term Matsubara frequency and the Fresnel coefficients take different values depending on whether we consider the plasma or the Drude model [43, 45]. For the Drude model, when $\zeta \to 0$, the Fresnel coefficients for either p- or s-polarized waves take the limit values

$$r_i^p = 1, \quad r_i^s = 0. \tag{37}$$

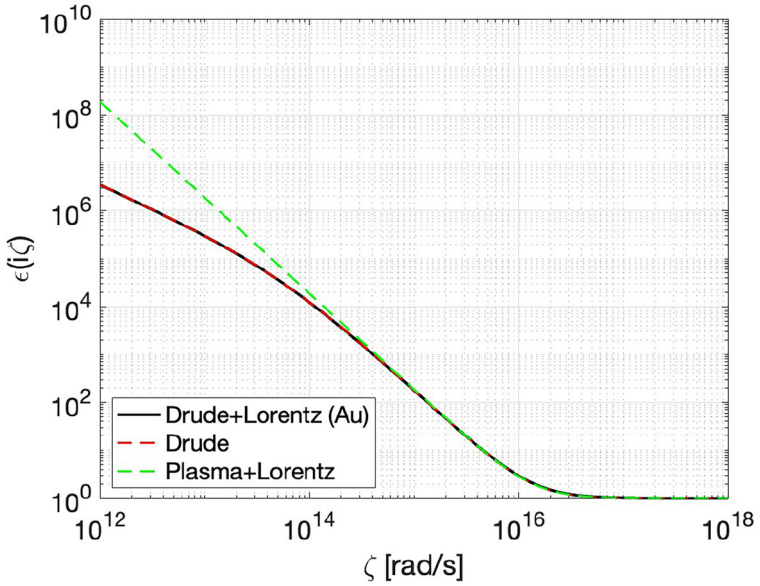


Fig. 13 Dielectric function of Au for the Drude model, Drude+Lorentz model, plasma+Lorentz. The data was an extrapolation of Palik data as reported in Ref. [45]

On the other hand, for the plasma model, the limit of the Fresnel coefficients at zero frequency is

$$r_i^p = 1, \quad r^s = \frac{cQ - \sqrt{c^2Q^2 + \omega_p^2}}{cQ + \sqrt{c^2Q^2 + \omega_p^2}}. \tag{38}$$

where Q was defined in Sect. 2. As mentioned before, one should favor the Drude model for being a more realistic model, and in principle consistent with Lifshitz theory. However, Nesterenko [171] has shown that this may not be the case. This author discusses a first-principles-inspired analysis of how the plasma and Drude permittivity models may or may not be consistently inserted into the Lifshitz formula via the fluctuation–dissipation theorem. He shows that the plasma model fits more cleanly into the formal Green-function relation framework, while the Drude model introduces problematic analytic behavior (extra poles, nonanalytic dependence) that obstructs direct use in exact fluctuation–dissipation relations. Nevertheless, since the damping in real metals is small, using the Drude model in the approximate fluctuation–dissipation theorem-based Lifshitz computations can be pragmatically acceptable. Ultimately, there is no decisive theoretical preference for plasma vs Drude, and the choice must be settled by comparison with experimental data. Although we centered the discussion on Au, several experiments have supported the plasma model with other materials such as Ni [172] and ferroelectrics [173], and discarded confounders such as the role of nonlocality [174], or subtle electrostatic contributions [175].

As a final remark, it is important to emphasize that the long-standing discussion surrounding the Drude and plasma models is no longer understood as a question of

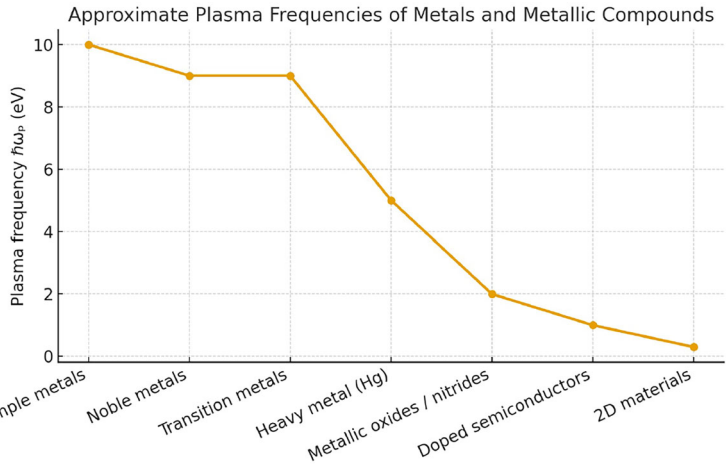


Fig. 14 Chart showing typical values of the plasma frequency for different sets of materials

which model is "correct" in general. The Drude model correctly describes the response of metals to low-frequency propagating electromagnetic waves of both polarizations, as well as to evanescent waves with TM (p) polarization. However, it fails to describe the contribution of evanescent TE (s) modes to the Casimir interaction, as analyzed in detail by Klimchitskaya and Mostepanenko [170]. Recent experimental measurements of the lateral component of the oscillating magnetic field reflected from metallic surfaces [116] provide direct confirmation of this polarization-dependent behavior. This refined understanding has shifted the focus from a model-selection dilemma to the role of mode polarization and dissipation in Casimir physics.

5.4 Casimir force for various metals

In this section, we present the behavior of the Casimir force for various plasmonic materials. The metal with the highest plasma frequency is Al ($\omega_p = 15.3$ eV) and the lowest one K ($\omega_p = 3.72$ eV). Lower plasma frequencies can be achieved with semi-metals, perovskites or narrow-band metallic oxides. In Fig. 14, we show the typical plasma frequency for different classes of materials. For many metals, a good fit that includes the Drude plus the Lorentz–Drude contributions from experimental data can be found in the MATLAB code (https://la.mathworks.com/matlabcentral/fileexchange/18040-drude-lorentz-and-debye-lorentz-models-for-the-dielectric-constant-of-metals-and-water) developed by Ung and Sheng [176].

To see the behavior of the Casimir force for different metals, we have calculated the force between two parallel plates at a temperature of 300 K for Al, Ag, Au, Na, K, VSi_2, and H_fSi_2. The results for η are presented in Fig. 12. The overall behavior of η is similar for all the metals considered, except that the lower the plasma frequency in the Drude extrapolation, the lower the η at a given separation d and, consequently, the Casimir force.

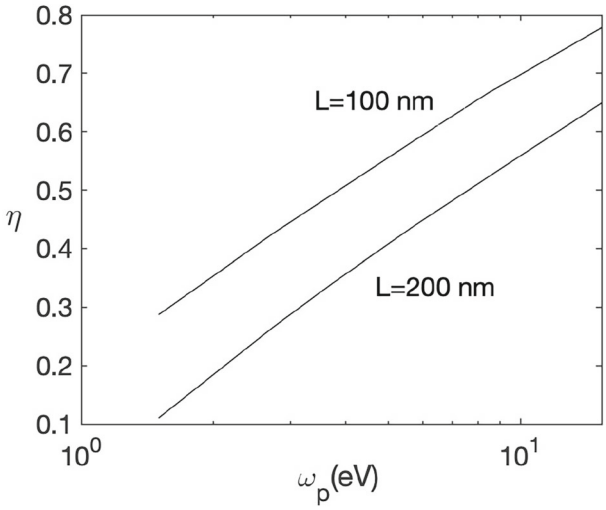


Fig. 15 For two different separations, the reduction factor η scales linearly with the log of the plasma frequency. The equation is of the form $\eta = 0.2\omega_p + b$

Interestingly, at a fixed separation, the reduction factor η scales linearly in a semi-log plot as function of the plasma frequency. This is shown in Fig. 15. For $d = 100$ nm, we have $\eta = 0.2\omega_p + .028$ and for $d = 200$ nm, $\eta = 0.2\omega_p + .21$. Given this behavior, by selecting the plasma frequency of a metal or metallic compound, the force can be adjusted.

5.5 2D materials

Two-dimensional (2D) materials such as graphene [21, 25, 127, 128, 177–179], hexagonal boron nitride (hBN), phosphorene [17, 180], and transition-metal dichalcogenides (TMDs) [181] have also been studied in the context of Casimir and van der Waals interactions [44]. Unlike bulk metals or dielectrics, 2D systems exhibit ultrathin confinement, reduced dimensionality, and tunable optical conductivity, which fundamentally modify the spectrum of electromagnetic fluctuations. These properties make 2D materials ideal testbeds for exploring nonlocal [182], anisotropic[183], and tunable Casimir forces at the nanoscale [17, 184].

The interaction between 2D layers can be described using the Lifshitz theory generalized to two-dimensional conductors, such as a graphene sheet. In the long-wave limit (< 3 eV), the response of graphene to quantum and thermal fluctuations of electromagnetic fields can be addressed in the framework of the Dirac model by a quantum field theoretical description for the reflectivity of graphene [185–190]. Within this

approach, the reflection coefficients can be written in terms of the polarization tensor $\Pi_{\mu\nu}(\omega, k_\perp; T, \Delta)$ in (2+1)-dimensional space-time as [187, 188]:

$$r_p(\omega, k_\perp; T, \Delta) = \frac{p\Pi_{00}}{p\Pi_{00} + 2ik_\perp^2}, \quad r_s(\omega, k_\perp; T, \Delta) = -\frac{p^2\Pi_{00} + k_\perp^2\Pi_{\rm tr}}{p^2\Pi_{00} + k_\perp^2\Pi_{\rm tr} - 2ik_\perp^2 p}, \tag{39}$$

where $p = (\omega^2 - k_\perp^2)^{1/2}$ and $k_\perp = |\boldsymbol{k}_\perp|$ can be interpreted as the wavevector components normal and parallel to the graphene sheet, respectively. Here, $\boldsymbol{k}_\perp = (k^1, k^2)$ is the transversal component of the 3-momentum of external photon $k^\mu = (\omega, \boldsymbol{k}_\perp)$. The quantities $\Pi_{00} \equiv \Pi_{00}(\omega, k_\perp; T, \Delta)$ and $\Pi_{\rm tr} \equiv \Pi_{\rm tr}(\omega, k_\perp; T, \Delta) = \Pi^\mu_\mu(\omega, k_\perp; T, \Delta)$, respectively, are the 00-component and the trace of the polarization tensor of graphene, which describes the response of graphene to quantum and thermal fluctuations of the electromagnetic field.

The previous expressions are valid for the Minkowskian region of real frequencies $\omega \geq k_\perp$. The corresponding reflection formulas $r_{p,s}(i\xi, k_\perp)$ in the Euclidean region can be obtained by replacing $p \to iq$ in formulas (39), with $q = (\xi^2 + k_\perp^2)^{1/2}$ obtained by substituting $\omega = i\xi$, being ξ the Matsubara frequency.

The dependence of polarization tensor on temperature T is usually described by means of a thermal correction $\Pi^{(1)}_{00/\rm tr}$ to the polarization tensor at zero temperature $\Pi^{(0)}_{00/\rm tr}$, that is, $\Pi_{00/\rm tr}(\omega, k_\perp; T, \Delta) = \Pi^{(0)}_{00/\rm tr}(\omega, k_\perp; \Delta) + \Pi^{(1)}_{00/\rm tr}(\omega, k_\perp; T, \Delta)$. In the case of non-pristine graphene, both contributions could also depend on graphene's mass-gap parameter, Δ, which arises from symmetry breaking due to factors such as substrate effects and external fields. The relevance of this parameter in the Casimir force in graphene-based systems has been discussed exhaustively in studies such as Refs. [191–194]. The explicit expressions for the quantities $\Pi^{(0)}_{00/\rm tr}(\omega, k_\perp; \Delta)$ and $\Pi^{(1)}_{00/\rm tr}(\omega, k_\perp; T, \Delta)$ for both imaginary Matsubara frequency and real frequency domain can be found for example in Refs.[187, 188].

On the other hand, the reflection coefficients of graphene in the Euclidean region can be rewritten in terms of its surface conductivity $\sigma(i\xi)$ as [189, 190]

$$r_p(i\xi, k_\perp; T, \Delta) = \frac{q\sigma(i\xi; T, \Delta)}{2\xi + q\sigma(i\xi; T, \Delta)} \quad r_s(i\xi, k_\perp; T, \Delta) = -\frac{\xi\sigma(i\xi; T, \Delta)}{2q + \xi\sigma(i\xi; T, \Delta)}. \tag{40}$$

Here $\sigma(\omega)$ captures the intraband and interband transitions of the 2D electron gas. Importantly, the inclusion of finite chemical potential, temperature, and nonlocal response allows continuous modulation of the force—from attraction to potential repulsion under specific conditions.

As discussed previously, the local surface conductivity of graphene can be written as

$$\sigma(i\xi_n, T, \mu, \Gamma) = \sigma_{\rm intra}(i\xi_n, T, \mu, \Gamma) + \sigma_{\rm inter}(i\xi_n, T, \mu), \tag{41}$$

where μ is the chemical potential (Fermi level) and $\Gamma = 1/\tau$ is the phenomenological scattering rate. The intraband contribution at finite temperature T, obtained from the

Kubo formula, for gapless graphene ($\Delta = 0$)[195] is

$$\sigma_{\text{intra}}(i\xi_n, T, \mu, \Gamma) = \frac{2e^2 k_B T}{\pi\hbar^2} \frac{\ln\left[2\cosh\left(\frac{\mu}{2k_B T}\right)\right]}{\xi_n + \Gamma}. \tag{42}$$

In the degenerate limit $k_B T \ll |\mu|$, this reduces to

$$\sigma_{\text{intra}}(i\xi_n) \simeq \frac{e^2|\mu|}{\pi\hbar^2} \frac{1}{\xi_n + \Gamma}. \tag{43}$$

For the interband transition, a numerically stable expression on the imaginary axis is [195]

$$\sigma_{\text{inter}}(i\xi_n, T, \mu) = \frac{e^2}{4\pi\hbar} \int_0^\infty d\varepsilon \, \frac{\tanh\left(\frac{\varepsilon+\mu}{2k_B T}\right) + \tanh\left(\frac{\varepsilon-\mu}{2k_B T}\right)}{\varepsilon} \frac{(\hbar\xi_n)^2}{(\hbar\xi_n)^2 + 4\varepsilon^2}. \tag{44}$$

This form is smooth for all n (including $n = 0$) and convenient for quadrature.

For $T \to 0$, one obtains

$$\sigma_{\text{inter}}(i\xi_n, 0, \mu) = \frac{e^2}{4\hbar}\left[1 - \frac{4|\mu|}{\hbar\xi_n} \arctan\left(\frac{\hbar\xi_n}{2|\mu|}\right)\right]. \tag{45}$$

In numerical implementations, the $n = 0$ term is obtained by taking the limit $\xi_n \to 0$:

$$\lim_{\xi_n \to 0} \sigma_{\text{intra}}(i\xi_n, T, \mu, \Gamma) = \frac{2e^2 k_B T}{\pi\hbar^2\Gamma} \ln\left[2\cosh\left(\frac{\mu}{2k_B T}\right)\right], \tag{46}$$

$$\lim_{\xi_n \to 0} \sigma_{\text{inter}}(i\xi_n, T, \mu) = \frac{e^2}{4\pi\hbar} \int_0^\infty d\varepsilon \, \frac{\tanh\left(\frac{\varepsilon+\mu}{2k_B T}\right) + \tanh\left(\frac{\varepsilon-\mu}{2k_B T}\right)}{\varepsilon} \lim_{\xi_n \to 0} \frac{(\hbar\xi_n)^2}{(\hbar\xi_n)^2 + 4\varepsilon^2} = 0, \tag{47}$$

so that the $n = 0$ contribution is purely intraband (Drude-like) in the local model. For $n \geq 1$, Eqs. (42)–(44) are to be used.

The dependence on the chemical potential μ in the conductivity allows for tuning or controlling the Casimir force. In Fig. 16, we plot the reduction factor η as a function of separation between two graphene sheets. The different curves correspond to different values of the chemical potentials.

The behavior of the reduction factor differs from metals or semiconductors, since it decreases with separation. The difference in the behavior is understood by considering that in realistic metallic systems described by the Drude or plasma model, the pressure remains close to P_0 at short separations due to high reflectivity, and may even increase slightly at larger separations as the zero-frequency contribution becomes dominant. In contrast, for graphene, the interaction is governed by surface conductivity rather than a bulk dielectric function. At low frequencies ($\xi \to 0$), $\sigma(i\xi)$ is finite but small

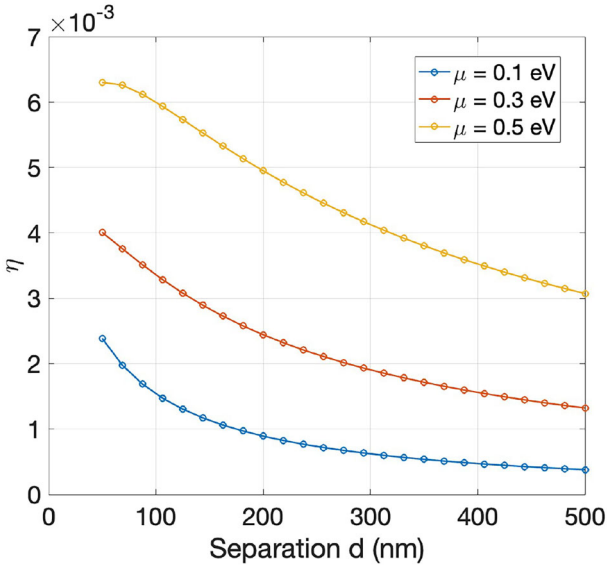


Fig. 16 Reduction factor η as a function of separation between two graphene sheets as a function of separation. The different curves correspond to distinct values of the chemical potential. These calculations are for gapless graphene

compared to the effective surface admittance of a metal, and it decays as $1/\xi$ for higher Matsubara frequencies.

Another interesting 2D material in Casimir force applications is phosphorene [17, 184]. Phosphorene is an example of a highly anisotropic two-dimensional conductor, whose electronic and optical properties differ markedly along the in-plane armchair (AC) and zigzag (ZZ) directions. This anisotropy leads to a tensorial surface conductivity, which directly modifies the Fresnel reflection coefficients entering the Lifshitz formula. As a consequence, the Casimir force between phosphorene sheets depends not only on their separation and carrier density, but also on their relative crystalline orientation. When two sheets are aligned, the interaction is maximized along the high-conductivity axis (armchair direction); when rotated, the mismatch of optical axes leads to Casimir torque and a reduction of the net attractive pressure [184]. The anisotropy of the conductivity arises from the direction-dependent effective masses and band dispersions. In the near-infrared and terahertz regimes, this anisotropic response produces directionally selective vacuum coupling, enabling potential control of fluctuation-induced forces through rotation or strain engineering [17]. Such behavior contrasts with isotropic 2D materials like graphene, where the Casimir force depends solely on carrier concentration and temperature. Consequently, phosphorene offers a unique platform to explore angle-dependent and polarization-resolved Casimir interactions, bridging van der Waals and torque-based Casimir physics within the broader class of low-symmetry 2D materials.

Table 3 Casimir-relevant implications by material class

Material class	Remarks for Casimir force
Metals	Highest reflectivity $\Rightarrow$ largest forces; damping γ reduces magnitude; interband contributions matter more at small d
Semimetals	Intermediate force levels; tunable via band topology/anisotropy; nonlocal corrections more relevant than in good metals
Semiconductors	Smaller forces; IR phonons grow in importance at larger d; doping (carrier density $\Rightarrow \omega_p$) enhances reflectivity and force
2D materials	Force highly tunable via chemical potential, external bias, and layer number; treat as sheets in Lifshitz with $\sigma(\omega)$; strong anisotropy and nonlocal effects

5.6 Effect of the medium between the interacting bodies

The Casimir force depends also on the optical properties of the medium existing between the interacting bodies. In the case of two interacting semispaces, it is the medium filling the gap between the two plane surfaces. In this last case, the presence of any usual material in the gap replacing the vacuum results in a weaking of the Casimir force between materials with ordinary dielectric functions [55, 65]. This is a consequence of the polarization of the medium which screens the electromagnetic interaction between any two bodies. The force can also turn repulsive when the material in the gap has a dielectric function along the imaginary-frequency axis that is intermediate between those of the two semispaces. While the consequences of the existence of a repulsive Casimir force have been observed for a long time [55], it was only in 2009 that Munday et al. [15] measured this force between a silica surface and a gold sphere separated by bromobenzene. Because the Casimir force is more relevant for systems separated by very small gaps, it is important to take into account that the dielectric properties of the thin layers within the gap can differ from those of the bulk, as we have noticed in Sect. 4. Surprisingly, such changes can also take place for confined thin films of liquids. In the particular case of water, it has been observed a huge decrease in its dielectric constant below 100 nm, going from $\varepsilon(0) \sim 80$ down to ~ 2 for films with thickness below 2 nm [196]. Recent experiments have shown that the dielectric function of water in the Terahertz range also decreases significantly below 20 nm [197]. The effect of the change of the dielectric constant of water with the confined film thickness on the Casimir force was investigated by Esquivel-Sirvent [123].

We summarize in Table 3 the Casimir force behavior for the different materials discussed in this section.

5.7 Theoretical materials

So far, the great majority of the works mentioned as examples of the calculation of the Casimir force, the dielectric function used for that purpose was measured experimentally. In the case of graphene, simple analytical models of the conductivity, including the interband and intraband contributions, have been developed (see Sect. 5.5). How-

ever, comparatively few materials have had their optical properties measured in a sufficiently large range of frequencies that can be used for accurate estimates of the Casimir force. Most of the data on metals, semiconductors, and insulators are available in the books by Palik [108] and Adachi [114]. While encompassing a good deal of known materials, the data in these references is restricted for the bulk. This data does not reflect the actual optical properties of sufficiently thin films of such materials as briefly discussed in Sect. 4.5.

In the case of new materials, experimental data may not be available for years. Furthermore, simple theoretical models, like that of graphene, are not readily available and may be not sufficiently accurate.

If the Casimir force is to be calculated between systems that comprised either thin films of known or new materials, or thick slabs of new materials, the optical properties required to calculate the force must be obtained from first-principles (*ab initio*) calculations, such as density functional theory (DFT).

The first such calculation using DFT was performed in Ref. [198] in which the authors determined the Casimir force between atomically thin gold films with thickness ranging from one up to 15 layers. The theoretical prediction of a distance-dependent sign-reversal in the Casimir–Lifshitz torque was illustrated using the calculated optical properties of both black phosphorous and its single-layer counterpart, phosphorene, in Ref. [184].

Using first-principles calculations, it has also been investigated: that a new class of materials, the gapped metals, can have the Casimir force tuned by changing the materials stoichiometry [183], the formation of ice and water films on soil particles [199], and the wettability of liquid iron on refractory oxides [200].

With the rapid increase in the number of new materials being proposed theoretically and discovered experimentally, it is reasonable to assume that calculations of the Casimir force for relevant physical systems are going to rely on optical properties obtained from first-principles calculations. Due to the increase in the recent number of works relying on DFT and the future perspective of its use in connection with the Casimir force, we present next a brief review on how the dielectric function can be obtained theoretically.

6 Dielectric response from *ab initio*

As we have discussed, the predictive power of Casimir–Lifshitz theory depends critically on the accuracy of the dielectric function used as input. While tabulated optical data exist for common bulk materials, they are often incomplete or unavailable for ultrathin films, two-dimensional materials, anisotropic crystals, doped semiconductors, and systems whose electronic structure evolves under strain, thickness reduction, or surface functionalization. In such regimes, experimental ellipsometry or optical spectrosrcopy may be impractical, and effective-medium models or simple Drude–Lorentz fits are insufficient to capture the dispersion across the imaginary-frequency axis required by the Lifshitz formalism.

First-principles approaches such as Density Functional Theory (DFT) and its extensions (GW, Bethe–Salpeter, TDDFT) [201–203] provide a route to compute the

dielectric tensor directly from the electronic band structure, including anisotropy, interband transitions, and thickness-dependent quantum confinement [204, 205]. For emerging materials—graphene, transition-metal dichalcogenides, phosphorene, or monolayer semi-metals—DFT, therefore, supplies the optical response that cannot be accessed otherwise, enabling quantitative Casimir calculations with no adjustable parameters. Although several studies have already used first-principles dielectric functions in Casimir-force calculations [155, 183, 198, 199], their implementation is still not common; for this reason, we present here a general procedure and road map for their use. An important detail when calculating the optical properties is that DFT yields a wavevector and frequency-dependent dielectric function $\varepsilon(\mathbf{q}, \omega)$, and in the limit $\mathbf{q} \to 0$, the local limit is obtained.

We emphasize that the dielectric function $\varepsilon(\omega)$ is a fundamental property of the material and does not depend on whether the electromagnetic excitation is propagating or evanescent. Lifshitz theory requires only that $\varepsilon(\omega)$ be known on the real axis so that its analytic continuation $\varepsilon(i\xi)$ can be obtained. Thus, even methods originally developed for propagating optical waves are fully applicable to Casimir-force calculations once this continuation is performed.

DFT is the most widely used ab initio method for studying the ground-state properties of materials and, combined with the Born–Oppenheimer approximation, pseudopotentials, and Bloch's theorem, allows one to compute electronic, optical, transport, and magnetic properties from first principles. As an illustrative guide, Fig. 17 presents a road map of the workflow for implementing ab initio techniques in Casimir-force calculations, from the material model to the final evaluation of the force.

6.1 System of N-particles

For many-body systems, any physical observable related to electronic structure under a non-relativistic scheme can be studied by means of the time-independent Schrödinger equation. The Schrödinger equation for a system composed of nuclei and electrons is

$$H\psi(\mathbf{r}, \mathbf{R}) = E\psi(\mathbf{r}, \mathbf{R}), \tag{48}$$

where H is the Hamiltonian operator and E is the eigenvalue of the total energy of the system. ψ is the wavefunction that depends on both the spatial and spin coordinates of the electrons, $\mathbf{r}$, and the nuclei of the system, $\mathbf{R}$. Assuming that the nuclei and electrons in a system behave as point objects and neglecting relativistic effects (such as spin-orbit coupling), then the Hamiltonian for a system with M nuclei and N electrons is

$$\begin{aligned} H &= T_n + T_e + V_{n,n} + V_{n,e} + V_{e,e} \\ &= -\sum_{A=1}^{M} \frac{1}{2m_A} \nabla_A^2 - \sum_{i=1}^{N} \frac{1}{2} \nabla_i^2 + \sum_{A=1}^{M} \sum_{B>A}^{M} \frac{Z_A Z_B}{R_{AB}} - \sum_{i=1}^{N} \sum_{A=1}^{M} \frac{Z_A}{r_{iA}} + \sum_{i=1}^{N} \sum_{j>1}^{N} \frac{1}{r_{ij}}, \end{aligned} \tag{49}$$

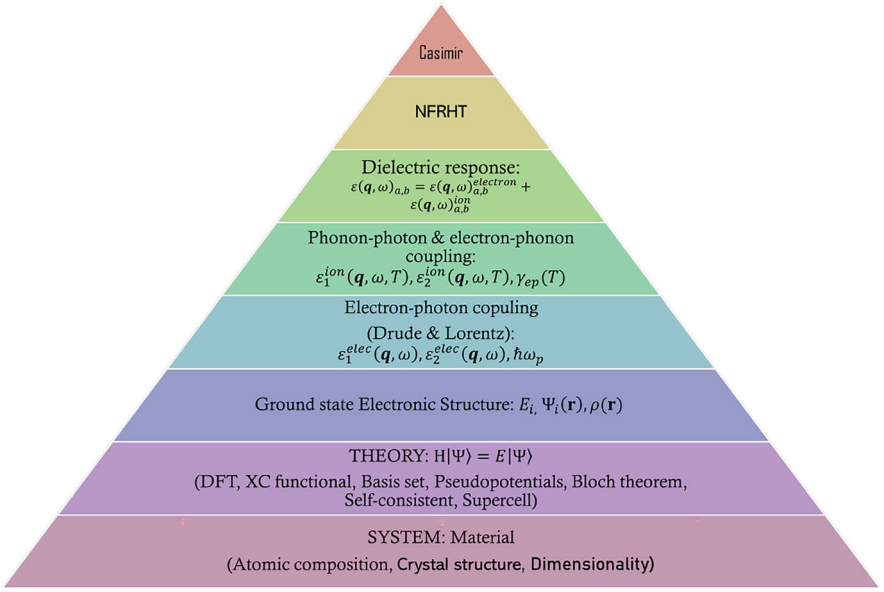


Fig. 17 Schematic representation of the steps towards the calculation of NFRHT or Casimir force from the materials properties and first-principles calculations

where A and B denote nuclei; i, j denote electrons; and ∇^2 is the Laplacian operator. The atomic units are assumed.[2]

The first and second terms T_n and T_e in Eq. (49) describe the kinetic energy of ions with individual mass m_A and electrons m_i, respectively; the third term V_{nn} is the electrostatic repulsion between ions with R_{AB} as the distance between nuclei A and B with atomic numbers Z_A and Z_B; the fourth term V_{ne} is the Coulomb potential energy for the ion–electron interaction, and the last term V_{ee} represents the electron–electron repulsion energy with r_{ij} as the distance between electrons i and j. A key observation when applying quantum mechanics to atoms is that the nuclei in atoms are much heavier than individual electrons, such that the mass ratio of nucleus to electron is $M_{P,N}/m_e \sim 1800$. This means that electrons respond much faster to perturbations than nuclei in a system. As a result, one can separate the problem into two parts. The division into separate mathematical problems for nuclei and electrons is known as the Born–Oppenheimer approximation. Specifically, Eq. (48) reduces to solving the electron dynamics in some fixed configuration of the nuclei. The lowest energy state is known as the ground state of the electrons. Thus, the wavefunction in Eq. (48) is

$$\psi(\mathbf{r}, \mathbf{R}) = \Psi_{e,R}(\mathbf{r})\Psi_n(\mathbf{R}), \tag{50}$$

[2] Atomic units have the electron mass m_e as their unit of mass, the proton charge e as their unit of charge, $\hbar$ as their unit of angular momentum, the Bohr radius as their unit of length, and the Hartree radius as their unit of energy. In this system of units, the absolute value of m_e, e, and $\hbar$ is equal to 1.

where $\Psi_{e,R}(\mathbf{r})$ is the electron wavefunction in some fixed configuration ($\boldsymbol{R}$) of the nuclei and $\Psi_n(\mathbf{R})$ is the nuclei wavefunction. With this, the T_n of the nuclei is neglected and the $V_{n,n}$ repulsion term between nuclei is treated as a constant. This gives the following Schrödinger equation for the electrons

$$H_{e,R}\left[\Psi_{e,R}(\mathbf{r})\right] = E_{e,R}\left[\Psi_{e,R}(\mathbf{r})\right], \tag{51}$$

with $H_{e,R}$ and $E_{e,R}$ as the Hamiltonian and electron energy, respectively, for a fixed-nuclei configuration. Here

$$H_{e,R} = \sum_{i=1}^{N}\sum_{j>1}^{N}\frac{1}{r_{ij}} + \sum_{i=1}^{N}\frac{1}{2}\nabla_i - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{Z_A}{r_{iA}}, \tag{52}$$

and the E_{tot} of the system in a fixed core configuration is

$$E_{tot} = E_{e,R} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{Z_A Z_B}{R_{AB}}. \tag{53}$$

Most of the physical properties of solids can be related to total energies or energy differences. However, even with this simplification, it is still not possible to solve the equation for any system, even for simple systems with one or two electrons. The root of this problem lies in the term V_{ee}, the electron–electron interaction, which contains all the physical information of the many-body interaction for an electronic structure problem and, furthermore, with $3N$ spatial coordinates of the electrons coupled by the operator V_{ee}.

A fundamental approach to solving the Schrödinger equation numerically is using the Hartree–Fock (HF) approximation, which transforms the many-body problem into a single-particle problem. In this HF approximation, the electron wavefunction $\Psi_{e,R}(\mathbf{r})$ is written as a Slater determinant of single-particle wavefunctions. However, the HF approximation does not consider the interaction between electrons of different spins or instantaneous interactions. Compared to the real interaction potential V_{ee}, the HF potential includes the classical Coulomb potential V_{ee}^{C}, and the exchange potential V_x, but does not take into account the electron correlation potential V_c. Although the V_c term is smaller than V_{ee}^{C} and V_x, it is relevant to obtain more accurate results compared to experimental data. More complex methods, such as Möller–Plesset perturbation theory and configuration interaction, have been developed to improve the HF approximation. These methods are more accurate but require more computing resources. An alternative is DFT, which replaces the complicated N electron wave function and the Schrödinger equation with a formulation based on an experimental observable, the electron density $\rho(\mathbf{r})$.

6.2 Density functional theory

The DFT has its origins in the Thomas–Fermi model proposed in 1927. However, modern DFT was established by Hohenberg and Kohn in 1964, giving some hope for a simple method to describe the exchange and correlation effects of an inhomogeneous electron gas. DFT is based on two theorems [201]:

I. For any system of interacting particles in an external potential $V_{ext}(\mathbf{r})$, the external potential $V_{ext}(\mathbf{r})$ is uniquely determined, except for one constant, by the ground-state electron density $\rho_0(\mathbf{r})$.
II. A universal functional $F_{HK}[\rho(\mathbf{r})]$ can be defined for the energy $E[\rho(\mathbf{r})]$ in terms of the density $\rho(\mathbf{r})$, valid for any $V_{ext}(\mathbf{r})$. For any particular $V_{ext}(\mathbf{r})$, the exact ground state of a system is the global minimum value of this functional, and the $\rho(\mathbf{r})$ that minimizes the functional is the exact ground-state density $\rho_0(\mathbf{r})$.

According to the above, the E_{tot} of an electronic configuration subject to an external potential $V_{ext}(\mathbf{r})$ can be written as

$$\begin{aligned} E[\rho(\mathbf{r})] &= T[\rho(\mathbf{r})] + V_{ee}[\rho(\mathbf{r})] + V_{ne}[\rho(\mathbf{r})] \\ &= F_{HK}[\rho(\mathbf{r})] + \int \rho(\mathbf{r}) V_{ne}[\rho(\mathbf{r})] d\mathbf{r} \geq E_0[\rho_0(\mathbf{r})], \end{aligned} \tag{54}$$

where

$$F_{HK}[\rho(\mathbf{r})] = T[\rho(\mathbf{r})] + V_{ee}[\rho(\mathbf{r})], \tag{55}$$

and

$$V_{ne}[\rho(\mathbf{r})] = \int \rho(\mathbf{r}) V_{ne}[\rho(\mathbf{r})] d\mathbf{r}, \tag{56}$$

with the constraint

$$\int \rho(\mathbf{r}) d\mathbf{r} = N. \tag{57}$$

The theorems of Hohenberg and Kohn show that if the exact functional $F_{HK}[\rho(\mathbf{r})]$ is known, the exact E_{tot} and the ground-state properties can be obtained from $\rho_0(\mathbf{r})$, using the variational principle. To date, an efficient scheme to obtain $\rho_0(\mathbf{r})$ and E_{tot} is not yet known. With the aforementioned, Kohn and Sham in 1965 [206] defined the universal functional $F_{HK}[\rho]$ using the electron orbital concept and approximating the kinetic energy of the electrons in a real system by the kinetic energy of non-interacting electrons in a fictitious system. This results in a one-electron equation of the Schrödinger type; which is the Kohn-Sham equation expressed as

$$\left[-\frac{1}{2}\nabla^2 + V_{ext}(\mathbf{r}) + V_H(\mathbf{r}) + V_{xc}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \varepsilon_i(r) \psi_i(\mathbf{r}), \tag{58}$$

where ψ_i is the wavefunction of the electronic state i, and the energy ε_i is the Kohn–Sham eigenvalue. In Eq. (58), the terms on the left-hand side are: the kinetic energy of the non-interacting electrons, the external potential, the Hartree potential, and

the exchange and correlation potential, respectively. The Hartree potential and the exchange and correlation potential are formally given by

$$V_H(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \tag{59}$$

$$V_{xc}(\mathbf{r}) = \frac{\delta E_{xc}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})}, \tag{60}$$

and so the electron density of the system constructed from the Kohn–Sham wave function is

$$\rho(\mathbf{r}) = \sum_{i}^{N} |\Psi_i(\mathbf{r})|^2, \tag{61}$$

Correspondingly, the total Kohn–Sham energy for an electron set of N occupied states is written as

$$\begin{aligned} E[\rho(\mathbf{r})] = -\frac{1}{2} \sum_{i}^{N} \int \Psi_i^*(\mathbf{r}) \nabla^2 \Psi_i(\mathbf{r}) d\mathbf{r} + \int V_{ext}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \\ + \frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' + E_{xc}[\rho(\mathbf{r})]. \end{aligned} \tag{62}$$

So far, the theory with the Born–Oppenheimer approximation is exact. If each term in the Kohn–Sham equation and the exchange and correlation energy functional are known, it is possible to obtain the exact electron density and total energy of the ground state. Unfortunately, the term that is completely unknown is the exchange and correlation energy, $E_{xc}[\rho(\mathbf{r})]$, which includes the non-classical aspects of the electron–electron interaction along with the kinetic energy component of the real system that is different from the fictitious system of non-interacting electrons. Therefore, it is necessary to resort to approximations to describe the term $E_{xc}[\rho(\mathbf{r})]$ if DFT results are to be obtained. Therefore, the central goal of DFT is to find accurate approximations for $E_{xc}[\rho(\mathbf{r})]$. It is worth mentioning that in Eq. (62), the total electron energy is not simply the summation of the Kohn–Sham eigenvalues. Equations. (58–62) are nonlinear and must be solved iteratively. The general procedure starts with an initial proposal of $\rho(\mathbf{r})$ that helps to build the potentials of Eq. (58), to then obtain the Kohn–Sham wavefunctions, and finally to calculate a new $\rho(\mathbf{r})$ and E_{Tot}. This cycle is repeated until a convergence criterion is met. This process is called *self-consistent*.

6.2.1 Exchange–correlation functional

i. **Local density approximation—LDA**
The theorems of Hohenberg and Kohn and the equations of Kohn and Sham provide motivation for the use of approximate methods to express the universal functional and the exchange and correlation energies as a function of $\rho(\mathbf{r})$. The simplest method to describe the exchange and correlation energies in an electron system is with the local density approximation (LDA) introduced by Kohn and Sham. In

LDA, the exchange and correlation energies per electron of the real system, $\varepsilon_{xc}(\mathbf{r})$, are equal to the exchange and correlation energies per electron of a homogeneous electron gas having the same density at point $\mathbf{r}$, $\varepsilon_{xc}^{hom}[\rho(\mathbf{r})]$

$$E_{xc}^{LDA}[\rho(\mathbf{r})] = \int \rho(\mathbf{r})\varepsilon_{xc}^{hom}[\rho(\mathbf{r})]\,d\mathbf{r}. \tag{63}$$

LDA is the most common local approximation for the exchange and correlation energies. The corresponding exchange and correlation potential is

$$V_{xc}^{LDA} = \frac{\delta E_{xc}^{LDA}[\rho(\mathbf{r})]}{\delta\rho(\mathbf{r})} = \varepsilon_{xc}^{hom}[\rho(\mathbf{r})] + [\rho(\mathbf{r})]\frac{\partial\varepsilon_{xc}^{hom}[\rho(\mathbf{r})]}{\partial\rho(\mathbf{r})}, \tag{64}$$

with $\frac{\partial}{\partial\rho(\mathbf{r})}$ as the functional derivative. $E_{xc}(\rho)$ can be split into an exchange contribution $E_x^{LDA}(\rho)$ and a correlation contribution $E_c^{LDA}(\rho)$. In LDA, $E_x^{LDA}(\rho)$ is known exactly, while $E_c^{LDA}(\rho)$ is obtained by fitting highly accurate quantum-mechanical Monte Carlo calculations for a free-electron gas.

ii. **Generalized gradient approximation—GGA**
Strictly speaking, LDA is valid for systems where the electron density varies slowly in space, giving good results for a wide range of real-world systems; however, LDA overestimates binding energies and underestimates bond lengths. For systems with less homogeneous electron density, LDA results are less accurate. The simplest correction for LDA is by including the density gradient $\nabla\rho(\mathbf{r})$. Thus, in the generalized gradient approximation (GGA), the exchange and correlation energies are treated as semi-local functionals of both the electron density $\rho(\mathbf{r})$ and its gradient $\nabla\rho(\mathbf{r})$ for each point $\mathbf{r}$ in space. The general form of the exchange and correlation functionals in GGA is

$$E_{xc}^{GGA}[\rho(\mathbf{r})] = \int \rho(\mathbf{r})\varepsilon_{xc}^{GGA}[\rho(\mathbf{r}), \nabla\rho(\mathbf{r})]\,d\mathbf{r}, \tag{65}$$

and the potential for exchange and correlation is

$$V_{xc} = \frac{\delta E_{xc}^{GGA}[\rho(\mathbf{r})]}{\delta\rho(\mathbf{r})} = \varepsilon_{xc} + \rho(\mathbf{r})\frac{\partial\varepsilon_{xc}}{\partial\rho(\mathbf{r})} + \rho(\mathbf{r})\frac{\partial\varepsilon_{xc}}{\partial\nabla\rho(\mathbf{r})}\nabla. \tag{66}$$

As in LDA, the exchange and correlation energy in GGA can be divided into two parts: $E_x^{GGA}(\rho)$ and $E_c^{GGA}(\rho)$. There are different versions of GGA based on different constructions of exchange and correlation functionals such as the Perdew–Burke–Ernzerhof (PBE) parameterization, which is widely used in the scientific community.

iii. **Van der Waals density functional—VDW-DF**
A van der Waals (vdW) interaction is a nonlocal quantum-mechanical phenomenon for which electronic charge fluctuations in one part of an atomic system are electrodynamically correlated with electronic charge fluctuations elsewhere [207].

vdW forces are indispensable for correctly describing not only biological systems and processes [208, 209], but also metal-organic interfaces [210–213], as well as molecules [214], clusters [215], and bulk systems [216, 217].
In general, there are two different schemes for treating van der Waals interactions. First, an additive potential for atom pairs of the form $C_6 R^{-6}$, where C_6 is a parameter for each specific atom that determines the magnitude of interaction between the two atoms with R being the separation distance between them. This contribution is added to the GGA functional to describe the expected asymptotic behavior [218]. It has been noted that this treatment can be challenging when studying adsorption processes on metallic surfaces and the same polarizability [219]. Second, a pure formulation, for an exchange and correlation functional that splits into short- and long-range interactions [220], which is consistent for both short- and long-range interactions. The proposal by Dion et al. [220] of a functional capable of describing dispersion forces is based on the Adiabatic-Connection Fluctuation-Dissipation Theorem, ACFDT, [206, 221] which allows the exchange and correlation functional to be written as a function of single-electron orbitals

$$E_{xc} = -\int d^3r d^3r' \int_0^\infty \frac{du}{2\pi} \int_0^1 \frac{d\lambda}{\lambda} \left(\chi_\lambda \left(\mathbf{r}, \mathbf{r}', iu\right) V_\lambda \left(\mathbf{r}, \mathbf{r}'\right) - \rho(\mathbf{r})\delta\left(\mathbf{r} - \mathbf{r}'\right)\right), \tag{67}$$

where $\rho(\mathbf{r})$ is the electron density and $V_\lambda\left(\mathbf{r}, \mathbf{r}'\right) = \dfrac{\lambda}{\mid \mathbf{r} - \mathbf{r}' \mid}$ is the electron–electron repulsion potential scaled by a coupling magnitude λ. The orbital dependence in Eq. (67) is via $\chi_\lambda\left(\mathbf{r}, \mathbf{r}', \omega\right)$, the nonlocal imaginary-frequency-dependent full Kubo response density function, $\omega = iu$.
The electron correlation energy follows from Eq. (67) if χ_λ is replaced by a non-interacting response function $\chi_{\lambda=0}$. In that case, Eq. (67) becomes equal to the exchange energy E_x. The correlation energy E_c is expressed by the difference $E_{xc} - E_x$ as

$$E_c = -\int d^3r d^3r' \int_0^\infty \frac{du}{2\pi} \int_0^1 \frac{d\lambda}{\lambda} \left(\chi_\lambda V_\lambda - \chi_0 V_\lambda\right). \tag{68}$$

The short-range part can be adequately described by a local or semi-local functional. Typically LDA is used for this purpose. The long-range part, however, is derived using a number of approximations such as: (*i*) the full potential approximation $\chi_\lambda \approx \chi_{\lambda=1}$, which is well justified for long-range interactions if they are dominated by terms included in RPA (Random Phase Approximation), (*ii*) the plasmon pole approximation which allows relating the linear response functions to the electron density, and (*iii*) the expansion of the correlation energy expression into powers of the dynamical structure factor. These simplifications are essential to remove the hidden orbital dependence in Eq. (67) and thus reduce its complexity. As a result, the long-range term of the correlation energy E_c^{nl} is expressed as

$$E_c^{nl}[\rho] = \frac{1}{2} \int \int \rho(\mathbf{r}) \varphi\left(\mathbf{r}, \mathbf{r}'\right) \rho(\mathbf{r}') d^3r d^3r', \tag{69}$$

where $\varphi\left(\mathbf{r}, \mathbf{r}'\right)$ is the exchange and correlation kernel.
This kernel function has an important property; its spatial dependence satisfies the relation $\varphi\left(\mathbf{r}, \mathbf{r}'\right) = \varphi\left(q_0 \mid \mathbf{r} - \mathbf{r}' \mid, q_0' \mid \mathbf{r} - \mathbf{r}' \mid\right)$ with local scaling parameters $q_0 = q_0(\mathbf{r})$ and $q_0' = q_0(\mathbf{r}')$. For practical calculations, it is convenient to express the kernel in terms of different variables; by renaming $\varphi = \varphi(D, \delta)$ with $D = \frac{q_0 + q_0'}{2} \mid \mathbf{r} - \mathbf{r}' \mid$ and $\delta = \left(\mid q_0 + q_0' \mid\right) / \left(\mid q_0 - q_0' \mid\right)$. This new redefinition makes storing and plotting the kernel much simpler. Interestingly, with these new coordinates (see Ref. [220]), it is clearly observed that the kernel decays as $1/D^6$ for large D.
The local scaling parameter is defined as an estimate of the short-range part of the exchange and correlation energy density $\varepsilon_{xc}^0 = -\frac{3}{4\pi} q_0$, where $\varepsilon_{xc}^0 = \varepsilon_{xc} - \varepsilon_c^{nl}$. Whereas, for total energy calculations, the short-range part is written as

$$\varepsilon_{xc}^0 = \varepsilon_x^{GGA} + \varepsilon_c^{LDA}, \tag{70}$$

The calculation of q_0 requires a different treatment. The GGA form of ε_{xc}^0 provides a relatively small correction to the LDA part in Eq. (70) and eventually leads to infinite static polarizabilities of atoms and molecules. This can be avoided using the following approximation (gradient expansion):

$$q_0(\mathbf{r}) = -\frac{4\pi}{3}\left(\varepsilon_{xc}^{LDA}(\mathbf{r}) - \varepsilon_x^{LDA}(\mathbf{r})\left[\frac{Z_{ab}}{9}\left(\frac{\nabla\rho(\mathbf{r})}{2k_F\rho(\mathbf{r})}\right)^2\right]\right), \tag{71}$$

where $k_F = \left(3\pi^2\rho\right)^{1/3}$ is the Fermi wavevector and $Z_{ab} = -0.8491$ is known as "screened exchange" [222].

6.2.2 Noble metals and semiconductors in bulk

In this review, we present the DFT calculations within the GGA(PBE) approximation of four representative bulk materials with same, the FCC (face-centered cubic) primitive lattice. Alluminium (Al), silver (Ag), silicon (Si), and germanium (Ge) in bulk have the same Bravais lattice but with one and two atoms in the basis for Al and Ag, and Si and Ge; respectively. Figure. 18 shows the calculated band structures along the same high-symmetry points defining the irreducible Brillouin zone. The calculations were performed with the Vienna *ab initio* Simulation Package (VASP) code [223]. DFT results show three metallic systems (Al, Ag, and Ge) and one semiconductor Si with a indirect transition from $\Gamma \rightarrow X$. Note that, the Al system shows a parabolic electronic band characteristic of a free-electron system where the Drude model and plasma frequency could be relevant for transport properties. The dielectric properties of these materials can be well described at low energy (intraband electronic transitions). The main difference between Al and Ag systems is the presence of low-dispersive 4*d* states around -5 eV in Fig. 18. With two atoms in the basis, the Si and Ge systems, having the same Bravais lattice as Al and Ag, can develop an electronic

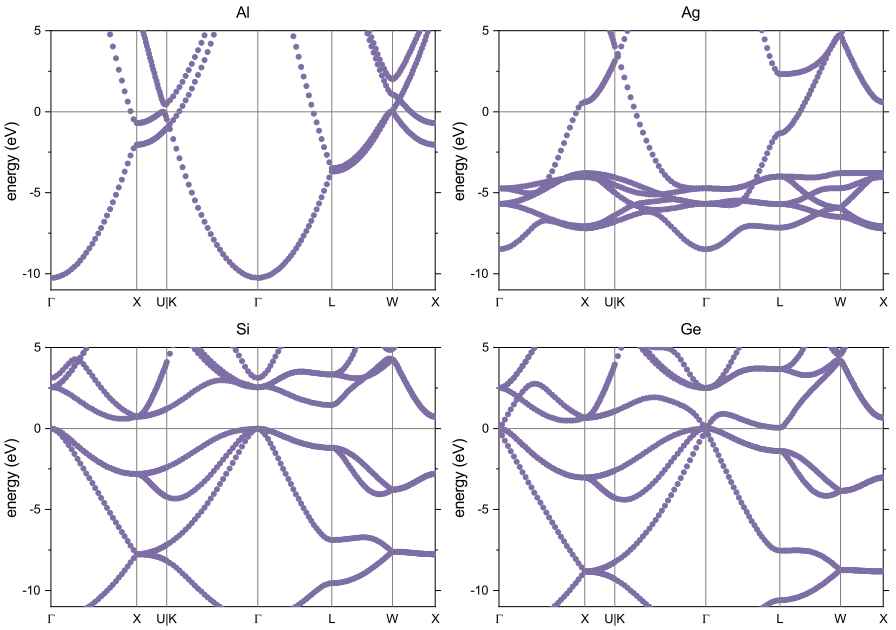


Fig. 18 Calculated band structures of two metallic crystalline systems, Al and Ag, and two semiconducting bulk systems, Si and Ge. The Fermi energy is 0 eV as reference for occupied (below 0) and unoccupied (above 0) states. The absence of an electronic bandgap in the Ge system is due to the exchange–correlation approximation employed

bandgap. Si and Ge systems can de considered as materials with bound electrons to study dielectric properties with the Lorentz model (interband electronic transitions).

6.2.3 Microscopic connection

The response function and optical constants of crystalline solids in the presence of transverse electromagnetic fields are of interest for semiconductors and insulators. Considering the next general electronic Hamiltonian

$$H_0 = \frac{\mathbf{p}^2}{2m} + V(\mathbf{r}), \tag{72}$$

where $V(\mathbf{r})$ is the potential felt by the electron in the system. Associated with this Hamiltonian are $\psi_n(\mathbf{r})$ and E_n.[3] If the electron system interacts with an electromagnetic field, described by a vector potential $\mathbf{A}(\mathbf{r}, t)$ with the Coulomb norm,[4] the Hamiltonian is

$$H = \frac{1}{2m}\left[\mathbf{p} + \frac{e}{c}\mathbf{A}(\mathbf{r}, t)\right]^2 + V(\mathbf{r}) = H_0 + \frac{e}{mc}\mathbf{A}(\mathbf{r}, t)\cdot\mathbf{p} + \frac{e^2}{mc^2}\mathbf{A}^2(\mathbf{r}, t), \tag{73}$$

[3] The electron spin degeneracy is taken into account by a factor of 2 if necessary.

[4] The scalar potential is zero as is the divergence of the vector potential.

the last two terms on the right-hand side of Eq. (73) being those that represent the quantum-mechanical interaction between the radiation and the electron charge. Considering a *plane electromagnetic and transverse wave*, (in addition to discarding the term $\mathbf{A}^2$ for the moment), of frequency ω, wave vector $\mathbf{q}$ and polarization vector $\hat{\mathbf{e}}$ described by the vector potential $\mathbf{A}(\mathbf{r}, t) = A_0 e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}\hat{\mathbf{e}} + c.c.$ (with $\hat{\mathbf{e}} \perp \mathbf{q}$); and also by (simplicity) A_0 is real and ω is positive, the resulting Hamiltonian is

$$H = H_0 + \frac{eA_0}{mc} e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}\hat{\mathbf{e}}\cdot\mathbf{p} + \frac{eA_0}{mc} e^{-i(\mathbf{q}\cdot\mathbf{r}-\omega t)}\hat{\mathbf{e}}\cdot\mathbf{p}. \tag{74}$$

The t-dependent terms on the right-hand side of Eq. (74) induce transitions between states of H_0 and can be treated with t-dependent perturbative theory. The first term in Eq. (74) gives rise to the radiative *absorption* mechanism, while the second term describes the *emission* mechanism. The probability per unit time that an electron initially in the state $|\psi_i\rangle$ (filled state) transitions to a final state $|\psi_j\rangle$ (empty state) upon the *absorption* of a photon of energy $\hbar\omega$ is given by the Fermi golden rule:

$$P_{i\to j} = \frac{2\pi}{\hbar}\left(\frac{eA_0}{mc}\right)^2 \left|\langle\psi_j| e^{i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}\,|\psi_i\rangle\right|^2 \delta\left(E_j - E_i - \hbar\omega\right). \tag{75}$$

In a similar way, one can define the transition probability that an electron initially in the state $|\psi_j\rangle$ (full state) is transferred to the final state $|\psi_i\rangle$ (empty state) with the *emission* of a photon of energy $\hbar\omega$

$$P_{j\to i} = \frac{2\pi}{\hbar}\left(\frac{eA_0}{mc}\right)^2 \left|\langle\psi_i| e^{-i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}\,|\psi_j\rangle\right|^2 \delta\left(E_i - E_j + \hbar\omega\right). \tag{76}$$

For a system in thermal equilibrium, the electronic states are occupied following a Fermi–Dirac distribution $f(E)$, such that the net number of transitions per unit time for the energy $\hbar\omega$ is given by

$$W(\mathbf{q}, \omega) = 2\sum_{ij} P_{i\to j}\left[f(E_i) - f(E_j)\right], \tag{77}$$

where the factor 2 in front of the sum accounts for electron spin degeneracy. Equation. (77) is useful for the microscopic expression of various optical response functions. For example, the dielectric function, $\varepsilon(\mathbf{q}, \omega)$, of a material which relates the electric displacement $\mathbf{D}$ to the electric field $\mathbf{E}$. The connection between the imaginary part of $\varepsilon(\mathbf{q}, \omega)$, $\varepsilon_2(\mathbf{q}, \omega)$, and the microscopic transition rate $W(\mathbf{q}, \omega)$ is

$$\varepsilon_2(q, \omega) = \frac{2\pi\hbar c^2}{\omega^2}\frac{1}{V}\frac{W(\mathbf{q}, \omega)}{A_0^2}. \tag{78}$$

The combination of Eqs. (77) and (78), the microscopic of the imaginary part of the transverse dielectric function results

$$\varepsilon_2(\mathbf{q},\omega)=\frac{8\pi e^2}{m^2\omega^2 V}\sum_{ij}\left|\langle\psi_j|e^{i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}|\psi_i\rangle\right|^2\delta(E_j-E_i-\hbar\omega)[f(E_i)-f(E_j)]. \quad (79)$$

Notice that the relation $\varepsilon_2(\mathbf{q},-\omega)=-\varepsilon_2(\mathbf{q},\omega)$ is valid. The full expresion of the complex dielectric function, $\varepsilon=\varepsilon_1+\varepsilon_2$, can be written in the form

$$\varepsilon(\mathbf{q},\omega)=1+\frac{8\pi e^2}{m^2 V}\sum_{ij}\frac{\left|\langle\psi_j|e^{i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}|\psi_i\rangle\right|^2}{(E_j-E_i)^2/\hbar^2}\frac{[f(E_i)-f(E_j)]}{E_j-E_i-\hbar\omega-i\eta}. \quad (80)$$

Equation. (80) is the fundamental expression of the transverse dielectric function of an electronic system described by a one-electron Hamiltonian H_0, with eigenvectors $\psi_i(\mathbf{r})$, and eigenvalues E_i.

Once $\varepsilon_2(\mathbf{q},\omega)$ is determined, it is possible to find other quantities such as the optical absorption coefficient, α,[5] which is related to $\varepsilon_2(\mathbf{q},\omega)$ and the refractive index $n(\omega)$ by

$$\alpha(\omega)=\frac{\omega}{cn(\omega)}\varepsilon_2(\mathbf{q},\omega). \quad (81)$$

Other linear optical properties such as the extinction coefficient $\kappa(\omega)$, the reflectivity $R(\omega)$, the energy loss function $L(\omega)$ and $n(\omega)$ can be calculated if the real part of $\varepsilon(\mathbf{q},\omega)$, $\varepsilon_1(\mathbf{q},\omega)$ is also known (this is obtained using the Kramers–Kronig relations) [224].

6.3 Electronic contribution to $\varepsilon(\omega)$

In this section, we briefly describe the optical properties of insulators and semiconductors with the constraint of $T=0$, this means fully occupied or fully empty one electron state. For this case, the Fermi–Dirac distribution can be dropped from Eq. (79). Thus, Eq. (79) has the next form

$$\varepsilon_2(\mathbf{q},\omega)=\frac{8\pi e^2}{m^2\omega^2 V}\sum_{i}^{occ}\sum_{j}^{unocc}\left|\langle\psi_j|e^{i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}|\psi_i\rangle\right|^2\delta(E_j-E_i-\hbar\omega). \quad (82)$$

Considering a radiation field with wavevector $\mathbf{q}$ and frequency ω, we now examine the matrix elements' electronic transitions from an initial valence state $\psi_{v\mathbf{k}_i}$ (for a given spin) into a final conduction state $\psi_{c\mathbf{k}_j}$ (with the same spin), this is

$$\langle\psi_{c\mathbf{k}_j}|e^{i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}|\psi_{v\mathbf{k}_i}\rangle=\int\psi_{c\mathbf{k}_j}(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}}\hat{\mathbf{e}}\cdot\mathbf{p}\psi_{v\mathbf{k}_i}(\mathbf{r})d\mathbf{r}. \quad (83)$$

[5] This is neglecting momentum transfer.

Since electronic states $\psi^n_{\mathbf{k}}(\mathbf{r})$ in a crystal can be expressed in a normalized $(1/V)$ Bloch form, with $u^n_{\mathbf{k}}(\mathbf{r})$ written as a Fourier series, then

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} u_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} \sum_{\mathbf{G}} \alpha_{\mathbf{k}}(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}}, \tag{84}$$

being $\alpha_{\mathbf{k}}(\mathbf{G})$ the complex Fourier coefficients in a plane-wave expansion, and

$$\langle \psi_{c\mathbf{k}_j} | e^{i\mathbf{q}\cdot\mathbf{r}} \hat{\mathbf{e}}\cdot\mathbf{p} | \psi_{v\mathbf{k}_i} \rangle = \sum_{\mathbf{G}_j,\mathbf{G}_i} \left[\alpha_{c\mathbf{k}_j}(\mathbf{G}_j)\right]^* \alpha_{v\mathbf{k}_i}(\mathbf{G}_i) \left(\frac{\hat{\mathbf{e}}\cdot\mathbf{p}}{V}\right) \int e^{(\mathbf{k}_i - \mathbf{k}_j + \mathbf{q} + \mathbf{G}_i - \mathbf{G}_j)\cdot\mathbf{r}} d\mathbf{r}, \tag{85}$$

On one hand, it is noteworthy to mention that Eq. (85) has a sum of plane-wave coefficients $\alpha_{c\mathbf{k}_{j,i}}(\mathbf{G}_{j,i})$. This first term is relevant for the magnitude of electric transition dipole moment calculation from *ab initio* [225]. On the other hand, for the (second) integral term, remember that the derivative $\mathbf{p}\psi_{v\mathbf{k}_i}(\mathbf{r}) = -i\hbar\partial\psi_{v\mathbf{k}_i}(\mathbf{r})/\partial\mathbf{r}$ of the Bloch function of vector $\mathbf{k}_i$ results a Bloch function of wavevector $\mathbf{k}_i$. Second, the multiplication by $\exp(i\mathbf{q}\cdot\mathbf{r})$ also of a Bloch function of wavevector $\mathbf{k}_i$ produces a Bloch function of wavevector $\mathbf{q}+\mathbf{k}_i$. Such that the last integral produces a δ function in reciprocal space vectors

$$\left(\frac{1}{V}\right) \int e^{(\mathbf{k}_i - \mathbf{k}_j + \mathbf{q} + \mathbf{G}_i - \mathbf{G}_j)\cdot\mathbf{r}} d\mathbf{r} = \delta(\mathbf{k}_i - \mathbf{k}_j + \mathbf{q} + \mathbf{G}_i - \mathbf{G}_j), \tag{86}$$

and this, in turn, results in the matrix element in Eq. (83) different from zero solely if

$$\mathbf{k}_j = \mathbf{q} + \mathbf{k}_i + \mathbf{G}_i - \mathbf{G}_j = \mathbf{q} + \mathbf{k}_i + \mathbf{G}, \tag{87}$$

with $\mathbf{G} = \mathbf{G}_i - \mathbf{G}_j$ a vector of the reciprocal lattice including $\mathbf{G} = \mathbf{0}$, the Γ-point. If we consider and experiment where the wavelength of the incident electromagnetic wave is much larger than the lattice parameter, then the photon wave vector $\mathbf{q}$ of the incident radiation is very small compared to the range of values of $\mathbf{k}$ within the first Brillouin zone.[6] Therefore, we can take $\mathbf{q}\to 0$ in Eq. (87) (dipole approximation) and choice $\mathbf{k}_j \approx \mathbf{k}_i$ (vertical transitions). Using Eq. (82) for insulator or semiconductor materials with fully occupied valence bands and fully empty conduction bands, we denote $\varepsilon_2(\mathbf{q}=0,\omega)$ by $\varepsilon(\omega)$, with only vertical transitions, and we have

$$\begin{aligned} \varepsilon_2(\omega) &= \frac{8\pi^2 e^2}{m^2\omega^2} \frac{1}{V} \sum_{cv} \sum_{\mathbf{k}} \left| \langle \psi_{c\mathbf{k}} | \hat{\mathbf{e}}\cdot\mathbf{p} | \psi_{v\mathbf{k}} \rangle \right|^2 \delta(E_{c\mathbf{k}} - E_{v\mathbf{k}} - \hbar\omega) \\ &= \frac{8\pi^2 e^2}{m^2\omega^2} \sum_{cv} \int_{B.Z.} \frac{d\mathbf{k}}{(2\pi)^3} \left| \hat{\mathbf{e}}\cdot\mathbf{M}_{cv}(\mathbf{k}) \right|^2 \delta(E_{c\mathbf{k}} - E_{v\mathbf{k}} - \hbar\omega), \end{aligned} \tag{88}$$

[6] Since the radiation momentum $|\mathbf{q}| = (2\pi/\lambda)$, then in the limit of long wavelength $\mathbf{q}\to 0$ and $e^{i\mathbf{q}\cdot\mathbf{r}} \approx 1 + i\mathbf{q}\cdot\mathbf{r} \simeq 1$. Usually, the radiation wavelength is $\lambda \sim 10^4$ Å, while the crystal wave vectors have wavelength values in the order of interatomic distances, ~ 1 Å.

with ω positive, the sum runs over each pair of valence and conduction bands, and $\mathbf{M}_{cv}(\mathbf{k})$ denotes the dipole matrix element $\langle\psi_{c\mathbf{k}}|\mathbf{p}|\psi_{v\mathbf{k}}\rangle$. It is worth to mention that, for periodic systems, the absorption spectrum due to band-to-band transitions can show a sharp behavior, which correspond to critical points in the Joint Density Of States (JDOS). Given a couple of valence and conduction bands, if the dipole matrix element $\hat{\mathbf{e}}\cdot\mathbf{M}_{cv}(\mathbf{k})$ changes slowly (or is a smooth function) of $\mathbf{k}$ over the whole first Brillouin zone, the its average value can be factorized from Eq. (88). Thus, the contribution to the optical constants from these couple of valence and conduction bands is determined by the so called JDOS

$$J_{cv}(\omega) = \int_{BZ} \frac{d\mathbf{k}}{(2\pi)^3}\delta(E_{c\mathbf{k}} - E_{v\mathbf{k}} - \hbar\omega). \tag{89}$$

The critical points in the JDOS are those for which $\nabla_{\mathbf{k}}(E_{c\mathbf{k}} - E_{v\mathbf{k}}) \equiv 0$.

Figure 19 shows the calculated real and imaginary part of the dielectric function for bound electrons, ε_1^b and ε_2^b, in a wide energy range to shows the obtained plasma frequency $\hbar\omega_p$. Clear differences in the are shown between Al and Ag systems for all the calculated energy values, but also for Si and Ge systems below energy lower than 10 eV. The allowed direct interband transitions leading to a non-zero terms in the dielectric function for Ag can be ascribed to d electron states. For the low-energy region, below -5 eV, several peaks for the Al and a broad peak for Ag are present. In particular, the broad peak for ε_2 in Ag around 4 eV is due to electron transitions from occupied states (belowe the Fermi energy) to unoccupied states (above the Fermi energy) just at the high-symmetry points X and L in the band structure of Fig. 18. In the same Fig. 18, other possible transitions, to give contributions to dielectric properties below 2 eV for Al, are found around $U|K$ and W points. Regarding the Si semiconductor, the interband transitions around X, Γ, and L for Si are responsible for the peaks below 5 eV in the component ε_2^b in Fig. 19. Finally, for the Ge system, within the exchange–correlation approximation used, the two peaks in 2 and 4 eV could be assigned to L and X points, respectively, in the band structure. This analysis is in good agreement with regions in reciprocal space where the electronic band dispersion changes very low giving to a high density of states (DOS). To end this subsection, these first-principles results are in fair agreement with experimental results of Ehrenreich et al. [226, 227]. Once the ground-state electronic structure properties are found, the plasma frequency can be obtained by tracing the following tensor [228]

$$\omega^2_{p,(\alpha,\beta)} = -\frac{4\pi e^2}{\Omega}\sum_{v,\mathbf{k}} 2\frac{\partial w_{\mathbf{k}}(E_v)}{\partial E_v}\left(\frac{\partial E_{v,\mathbf{k}}}{\partial \mathbf{k}}\mathbf{e}_\alpha\right)\left(\frac{\partial E_{v,\mathbf{k}}}{\partial \mathbf{k}}\mathbf{e}_\beta\right); \tag{90}$$

thus, the calculated plasma frequency for each system is shown in Fig. 19, as well. On the other hand, the electron scattering rate, γ, is obtained from the imaginary part of the electron–phonon self-energy, $\Sigma_{n,\mathbf{k}}(T)$ [228, 229]. Using the electronic and the

phonon band dispersions, the self-energy is calculated as

$$\Sigma_{n,\mathbf{k}}(T) = \frac{2\pi}{\hbar} \sum_{\mathbf{q},\mathbf{vm}} w_{\mathbf{q}} \left| g^{SE}_{mn,\nu}(\mathbf{k},\mathbf{q}) \right|^2 \times \left[\frac{n_{\mathbf{q},\nu}(T) + f_{m,\mathbf{k}+\mathbf{q}}(T)}{E_{n,k} - E_{m,k'} + \omega_{\mathbf{q}\nu} - i\eta} + \frac{n_{\mathbf{q},\nu}(T) + 1 + f_{m,\mathbf{k}+\mathbf{q}}(T)}{E_{n,k} - E_{m,k'} - \omega_{\mathbf{q}\nu} - i\eta} \right], \tag{91}$$

with $w_{\mathbf{q}\nu}$ as the phonon eigen frequency of band ν and wavevector $\mathbf{q}$, $n_{\mathbf{q}\nu}(T)$ is the Bose–Einstein distribution function, $f_{m,\mathbf{k}+\mathbf{T}}(T)$ as the Fermi–Dirac distribution function, and $g^{SE}_{mn,\nu}(\mathbf{k},\mathbf{q})$ the electron–phonon coupling matrix term. Therefore, the scattering rate term of the electron for the band n and wavevector $\mathbf{k}$ is given by $\gamma_{ep}(n\mathbf{k},T) = 2\mathrm{Im}\Sigma_{n\mathbf{k}}(T)$. In fact, the electron–phonon coupling is key to determining the T-dependence of electron lifetime, $\tau = 1/\gamma_{ep}$, and intraband dielectric function, $\varepsilon^f(\omega)$, described by the Drude model in Eq. (23). Experimentally, ellipsometry experiments are useful to study finite-temperature dielectric properties of noble metals such as Ag, Cu, and Au [228]. The calculated first-principles plasma frequency and scattering rate term allow to define the intraband dielectric function with the Drude model of Eq. (23). Besides the free-electron contribution to the dielectric properties for intraband transitions, the optical phonons are further important to describe the dielectric properties. For this reason, the next section presents briefly the general aspects of the contribution of the photon–phonon interaction to the total dielectric function, $\varepsilon(\omega)$, of the material.

6.4 Ionic contribution to $\varepsilon(\omega)$

The dynamical matrix formalism or "interatomic force constant method" is useful to determine the phonon dissension curves in crystals. In the case of polar crystals such as ionic crystals or heteropolar semiconductors, optical phonons are electric dipole-active. This means that, quantized mechanical waves (phonons) and electromagnetic waves (photons) couple each other to give mixed phonon–photon modes. Along high-symmetry directions of wavevector $\mathbf{q}$, in 3D cubic crystal systems, there exist six phononic branches: two degenerate transverse acoustic (TA) and one longitudinal acoustic (LA) branches, and two degenerate transversal optical (TO) modes and one optical longitudinal (LO) branch. These LO and TO branches converge to finite values of frequency ω_{LO} and ω_{TO} in the long wavelength limit ($\mathbf{q}\to 0$) [230].[7] For the case of nonpolar diatomic crystals, the optical modes around the Γ point are degenerate, this means $\omega_{LO} \equiv \omega_{TO}$; conversely, in diatomic polar crystals, the $\omega_{LO} \neq \omega_{TO}$ if retardation effects are discarded in addition to the long-range and anisotropy of the electrostatic dipole–dipole interactions. This TO-LO energy splitting gives a phononic band gap with implications on the infrared regime of dielectric function for polar semiconductors and insulators. Hence, materials reflect those wavelengths within the region $\omega_{TO} < \omega < \omega_{LO}$. Moreover, the frequencies ω_{LO} and ω_{TO}

[7] For $\mathbf{q}$ values around the Γ point in the phonon dispersion relation.

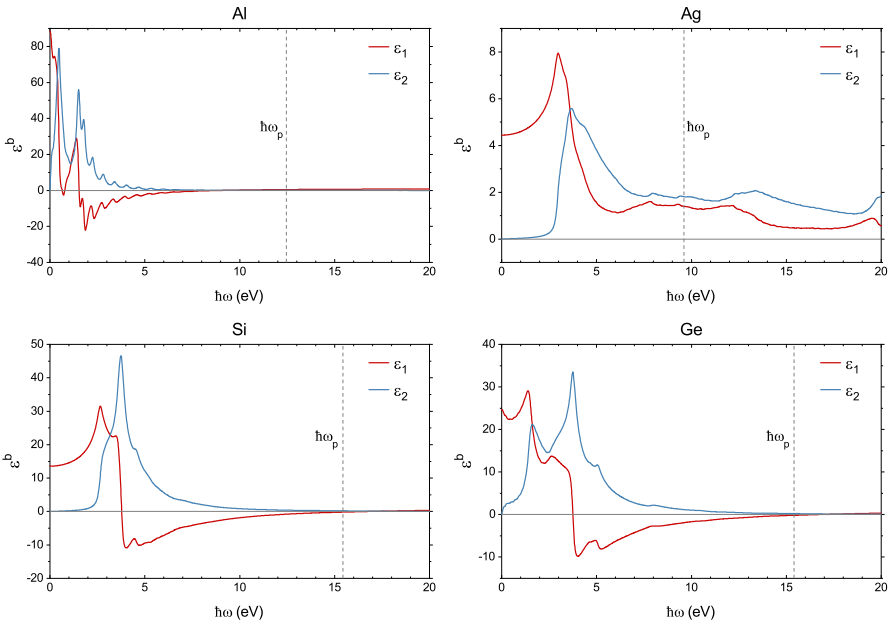


Fig. 19 Calculated dielectric properties for Al, Ag, Si, and Ge systems for bound electrons. Only direct electronic transitions were accounted for real (ε_1) and imaginary (ε_2) terms that are colored with red and blue, respectively. The calculated plasma frequencies are $\hbar\omega_p = 12.45$, 9.6, 15.44, and 15.4 eV for Al, Ag, Si, and Ge, respectively

satisfy the Lyddane–Sachs–Teller relation [231]

$$\frac{\omega_{LO}^2}{\omega_{TO}^2} = \frac{\varepsilon_s}{\varepsilon_\infty}, \tag{92}$$

with ε_s as the static dielectric constant (also known as ε_0), and ε_∞ as the high-frequency dielectric constant. ε_∞ is determined by the electronic contribution to the static dielectric constant. It is clear from Eq. (92) that if the ratio between TO and LO frequencies changes, the ratio between dielectric constants also does.

In general, ε is not a dielectric constant but rather a dielectric tensor $\varepsilon_{\alpha\beta}(\omega)$. The ionic contribution to the macroscopic low-frequency dielectric function $\varepsilon_{\alpha\beta}(\omega)$ leads to [232]

$$\varepsilon_{\alpha\beta}(\omega) = \varepsilon_{\alpha\beta}^{\infty} + \frac{4\pi e^2}{\Omega_0} \sum_{\nu} \frac{S_{\alpha\beta,\nu}}{\omega_\nu^2 - (\omega + i\eta)^2}, \tag{93}$$

with ω_ν as the phonon eigen frequency of mode ν, and $S_{\alpha\beta,\nu}$ as the mode-oscillator strength

$$S_{\alpha\beta,\nu} = \left(\sum_{I,\delta} Z_{I\alpha\delta}^* \varepsilon_{I\delta,\nu}^*(\mathbf{q} = \mathbf{0}) \right) \left(\sum_{J,\delta'} Z_{I\beta\delta'}^* \varepsilon_{J\delta',\nu}^*(\mathbf{q} = \mathbf{0}) \right). \tag{94}$$

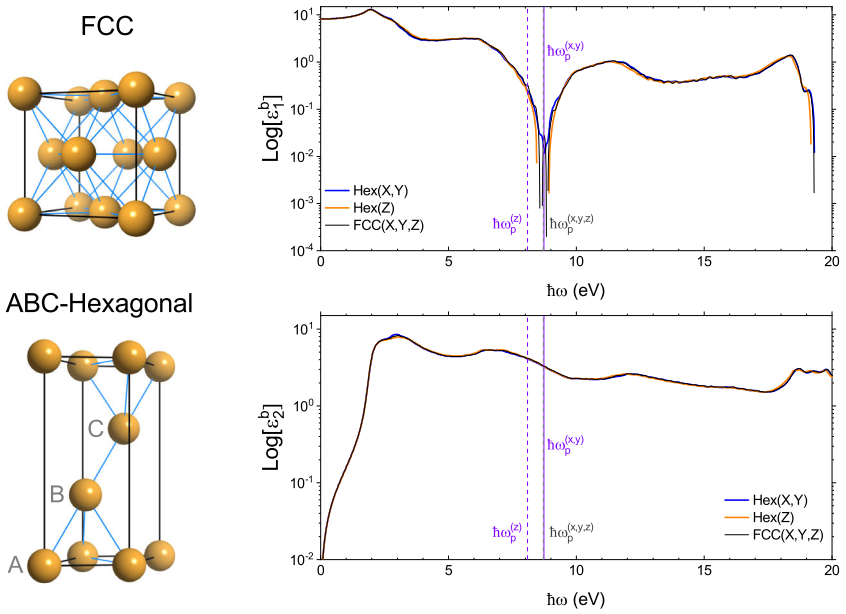


Fig. 20 Dielectric properties for bound electrons of Au in bulk with two different crystal structures: FCC and ABC-Hexagonal. Numerical calculations of optical properties were done with a fine sampling of Brillouin zone to achieve the same resolution for in-plane $\varepsilon_2^b(x, y)$ components

The $Z_{J\beta\delta'}$ are the Born effective charges and $\varepsilon_{J\delta',\nu}(\mathbf{q})$ are the eigen displacements related to the phonon mode ν for atom J along the direction δ'.

7 Casimir force in microdevices and nanodevices

Microdevices, in the form of microelectromechanical systems (MEMS) [233], have been present in our daily life for a long time in the form of microchips containing accelerometers, gyroscopes, pressure sensors, and compasses. Most of these sensors have been incorporated in smartphones but can also be found, for instance, in cars and wrist watches. Many other microdevices have been developed like microopto-electromechanical systems (MOEMS) [234] and microfluidic systems [235] which are in different stages of research and development (Fig. 21).

As the technology evolves, it is expected that the dimensions of microdevices decrease, to save space and energy. Smaller versions MEMS and MOEMS, with many features at the nanoscale (nanoelectromechanical systems, NEMS, and nanooptoeletromechanical systems, NOEMS), have also been developed, but are not yet commercially available. In going from microscale to the nanoscale, many new physical phenomena start playing a role in the functioning of the systems. The Casimir force is one example.

Many microdevices and nanodevices present complex moveable parts. As these parts get closer, the Casimir force can become comparable to other actuating forces,

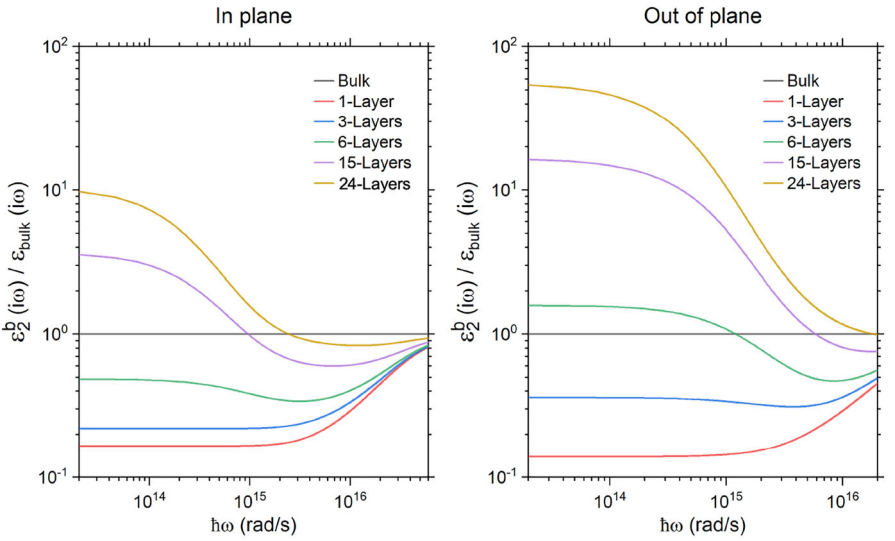


Fig. 21 Numerical results for the calculation of $\varepsilon(i\zeta)$ with the imaginary part of the dielectric function $\varepsilon_2(\omega)$ using Eq. (7) for the ABC-Hexagonal Ag bulk and slab systems, see Fig. 20

like the electrostatic force. It is in this context that it is relevant to know when and how to consider the Casimir force in microsystems and nanosystems.

Starting with the work of Serry et al. in 1995 [236], there is now a great number of theoretical works that have incorporated the Casimir force in the analysis and design of a great variety of microdevices and nanodevices. Some experiments involving MEMS on which the Casimir force was either measured or its consequences were investigated have also been performed. We discuss separately the purely theoretical analysis and experimental works in the next subsections.

7.1 Theoretical analysis

The Casimir force was first incorporated in the theoretical analysis of MEMS by Serry et al. in 1995 [236]. In this work, the MEMS analyzed was an anharmonic oscillator. Its mechanical behavior was modeled as a simple spring-mass system. The model assumed that a plate was held by a spring and was close to an electrode. It was, thus, subject to three forces: the elastic force from the spring, and the electrostatic and the Casimir forces. The anharmonicity originated from the fact that these two last forces are nonlinear functions of the displacement of the plate, as it is going to be exemplified later in this Sect. 7.3. Serry et al. [236] made no reference to a particular kind of MEMS; however, its model could describe accurately a MEMS of the kind depicted in Fig. 22(a), in which a large plate is supported by four tethers, that act like springs. Later in 1998, the same authors investigated the deflection and stiction of membrane strips caused by the Casimir force [237]. In this case, the simple parallel plate model was not adequate, and the membranes were modeled using a continuum model and taking into account the curvature of the membrane, which is illustrated for a similar system

in Fig. 22(b). These two works employed the two approaches most widely used later by other authors in the modeling of many microdevices and nanodevices.

Most of the theoretical works analyzed micro- and nanoswitches or micro- and nano-oscillators of different types. Lin and Zhao [238, 239] modeled cantilever and bridge nanobeams as simple movable parallel plates, ignoring the curvature that accompanies the beam displacement (See Fig. 22b). With this simple model, they analyzed the dynamics and pull-in parameters of the beams, incorporating the van der Waals [238] and, separately, the Casimir force [239]. The same authors have later analyzed an electrostatic torsional actuator subject, separately, to the van der Waals and Casimir force [240]. In the works mentioned so far, the van der Waals or Casimir forces were modeled simply as forces proportional to d^{-3} and d^{-4}, respectively. However, as we have seen in Sect. 5, the Casimir force between large planar bodies is corrected, at least, for the finite conductivity. As a consequence, the effective exponent at separations ranging from a few nanometers up to hundreds of nanometers is going to vary with d between 3 and 4. In spite of this change in the actual effective exponent for the dependence of the force with d, most of the hundreds of theoretical works on the Casimir force on M/NEMS used the force between ideal conductors or the limit of the van der Waals force in their calculations [238–248]. Only in a few works, the force corrected to the finite conductivity was employed [62, 63, 249, 250].

It is important to stress that the use of the force corrected to the finite conductivity is necessary to get realistic results in theoretical simulations of microdevices and nanodevices. Let us note, for instance, that the correction factor for silicon, the most used material in M/NEMS, is quite analogous to that for GaAs depicted in Fig. 1(a) (See Ref. [60]). The η for GaAs shows that while the Casimir force goes approximately as d^{-4} for $d \gtrsim 100$ nm, it is reduced by a factor of three compared to that between perfectly conducting plates, a result that holds for silicon. For $d \lesssim 3$ nm, $\eta(d)$ varies linearly, which implies that only at such small separations, the force has turned into the van der Waals force, varying as d^{-3}. In the range between approximately 3 and 100 nm, the Casimir force is changing as d^{α} with $3 < \alpha < 4$. Other materials behave differently in some aspects. In the case of SiO_2, as seen in Fig. 1(b), η does not have a plateau above 100 nm due to the IR-active phonons, and the effective exponent varies continuously without plateauing up to, at least, one micrometer. This trend is found for other materials used to fabricate microdevices and nanodevices which are IR-active, like silicon carbide and silicon nitride [60]. It is interesting to note that, even for a good conductor like gold, the effective exponent does not reach the value corresponding to that for an ideal conductor for d below a few micrometers, as evidenced by the continuous variation of η in Fig. 12.

From the analysis of the Casimir force for several materials frequently used in microdevices and nanodevices presented by Gusso and Delben [60], it is possible to conclude that: (i) in the limit of small separations, below a few nanometers, the force is going to behave as expected for the van der Waals force for all materials, varying as d^{-3}; (ii) in the intermediate range between a few nanometers and a few tens of nanometers, the correction factor increases monotonically; (iii) for metals, it follows increasing monotonically up to, at least, a few hundreds of nanometers, after which an inflection occurs towards a constant value close to one; (iv) for semiconductors or insulators without IR activity, η tends to a constant value above about 100 nm,

indicating that the force goes as d^{-4}. (v) for IR-active semiconductors and insulators η above $\sim$ 100 nm has a more complex behavior, increasing at significantly different rates as d varies.

In spite of the fact that the Casimir force between perfect conductors and what is usually refered to as the van der Waals force are just limiting cases of the same force, many authors have considered them as being two different and independent forces. In some cases, this has led the authors to consider these two forces as two distinct contributions, in some cases acting separately in the same or similar range of distances [244, 245, 247, 248] and in others acting simultaneously at any separation [246, 251].

Other factors can influence the Casimir force besides the finite conductivity of the materials. The geometry, for instance, is also quite relevant, as already discussed in Sect. 3. In the case of the planar geometry that was considered so far and found in suspended plates, membranes, and wide beams, a correction comes from the fact that these movable parts in microdevices and nanodevices are always finite in thickness. The reflection properties of a plate of finite thickness are changed compared to those of a semispace. The critical thickness that makes this correction relevant is going to depend upon the materials involved and the configuration (a plate interacting with a semispace or two interacting plates, for instance) [48, 59]. The finite thickness correction for metals is much smaller than that for semiconductors and insulators [59]. In case of metals, plates with thickness larger than a few tens of nanometers can have their reflectivity approximated by that of a semispace. However, semiconductors like silicon can only be approximated as semispaces for thickness almost one order of magnitude larger [48, 59]. The finite thickness effect has been generally ignored in the modeling of microdevices and nanodevices. One of the few works which included this effect is that of Rodrigues et al. [64], in which a MOEMS comprised by a 110 nm thick silicon beam on top of a SiO_2 substrate was analyzed. The finite thickness effect was considered approximating the beam with a width of 500 nm by an infinite plate and using generalized Fresnel reflection coefficients.

More complex geometries including finite structures, like small discs or beams, are also of interest. However, the calculation of the Casimir force in these cases is much more complex as seen in Sect. 3. Maybe because of this complexity, there have been very few theoretical analysis of microdevices and nanodevices on which nontrivial effects of the geometry have been included. The stress-tensor method calculated via FDTD was used to go beyond the infinitely extended plane approximation in the modeling of MOEMS in the case of the interaction between two rectangular beams [254] and a rectangular beam and a semispace [64].

Other methods have also been used to calculate the Casimir or van der Waals force for more complex geometries. Dequesnes et al. [255] have modeled carbon-nanotube-based nanoswitches, taking into account molecular forces adopting the pairwise summation method, within both a continuum and a atomistic model of the nanotube. In this work, the intermolecular force was given by the van der Waals potential (the potential energy between two atoms or molecules varies as d^{-6}). A Casimir–Polder potential energy (varies as d^{-7}), valid for larger intermolecular separations, was used in Ref. [256] to model the force between circular and rectangular beams and a plane substrate.

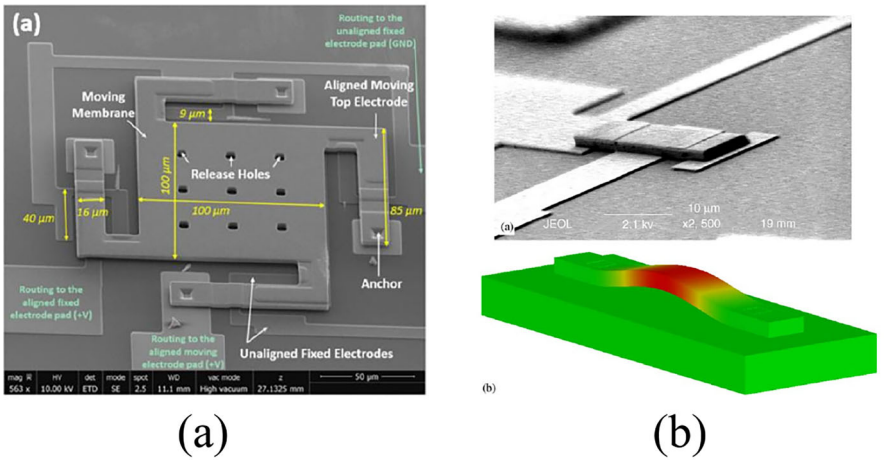


Fig. 22 (**a**) Image of a MEMS varactor comprised by a movable plate suspended by four springs. Reproduced from Ref. [252]. (**b**) On the top panel, it is presented the image of a suspended doubly clamped (bridge) MEMS resonator, while the bottom panel shows a simulated vibrational mode. Reproduced from Ref. [253]

Another geometrical characteristic, that can be treated as a separate geometrical effect on the Casimir force, is that of surface roughness. The roughness effect was first considered by Palasantzas and De Hosson [257] who modeled metallic switches as simple spring-mass systems. In this work, the roughness effect was calculated using the method of Genet et al. [258] based on the PFA. The effect of roughness was also included in the analysis of a torsional actuator made from silicon by Gusso and Delben [62]. In this work, the roughness correction was calculated using the perturbative PWS [91].

Besides the direct effect of the Casimir force on microdevices and nanodevices, it has been predicted theoretically that this force can affect the dissipation through noncontact friction [259, 260]. In Ref. [259], the Casimir force between a vibrating beam and the nearby underlying substrate is predicted to generate a variable pressure on top of the substrate that results in the conversion of mechanical energy of the beam into acoustic waves. The Casimir force can also be behind the large tip-sample dissipation observed in experiments with atomic force microscopes. If that is the case, the observed dissipation can also be present in the interaction of a vibrating beam and the nearby substrate [260].

7.2 Experiments in Casimir-driven MEMS

In 2001, Chan et al. [9] developed a MEMS based on a torsional balance that was used to measure the Casimir force. A large gold-plated sphere was brought in close proximity to one side of the plate of the torsional balance, causing an attractive force that tilted the plate. The force was inferred by measuring the tilting angle through the change in the capacitance of the other side of the plate. This experiment demonstrated

an alternative manner to measure the Casimir force and, at the same time, demonstrated that a MEMS could be actuated by this force. The nonlinear dynamics of the same device was later studied experimentally [261]. In this case, the interaction between the plate and the sphere results in a nonlinear force. It corresponds to the softening of the spring constant with the increase in the amplitude of motion of the plate. The nonlinear force resulted in the bistability in the frequency response curve, characteristic of a nonlinear oscillator, which was measured in the experiment.

A MEMS was custom built as a platform to measure the Casimir force between between two planar structures [262] and the interpenetration of trenches with non-planar [74] and planar geometries [75]. The last two setups have allowed the measurement of nontrivial effects of the geometry on the Casimir force.

Using a commercial MEMS accelerometer, a metrology platform was developed by Stange et al. [263]. The system comprised a silver-coated sphere that was glued on top of the proof mass of the accelerometer. By bringing a gold-coated parallel plate in close proximity, the Casimir force was measured. This platform was investigated theoretically for application as a magnetic gradiometer [264] or a system with zeptometer positional resolution [265].

7.3 Example theoretical modeling

In this section, we discuss how the Casimir force should be included in the modeling of MEMS and NEMS. The simplest M/NEMS to model are those that are described by one generalized coordinate. For example, a parallel plate capacitive switch or oscillator or a torsional balance.

M/NEMS as the one depicted in Fig. 22 can be adequately modeled as a simple lumped system. This device can be depicted by a mass-spring system consisting on a parallel plate of mass m held by a spring with elastic constant k on top of a large substrate. Let us assume that the plate can move along the z axis with $z = 0$ at its rest position and increasing in the direction opposite to that of the substrate. The initial gap between plate and substrate is $d = d_0$ in such a way that as the plate moves, the separation becomes $d = d_0 + z$.

The total force acting on the plate is

$$F = F_{elastic} + \sum_{i=1}^{N} F_i, \tag{95}$$

where F_i can be any other force besides the elastic, acting on the plate. For example, electrostatic, van der Waals/Casimir, capillary, etc. In the presence of electrostatic and the Casimir forces, the equation of motion for the plate assumes the form

$$m\frac{d^2z}{dt^2} = -kz - \frac{\varepsilon_0 A V^2}{2}\frac{1}{(d_0+z)^2} - \frac{\pi^2\hbar c A}{240}\frac{\eta(d_0+z)}{(d_0+z)^4}, \tag{96}$$

where m is the mass of the plate, A its area, and V the voltage applied across the gap. The correction factor to the Casimir force, η, can include the finite conductivity

correction, the effect of finite thickness, that of surface roughness, as well as any other factor contributing significantly to determine the actual Casimir force. The electrostatic force can also be corrected to account, for instance, for the fringing fields.

Nonlinear ordinary differential equations similar to Eq. (96) have been obtained and solved numerically in several of the first works which included the Casimir force in the modeling of microdevices and nanodevices [236, 238, 239, 241].

For an accurate analysis of the statics and dynamics of devices corresponding to suspended beams or membranes, like the one depicted in Fig. 22(b), the use of a continuum elastic model is required. A M/NEMS doubly clamped suspended beam actuated by a one-sided electrode would be described by a nonlinear partial differential equation of the form [63]

$$EIw'''' + \rho A\ddot{w} + c\dot{w} - \frac{EA}{2l}w'' \int_0^l w'^2 dx = -\frac{\varepsilon_0 V(t)^2 b}{2(d_0+w)^2} - \frac{\pi^2 \hbar c\, \eta (d_0+w) b}{240(d_0+w)^4} \quad (97)$$

In this equation, E denotes de Young modulus of the beam material and ρ its density, A its cross-sectional area, I the bending moment of inertia, and c the damping coefficient. The function $w(x,t)$ represents the transversal displacement of the beam, that has width b and spans a length l along the x axis. The two terms in the right side of the equation correspond to the electrostatic and Casimir force per unit length. As it has been generally the case in the continuum modeling of beams, membranes, or torsional actuators, these two forces are computed assuming small curvature or inclination of the structures, which allows the use of the piecewise plane approximation [241–248, 251].

More elaborate elastic models of the beam or membranes have been adopted in the literature in some cases with the inclusion, for instance, of the surface [266, 267] or size effects [243]. However, the Casimir force acting along the beam is introduced as in Eq. (97). Alternatives to this method would include the calculation of the elastic, electrostatic, and the Casimir force using multiphysics models, in which the force along the structure would be calculated using some of the methods described in Sect. 3.2.

7.4 Casimir forces in self-assembly of nanosystems

As we have seen, the Casimir–Lifshitz interactions can do more than produce static attractions: in appropriately designed geometries and media, they can select configurations, impart torques, and even trap nanoscale objects in metastable states. This section surveys recent demonstrations of Casimir-driven or Casimir-assisted self-assembly, emphasizing (i) rotational self-alignment in templated microcavities, (ii) liquid-phase "Casimir self-assembly" as a measurement platform.

Self-assembly can be defined as an autonomous process by which components—such as atoms, molecules, macromolecules, or colloids—organize into stable, structurally well-defined arrangements through local interactions [268, 269]. These interactions can be, for instance, the screened electrostatic interaction, depletion forces due to the presence of micelles, entropic forces [270] and the Casimir–Lifshitz forces.

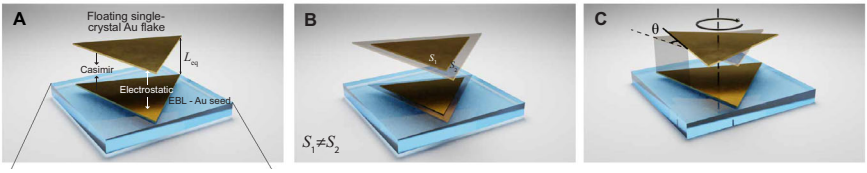


Fig. 23 Diagram of the self-assembly process of Au nanoprisms suspended in a solution to form stable microcavities. The self-assembly involves a lateral and rotational self-assembly. Figure taken from Ref. [273], published under Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC)

Young et al. [271] reported reconfigurable 1D superlattices of triangular Au nanoprisms whose equilibrium spacings and symmetries result from a synergy of depletion attraction and screened electrostatic repulsion [271]. Although the driving forces emphasized there were classical (depletion/DLVO) [272], the study established two design rules still pertinent for Casimir-assisted assembly: (i) anisotropy selects orientation and registry, and (ii) soft, long-range competitors (screened Coulomb, depletion, or fluid-mediated dispersion forces) can stabilize large interparticle spacings compared to the thickness of the Au naonoprisms. The spacing of the self-assembled structure was controlled by varying salt-concentration.

Improvements on self-assembly, on what can be considered Casimir-driven self-assembly, was done by Küçüköz et al. [273] who realized triangular Casimir microcavities in which a floating Au nanoflake, separated from a templated Au triangle by a liquid spacer, undergoes lateral and rotational self-alignment at true Casimir separations ($\sim$ 100–200 nm). The stability arises from a quantum trap created by the competition of a screened electrostatic repulsion and an attractive Casimir potential. At fixed ionic strength (Debye length κ^{-1}), the total free energy,

$$\mathcal{F}(\theta, d) \simeq \mathcal{F}_{\text{Casimir}}(\theta, d) + \mathcal{F}_{\text{DLVO}}(d), \tag{98}$$

develops a minimum at a separation d^* and orientation θ^* that maximize the overlap area of the opposing triangles, as shown in Fig. 23. The angular dependence comes from the anisotropic spectral response of the triangular cavity (effectively a tunable nanoscale Fabry–Pérot cavity), which gives rise to a Casimir torque $\tau = -\partial\mathcal{F}/\partial\theta$ that drives rotational self-alignment. Notably, electrostatic screening tunes both d^* and θ^*, providing an external knob for programmable assembly and on-chip reconfiguration [273].

Improvements on the experimental setups led Hošková et al. to introduce Casimir self-assembly (CaSA) as an *in situ* platform in liquids to map nanoscale interaction landscapes while assemblies form [274]. By varying ionic strength and surface functionalization, they delineate stability regimes where thermal fluctuations and Casimir–electrostatic competition select repeatable configurations. Beyond structure control, CaSA enables metrology since individual-particle surface charge densities with sub-e/nm^2 precision and locate aggregation thresholds in the space of salt, potential, geometry. Because the fluid permittivity and dispersion strongly influence Lifshitz spectra, the same setup can, in principle, probe repulsive regimes (via appro-

priate three-medium dielectric ordering), making CaSA broadly useful for testing fluctuation-induced forces in complex media [274].

8 Nonequilibrium Casimir force

Lifshitz theory assumes thermal equilibrium. In this situation, the Casimir–Lifshitz force at a finite temperature has two contributions: one from quantum zero-point fluctuations at $T = 0$, which dominate at small separations, and the other due to thermal fluctuations of the electromagnetic fields. The thermal component of the Casimir force is relevant at length scales comparable to the wavelength of thermal photons, $\lambda_{th} = \hbar c/k_B T$, becoming dominant over the quantum contribution. While this is a good estimate for typical 3D materials, Gomez-Santos [275] showed that in graphene, the thermal length scale is shorter and is given by $\hbar v_F/k_B T$, where $v_F \approx 10^6$ m/s is the Fermi velocity in graphene.

When a system is driven out of thermal equilibrium, the Casimir force can acquire entirely new features. In such regimes, repulsive interactions, oscillatory behaviors, and even self-propelling effects—where two bodies experience forces of equal magnitude and direction—have been predicted. Building on the fluctuation electrodynamics framework originally developed by Lifshitz, Henkel et al. [276] carried out one of the first analyses of the non-equilibrium Casimir–Polder force on a dipolar particle near a substrate at finite temperature T, embedded in an environment held at zero temperature. Extending this scenario to finite environmental temperatures, Antezza et al. [277] showed that the force exhibits a slower asymptotic decay ($\propto 1/z^3$) at large separations, and that its sign depends sensitively on relative temperatures: it becomes repulsive when the environment is hotter than the substrate, and attractive in the opposite case. The theoretical predictions of Ref. [277] were later confirmed experimentally by the JILA group, who measured the temperature-dependent Casimir–Polder interaction between a ^{87}Rb Bose–Einstein condensate and a dielectric substrate [278]. A generalization of the Casimir–Lifshitz theory for surface–surface, surface–atom, and surface-rarified body configurations, as well as between planar plates with thick and thin thickness, was presented in the pioneering papers [279, 280] in out-of-thermal-equilibrium conditions where each body is assumed at a different local temperature. For the simplest configuration of two parallel half-spaces at two different temperatures T_1 and T_2 and separated by a distance d, the total nonequilibrium Casimir pressure $P^{neq}(T_1, T_2, d)$ can be expressed in terms of corrections $\Delta P(T, d)$ about the quantum pressure, $P_0(d)$, and the mean equilibrium pressure $\overline{P_{th}^{eq}} = [P_{th}^{eq}(T_1, d) + P_{th}^{eq}(T_2, d)]/2$ originating, respectively, from vacuum and thermal fluctuations of the electromagnetic fields,

$$P^{neq}(T_1, T_2, d) = P_0(d) + \overline{P_{th}^{eq}} + \Delta P(T_1, d) - \Delta P(T_2, d), \tag{99}$$

The Lifshitz pressure associated with thermal fluctuations at equilibrium, $P_{th}^{eq}(T, d)$, can be split into contributions of propagating ($P_{th,PW}^{eq}$) and evanescent ($P_{th,EW}^{eq}$) waves:

$$P_{th,PW}^{eq} = -\frac{\hbar}{\pi^2}\int_0^\infty d\omega f(\omega,T)\int_0^{\omega/c} dQQq_z \sum_{\alpha=s,p}\frac{(r_1^\alpha r_2^\alpha e^{2iq_z d})' - |r_1^\alpha r_2^\alpha|^2}{|D_\alpha(Q,\omega)|^2}, \quad (100)$$

$$P_{th,EW}^{eq} = \frac{\hbar}{\pi^2}\int_0^\infty d\omega f(\omega,T)\int_{\omega/c}^\infty dQQq_z'' e^{-2q_z'' d} \sum_{\alpha=s,p}\frac{(r_1^\alpha r_2^\alpha)''}{|D_\alpha(Q,\omega)|^2}. \quad (101)$$

Here $f(\omega,T) = (e^{\hbar\omega/k_B T}-1)^{-1}$ is the Planckian distribution, Q and $q_z = q_z' + iq_z'' = (\omega^2/c^2 - Q^2)^{1/2}$, respectively, are the in- and out-plane components of the wavevector in the vacuum, and $D_\alpha(Q,\omega) = 1 - r_1^\alpha r_2^\alpha e^{2iq_z d}$. Also, it is possible to write the quantity $\Delta P(T,d)$ as the sum, $\Delta P(T,d) = \Delta P_{PW}(T,d) + \Delta P_{EW}(T,d)$ with

$$\Delta P_{PW}(T,d) = -\frac{\hbar}{4\pi^2}\int_0^\infty d\omega f(\omega,T)\int_0^{\omega/c} dQQq_z \sum_{\alpha=s,p}\frac{|r_2^\alpha|^2 - |r_1^\alpha|^2}{|D_\alpha(Q,\omega)|^2}, \quad (102)$$

$$\Delta P_{EW}(T,d) = \frac{\hbar}{2\pi^2}\int_0^\infty d\omega f(\omega,T)\int_{\omega/c}^\infty dQQq_z'' e^{-2q_z'' d} \sum_{\alpha=s,p}\frac{r_1^{\alpha\prime\prime} r_2^{\alpha\prime} - r_2^{\alpha\prime\prime} r_1^{\alpha\prime}}{|D_\alpha(Q,\omega)|^2}, \quad (103)$$

For more details of Eqs. (99)–(103), see Refs. [279, 280].

Note that for two identical bodies, the non-equilibrium pressure is just the average of the pressure at equilibrium, $\overline{P_{th}^{eq}}$. In the general case of different bodies, an additional contribution antisymmetric with respect to the interchange of materials arises, $\Delta P(T_1,d) - \Delta P(T_2,d)$, allowing tuning the force both in magnitude and sign.

Also, in contrast with the equilibrium case, the out-of-equilibrium pressure $P^{neq}(T_1,T_2,d)$ should be obtained within the real frequency domain, where the contribution of propagating and evanescent waves can be described using the different formulas (100)–(103) [280]. Since the evanescent part of the pressure disappears in the limit $d \to \infty$, the propagating part of the force still contributes with an d-independent component originating from a momentum transfer between the bodies in the non-equilibrium situation. Another d-independent contribution to the Casimir pressure given by $2\sigma(T_1^4 + T_2^4)/3c$ should be considered due to the pressure exerted on the most external surfaces of finite thickness bodies. Thus, the competing role played by propagating and evanescent modes, with contributions opposite in signs, shapes the resulting total Casimir pressure. This highlights the relevance of the dielectric response of materials since it determines the light–matter interaction that gives rise to different kind of modes that contributes to the thermal component of the Casimir force. Then tuning or switching the sense of the non-equilibrium Casimir force is possible not only by means of the temperature gradient, but also by the management of the electromagnetic response of the system, as observed in the following works.

Reference. [281] proposed the possibility of achieving repulsive Casimir forces by utilising an out-of-thermal-equilibrium configuration of two slabs made of aerogels that allows the dilution of their dielectric character and the tuning of their optical resonances within the thermal region by means of adjusting the porosity of the material.

In a plasmonic system comprising two parallel metal plates with permittivity described by a plasma model, a repulsive non-equilibrium Casimir force can be cre-

ated by a selective excitation of surface plasmons [282]. Other plasmonic materials with dielectric functions with a temperature dependence have been used to elucidate the non-equilibrium effects in the Casimir force. In Ref. [283], Ingold et al. considered two metallic plates characterized by temperature-dependent electronic relaxation rates, and observed different behavior in the non-equilibrium Casimir force compared to the case where temperature-independent dielectric functions were used. They predicted an out-of-equilibrium contribution to the Casimir pressure of up to 4% of the total. See Fig. 24a. Another system where dielectric response drastically changes with temperature was studied by Castillo et al. [284] by addressing the scenario of two high-T_c superconducting substates at two different temperatures. Here, a detailed study of the effects of the gradient temperature on the non-equilibrium Casimir pressure was presented for a wide range of separation distances; an abrupt change in the pressure was predicted at the superconducting transition as shown in Fig. 24b. In Ref. [285], for a setup of two slabs at the same temperature surrounded by an environment at a second temperature as Fig. 24c illustrates, Iizuka et al. showed that by placing a thin film of VO_2 on the top of the exterior surface of one of the slabs, the non-equilibrium Casimir force can be abruptly switched when the temperature of the slabs reaches the transition temperature, see Fig. 24d. This is an entirely out-of-equilibrium effect observed only when the temperature of the environment differs from that of the plates. Also, a study of the Casimir–Polder force between atoms and cavity walls coated with VO_2 films shows that the use of phase change materials significantly increases the strengths of the non-equilibrium effects, especially when the temperature of the walls increases by maintaining a constant temperature of the environment [286]. Thus, the pronounced change in the non-equilibrium Casimir force achieved using phase change materials may be used to elucidate the differentiation between equilibrium and non-equilibrium contributions to the force.

On the other hand, graphene-based structures have been used as a convenient platform to tune *in situ* the Casimir force by leveraging the graphene conductivity dependence on temperature and chemical potential. The Casimir pressure between two graphene layers at two different temperatures immersed in a thermal bath at a third temperature was investigated in Ref. [287], where the conductivity of graphene was described using the Kubo formula. This study shows that in non-equilibrium situations, the Casimir pressure can be tuned to exhibit a repulsive character or even a non-monotonic behavior with respect to chemical potential variations, which cannot be observed in the equilibrium configuration, see Fig. 25a–c. An alternative description of the electromagnetic response of graphene by means of the polarization tensor in the framework of the Dirac model was employed to study the Casimir–Polder interaction between a nanoparticle and a graphene layer, each at different temperatures [288]. When the temperature of the graphene layer is above that of the nanoparticle (the same as that of the environment), the non-equilibrium Casimir–Polder force results in an attractive interaction, which at a given separation distance vanishes when the temperature of graphene is below that of the nanoparticle, and becomes repulsive for larger separations. See Fig. 25b. Also, using the formalism of the polarization tensor, the non-equilibrium Casimir–Polder force between nanoparticles and plates coated with gapped graphene was considered [193, 194, 289, 290]. It is shown that the non-equilibrium force remains attractive in contrast with the case of pristine graphene

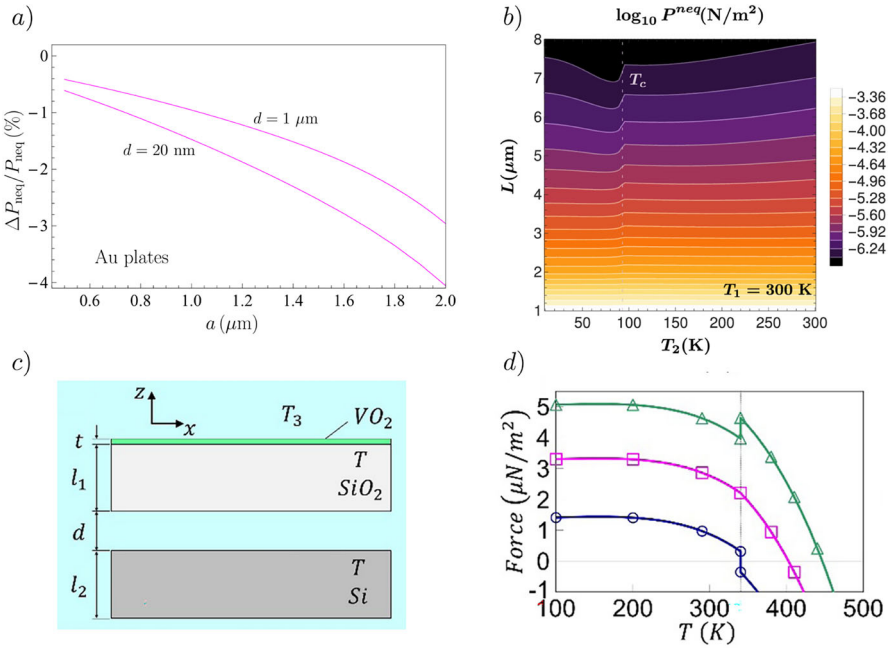


Fig. 24 (**a**) The ratio of the non-equilibrium contribution and the total Casimir non-equilibrium pressure is shown in percentages as a function of the separation for two Au plates with $d = 1\ \mu$m and 20 nm thicknesses. Reprinted with permission from G.-L. Ingold et al., Physical Review A 101,032506(2020). Copyright 2020 the American Physical Society. [283]. (**b**) Contour plot of the modulus of the non-equilibrium Casimir pressure (in logarithmic scale) P^{neq} between two plates made of high-T_c superconductors as a function of separation between plates L and the temperature of plate-2, T_2, for a fixed temperature of plate-1, $T_1 = 300$ K. Reprinted with permission from SG Castillo et al., Sci. Rep. (2022) 12:2905. [284] (**c**) Geometry of a system consisting of a SiO_2 slab and a silicon slab separated by a vacuum gap. A VO_2 thin film is attached to the exterior surface of the SiO_2 slab. The two bodies have the same temperature T and are immersed in the environment with the temperature T_3 ($d = 3.6\ \mu$m, $t = 100$ nm, and $l_1 = l_2 = 1$ mm). (**d**) Casimir force in the system of (**c**) in non-equilibrium as a function of the body temperature T at environment temperatures $T_3 = 100\ K$ (blue circles), 340 K (pink squares), and 400 K (green triangles), respectively. Reprinted with permission from [285] © Optical Society of America

[193], and that this interaction decreases with increasing energy gap [289, 290]. The combined effect of the energy gap, temperature, and a non-zero chemical potential of gapped graphene on the non-equilibrium Casimir force was presented in Ref. [194], observing an increase in the out-of-equilibrium Casimir–Polder with the chemical potential and the temperature, while the impact of the energy gap becomes favorable at lower temperatures. Subsequently, the spatial dispersion in the dielectric response of graphene was taken into account by the analytic form of the polarization tensor developed in Ref. [291], where the corrections to the non-equilibrium Casimir–Polder force were up to 9% where the spatial dispersion effects are considered as Fig. 25c demonstrates.

On the other hand, mathematical techniques have been developed to allow the study of the out-of-equilibrium Casimir force between two bodies of arbitrary geometries and dielectric properties, each placed at different temperatures different from those

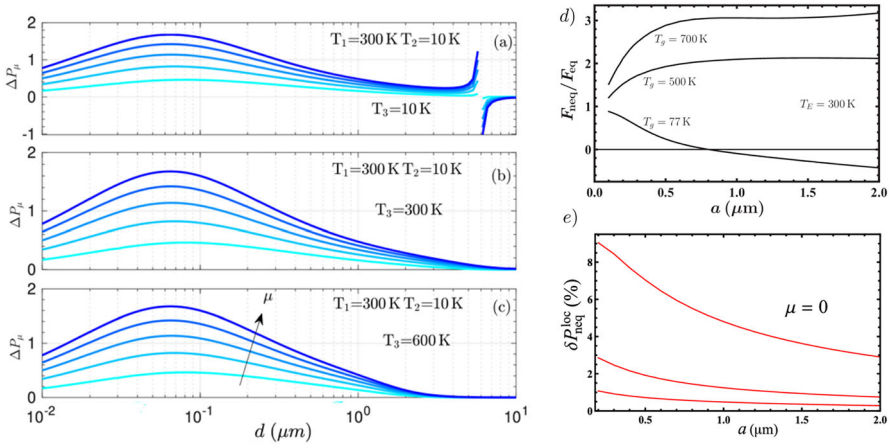


Fig. 25 (**a**)–(**c**) Relative variation of the Casimir pressure $\Delta P_\mu = [P_{1z}(\mu) - P_{1z}(\mu = 0)]/P_{1z}(\mu = 0)$ with respect to the chemical potential μ as a function of the distance between the two suspended graphene layers d with $T_1 = T_2$ indicated in the figure. Reprinted with permission from Y. Jeyar et al., Phys. Rev. B 108, 115412 (2023). Copyright 2023 the American Physical Society.[287]. (**d**) The ratio of non-equilibrium to equilibrium Casimir–Polder forces between a nanoparticle and a graphene sheet is shown as a function of separation by the three lines for different graphene temperatures T_g. In all cases, the temperatures of a nanoparticle and of the environment are equal to the temperature at thermal equilibrium T_E. Reprinted with permission from G. L. Klimchitskaya et al., Phys. Rev. B 105, 195430 (2022). Copyright 2022 the American Physical Society.[288] (**e**) The relative error in the non-equilibrium Casimir pressure for the graphene-coated plate heated to 500 K, which arises when using the spatially local approximation in the dielectric response of graphene coatings, is shown as the function of separation by the three lines counted from bottom to top for the values of the energy gap $\delta = 0.1,\ 0.2$, and $0.3\ eV$, respectively, and of the chemical potential $\mu = 0.25\ eV$. Reprinted with permission from G. L. Klimchitskaya et al., Phys. Rev. A 111, 012812 (2025). Copyright 2025 the American Physical Society [291]

of the surrounding environment [292–297]. Based on the scattering matrix approach, these formalisms are applied to study the nonequilibrium Casimir force between more complex structures such as nanostructured surfaces [292], cylinders [298], and gratings [299]. The underlying theory was also generalized to address the calculation of the Casimir force between more than two bodies with arbitrary shapes out of thermal equilibrium [297, 300], and the study of the force in a many-body system with planar geometry [301].

The studies mentioned above have been conducted under out-of-equilibrium conditions in stationary configurations. In addition, the influence of the thermal effects on the dynamical behavior of MEMS and NEMS subject to the omnipresent Casimir force has been investigated in the nonequilibrium situation where the interacting components have different temperatures [302, 303]. It has been predicted that nonequilibrium Casimir forces can push device actuation toward unstable and even chaotic behavior, in close correlation with the optical properties of their components, which are demonstrated by studying the dynamics of microdevices made of good (Au) and poor (doped SiC) conductive materials, both described by a Drude dielectric function [302]. Also, the sensitivity of out-of-equilibrium Casimir forces, and thus their influence on the

actuation of devices, on the optical properties of the materials at low frequencies was investigated by comparing results obtained with Drude and Plasma models [303].

From a technological point of view, it is interesting to create a configuration of multiple bodies that are subjected to nonequilibrium Casimir forces, exhibiting a stable mechanical equilibrium that prevents unstable or chaotic effects such as the stiction phenomenon (permanent adhesion), which ultimately hinders the operation of a device. Between two planar bodies, the out-of-equilibrium Casimir force typically leads the system to unstable mechanical states [285, 296]. However, as shown theoretically and numerically in Ref. [304], and extended to the case of multiple plates system [305], applying a feedback control processes that measures only one component of the physical state of a system (temperature, position, or velocity), it is possible to drive the system to a stable mechanical equilibrium state.

9 Casimir effect in time-varying media: photonic-time crystals

A time-varying medium is a material whose constitutive parameters depend explicitly on time, allowing the medium to exchange energy with electromagnetic (or other) fields. Temporal modulation breaks time-translation symmetry, enabling frequency conversion, non-reciprocity, and dynamic Casimir photon generation [306, 307]. In particular, we are interested in media with time-varying dielectric function $\varepsilon(t)$ and magnetic permeability $\mu(t)$. These systems are photonic analogs of the concept of time crystals proposed by Wilzcek [41]. In a time crystal, time-translational symmetry is broken unlike a normal crystal where space-symmetry is broken [308–310].

In such time-varying media, the electromagnetic constitutive relations are

$$\mathbf{D}(t) = \varepsilon_0 \int_{-\infty}^{t} \varepsilon(t,t')\,\mathbf{E}(t')\,dt', \qquad \mathbf{B}(t) = \mu_0 \int_{-\infty}^{t} \mu(t,t')\,\mathbf{H}(t')\,dt' \tag{104}$$

where the kernel $\varepsilon(t,t')$ and $\mu(t,t')$ that depend on both t and t', breaking time-translation symmetry, are usually represented by a kernel of the form $\varepsilon(t-t')$, for example. This leads to energy and momentum exchange between the field and the medium, producing phenomena such as frequency conversion, parametric amplification, and nonreciprocal wave propagation [311, 312]. When the modulation is periodic, $\varepsilon(t) = \varepsilon_0 + \Delta\varepsilon\cos(\Omega_m t)$, the material forms a temporal photonic crystal, whose band structure arises in the momentum domain rather than in frequency [313, 314].

Consider a Fourier transforms $\mathbf{E}(\omega) = \int_{-\infty}^{\infty} \mathbf{E}(t)\,e^{i\omega t}dt$ and similarly for $\mathbf{D}(\omega)$, and the two-time transform $\varepsilon(\omega,\Omega) = \iint \varepsilon(t,t')\,e^{i\omega t}e^{-i\Omega t'}\,dt\,dt'$. Applying these to Eq. (104)) and exchanging integrals gives

$$\mathbf{D}(\omega) = \varepsilon_0 \int_{-\infty}^{\infty} \varepsilon(\omega,\Omega)\,\mathbf{E}(\Omega)\,\frac{d\Omega}{2\pi}. \tag{105}$$

Equation (105) is a bilinear relation in frequency space: in general, an input at frequency Ω contributes to output at different frequency ω. The kernel $\varepsilon(\omega,\Omega)$ encodes frequency conversion.

For a linear, time-invariant medium, the response kernel depends only on the time difference, $\varepsilon(t, t') = \varepsilon(t - t') \equiv \varepsilon(\tau)$, and we implicitly refer to the retarded kernel, $\varepsilon(\tau) = 0$ for $\tau < 0$ (causality). In this case, the double-frequency representation is diagonal,

$$\varepsilon(\omega, \Omega) = 2\pi\, \varepsilon(\omega)\, \delta(\omega - \Omega). \tag{106}$$

This result is obtained by considering that if the medium is time-invariant (time-translation symmetric), the kernel depends only on the time difference,

$$\varepsilon(t, t') = \varepsilon_{ij}(t - t') \equiv \varepsilon(\tau). \tag{107}$$

We define the Fourier transform of the causal kernel by

$$\varepsilon(\omega) = \int_0^\infty d\tau\, \varepsilon(\tau)\, e^{i\omega\tau}, \tag{108}$$

where the lower limit 0 enforces causality. As a consequence, $\varepsilon(\omega)$ is analytic for $\mathrm{Im}\,\omega > 0$ and satisfies the Kramers–Kronig relations. For later use, it is sometimes convenient to introduce a double-frequency representation of the two-time kernel,

$$\varepsilon(\omega, \Omega) = \int_{-\infty}^\infty dt \int_{-\infty}^\infty dt'\, \varepsilon(t, t')\, e^{i\omega t}\, e^{-i\Omega t'}. \tag{109}$$

Substituting Eq. (107) and changing variables to $\tau = t - t'$ gives

$$\varepsilon(\omega, \Omega) = \int_{-\infty}^\infty dt'\, e^{i(\omega-\Omega)t'} \int_{-\infty}^\infty d\tau\, \varepsilon(\tau)\, e^{i\omega\tau} = 2\pi\, \varepsilon(\omega)\, \delta(\omega - \Omega), \tag{110}$$

which expresses frequency conservation in a time-invariant medium. Importantly, the $\delta(\omega - \Omega)$ factor follows solely from time-translation symmetry and does not conflict with causality; causality is encoded in the analyticity properties of $\varepsilon(\omega)$ through Eq. (108).

With the kernel given by Eq. (110), we recover from Eq. (105), to the familiar product

$$\mathbf{D}(\omega) = \varepsilon_0\, \varepsilon(\omega)\, \mathbf{E}(\omega), \tag{111}$$

for an isotropic medium, that we are considering in this section without loss of generality.

where $\varepsilon(\omega)$ is the Fourier transform of the causal kernel, and is therefore analytic in the upper half-plane (Kramers–Kronig consistency),

When the material is time-varying, time-translation symmetry is broken: $\varepsilon(t, t') \neq \varepsilon(t - t')$. The kernel $\varepsilon(\omega, \Omega)$ is no longer proportional to $\delta(\omega - \Omega)$, and thus couples different frequencies in (105). For a medium with an instantaneous but modulated permittivity,

$$\mathbf{D}(t) = \varepsilon_0\, \varepsilon(t)\, \mathbf{E}(t), \qquad \varepsilon(t) = \varepsilon_b + \Delta\varepsilon \cos(\Omega_m t), \tag{112}$$

where Ω_m is a constant frequency modulating the dielectric function. Fourier transforming the product (convolution theorem) gives

$$\mathbf{D}(\omega) = \varepsilon_0 \Big[\varepsilon_b \, \mathbf{E}(\omega) + \frac{\Delta\varepsilon}{2} \, \mathbf{E}(\omega - \Omega_m) + \frac{\Delta\varepsilon}{2} \, \mathbf{E}(\omega + \Omega_m) \Big]. \tag{113}$$

Thus, for a given frequency ω, modulation sidebands appear at $\omega_0 \pm \Omega_m$. More generally, for a periodic modulation $\varepsilon(t) = \sum_n \varepsilon_n e^{in\Omega_m t}$, one finds [313, 314]

$$\mathbf{D}(\omega) = \varepsilon_0 \sum_n \varepsilon_n \, \mathbf{E}(\omega - n\Omega_m), \tag{114}$$

which is precisely the frequency-mixing structure used in time-modulated media, temporal photonic crystals, and (in the quantum regime) the Hamiltonians underlying parametric amplification and the dynamical Casimir effect. When the modulation occurs on timescales comparable to or faster than the field oscillation period, the system enters the regime of parametric vacuum excitation, which is closely related to, but not identical with, the dynamic Casimir effect [315, 316].

A key open question is whether a time-dependent Casimir interaction can be observed directly between temporally modulated surfaces, without invoking the photon-pair creation mechanism that defines the dynamical Casimir effect (DCE). Thus, it is important to distinguish the two related but different scenarios:

1. **Casimir force in time-varying media:** The optical properties of the materials themselves (i.e., $\varepsilon(\omega, t)$) change in time due to external modulation or intrinsic dynamics, while the boundaries remain fixed. The resulting Casimir force $F_C(t)$ follows from the instantaneous or quasi-static fluctuation spectrum:

$$F_C(t) = \int_0^\infty \frac{d\zeta}{2\pi} \, \Phi[\varepsilon(i\zeta, t)], \tag{115}$$

where Φ denotes the standard Lifshitz kernel evaluated at the time-dependent dielectric function. Slow modulation ($\Omega_m \ll \omega_p$) yields an adiabatic Casimir response, while faster modulation produces retardation and non-equilibrium corrections [317]. In principle, such systems allow the measurement of a time-dependent Casimir pressure synchronized with the modulation of optical parameters, offering a route to probe dynamic vacuum susceptibilities.
An important aspect is that Eq.(115) is evaluated at the Matsubara frequency space, meaning that the Kramers–Kronig relations and, thus, Eq.(7) have to be satisfied. So we emphasize that Eq. (115) does not imply that a time-varying dielectric function $\varepsilon(\omega, t)$ automatically obeys the standard Kramers–Kronig relations. The usual Kramers–Kronig dispersion relations require the medium to be linear, causal, and time-invariant, so that the response function depends only on the time difference $\varepsilon(t, t') = \varepsilon(t - t')$ and can be represented by a single-frequency function $\varepsilon(\omega)$ that is analytic in the upper half of the complex frequency plane. In contrast, a time-varying medium breaks time-translation symmetry and the constitutive kernel becomes bi-temporal $\varepsilon(t, t')$, whose Fourier representation is a two-frequency

response $\varepsilon(\omega, \Omega)$ that mixes frequencies. In this case, the standard one-dimensional Kramers–Kronig relations no longer apply; instead, one must use the generalized causality relations developed for linear time-varying media by Sólis and Engheta [318]. Only in the quasi-static or adiabatic limit $\Omega_m \ll \omega$—when the modulation is slow compared to the optical response—can $\varepsilon(\omega, t)$ be treated at each instant as a stationary dielectric that locally satisfies the usual Kramers–Kronig relations. Thus, Eq. (115) should be interpreted as a time-parametrized Lifshitz formula rather than evidence that standard Kramers–Kronig relations hold in general for time-dependent materials.

2. **Dynamical Casimir (DCE):** Here, the time variation resides in the boundary conditions, such as a moving mirror or a rapidly modulated impedance, which alters the quantized mode structure itself. The DCE converts zero-point fluctuations into real photons when the boundary modulation frequency Ω satisfies the resonance condition. The essential signature is photon-pair creation, whereas the instantaneous Casimir pressure remains defined by static optical properties.

Hence, the DCE represents the quantum radiation limit of time-varying systems, while Casimir forces in time-varying media probe the non-equilibrium force response of fluctuating fields to parameter modulation. Both share a common mathematical framework (temporal Bogoliubov transformations), but differ in the physical observables: photon emission vs. modulated pressure.

Directly measuring the Casimir force between time-varying or time-crystalline materials remains an open challenge. Ultrafast pump-probe techniques, capable of modulating carrier densities or phonon populations on sub-picosecond scales, may enable a transient modification of $\varepsilon(\omega, t)$ in metals or semiconductors. Optomechanical force sensors could then track the synchronous variation of Casimir pressure as the dielectric environment evolves in time. Unlike the DCE, such experiments would not rely on photon emission but rather on force modulation—probing the instantaneous vacuum stress tensor in a temporally varying background. This regime constitutes a new frontier between static fluctuation forces and quantum field dynamics, linking the Casimir effect to the broader physics of non-equilibrium and time-symmetric quantum materials.

10 Conclusion and outlook

Over the past 7 decades, the study of Casimir forces has evolved from the idealized prediction between perfectly conducting plates to a broad and interdisciplinary field at the interface of quantum field theory, condensed matter physics, nanotechnology, and materials science. Advances in both experiment and theory have established Casimir forces as a sensitive probe of electromagnetic fluctuations, material response, and geometry at the nanoscale.

The historical trajectory—from early measurements with torsional pendulums and AFM-based techniques, to precision torsional oscillators and MEMS platforms—has provided benchmarks for testing the Lifshitz formalism and clarifying debates such as the Drude versus plasma model controversy. Meanwhile, theoretical frameworks have

matured, ranging from Lifshitz theory and its generalizations to numerical methods such as BEM and FDTD, which enable accurate predictions in complex geometries and materials.

Casimir forces have also become increasingly relevant for practical applications. In micro- and nanoelectromechanical systems, they can either pose challenges (stiction, dissipation) or be harnessed as actuation mechanisms. The possibility of tailoring dispersion forces via surface structuring, metamaterials, or intervening fluids has shifted the perspective from Casimir forces as unavoidable background interactions to controllable design parameters. The demonstration of Casimir–Lifshitz repulsion and the growing ability to engineer optical properties of media point toward functional Casimir engineering.

Perhaps most exciting are the frontiers that link Casimir physics with nonequilibrium and temporally modulated systems. The dynamical Casimir effect, periodically driven materials, and proposals involving time crystals suggest that quantum vacuum forces are not only tunable in magnitude and sign, but may also acquire nontrivial temporal structure. These directions merge fluctuation-induced forces with the broader themes of topological phases, quantum thermodynamics, and time-dependent quantum matter.

Several challenges remain. Experimentally, pushing measurements into new regimes—ultrashort distances, cryogenic conditions, strongly anisotropic or topological media—will be crucial. Theoretically, reconciling different approaches to nonlocality, roughness, and finite-temperature response remains an open task. On the applied side, integrating Casimir-aware design principles into nanosystems could unlock novel device functionalities.

In summary, Casimir forces stand as an example of how quantum fluctuations, once regarded as a theoretical curiosity, have become central to modern physics and engineering. From noble metals to time crystals, the field continues to reveal new connections, new controversies, and new opportunities, ensuring that Casimir physics will remain a vibrant area of research in the decades to come.

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Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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